

Factorization Algebras in Quantum Field Theory - Self Study

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Abstract

This document contains notes from my reading of Kevin Costello and Owen Gwilliam's *Factorization algebras in quantum field theory* ([CG16, CG20]). My readings of this book serve as a continuation of a course I took in Spring 2026, taught by Professor [Matiej Szczesny](#). It has been about 2 months since I actively worked on this content, so I will complete a brief review of material we covered in class, starting from Chapter 4 on free field theories. My goals are to finish volume 1 by the end of the summer, and preferably begin volume 2.

Notation. Given a smooth manifold M , the sheaf of smooth functions will be denoted C_M^∞ . If M is Riemannian, the metric will be denoted g , unless otherwise noted. The sheaf of k -forms will be denoted Ω_M^k . Given a fiber bundle E , we generally denote the sheaf of sections $\Gamma(-, E)$ by $\mathcal{E}$. Given a graded vector bundle E , we also denote the sheaf of sections $\Gamma(-, E)$ by $\mathcal{E}$. We use cohomological grading conventions. The action functional will be denoted S , and is considered to be a morphism of presheaves from an appropriate sheaf $\mathcal{E}$ of fields to the constant presheaf $\mathbb{R}$, and we denote by S_U the component of this morphism at the open set U . If X is a complex manifold, the sheaf of holomorphic functions is denoted $\mathcal{O}_X$.

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Part I

Prefactorization algebras

Chapter 3

Prefactorization algebras and basic examples

We covered the majority of chapter 3 in my spring 2026 course, but we skipped the machinery of differentiable vector spaces. Although this *can* be mostly treated as a technical point, I wanted to understand it more completely, so I have reviewed it here.

3.5 Some functional analysis

3.5.1 Differentiable vector spaces

We introduce the preliminary notion of a *site*. This is mostly a technical point required to properly formulate the notion of a sheaf on a category. The intuition is that we wish to view a category $\mathcal{C}$ in a similar way to the category $\text{Open}(X)$ of open sets on a space X , so that we may define sheaves on $\mathcal{C}$. Definitions 3.5.1 to 3.5.5 come from [wikipedia](#).

Definition 3.5.1. Let $\mathcal{C}$ be a category and $c \in \mathcal{C}$. A *sieve* on c is a subfunctor of the representable functor $\mathcal{C}(-, c)$. Given a sieve S on $x \in \mathcal{C}$, and a morphism $f : y \rightarrow x$ in $\mathcal{C}$, postcomposition yields a sieve f^*S on y , called the *pullback of S along f* . It is explicitly described as

$$(f^*S)(c) = \{g : c \rightarrow y \mid f \circ g \in S(c)\}.$$

In the category $\text{Open}(X)$, all hom-sets are singletons or empty. Thus a sieve S on U amounts to a collection $\{V_i\}_{i \in I}$ of subsets of U ; we then define $S(V_i)$ to be the only nonempty values of S . Pulling this sieve back along an inclusion $W \subseteq U$ corresponds to the sieve for the collection $\{V_i \cap W\}_{i \in I}$.

Using the fact that in $\text{Open}(X)$ sieves correspond to choices of subsets, we can try to view sieves as a more general notion of an open cover. Not all sieves in an arbitrary category are suitable for this purpose; we must specify a collection of *covering sieves* for some $c \in \mathcal{C}$. This is formalized by a Grothendieck topology.

Definition 3.5.2. A *Grothendieck topology* J on $\mathcal{C}$ is a collection $\{J(c)\}_{c \in \mathcal{C}}$ of distinguished sieves. We call $J(c)$ the *covering sieves* of c . This selection must satisfy the following three axioms.

- Pullbacks of covering sieves are covering sieves. In other words, if $f : y \rightarrow x$ is a morphism, then for any $S \in J(x)$, we have $f^*S \in J(y)$.
- Let S be a covering sieve on x , and let T be any sieve on x . If for any $f : y \rightarrow x$ in $S(y)$, the pullback f^*T is a covering sieve on y , then it follows that T is a covering sieve on x .
- The representable functor $\mathcal{C}(-, x)$ is a covering sieve for x .

A category $\mathcal{C}$ equipped with a Grothendieck topology J is called a *site*.

Following the intuition of $\text{Open}(X)$, we can write down what these axioms intuitively mean. The first axiom represents the idea that if $\{U_i\}_{i \in I}$ is a cover of U , then $\{U_i \cap V\}_{i \in I}$ covers V . The second represents the idea that if $\{U_i\}_{i \in I}$ is a cover for U and $\{V_{ij}\}_{j \in J}$ covers U_i for each i , then $\{V_{ij}\}_{i \in I, j \in J}$ covers U . The third axiom simply states that U covers U . We can write this data more topologically in some cases.

Definition 3.5.3. A covering family for $c \in \mathcal{C}$ is a collection of morphisms which all map to c , but need not have the same domain. A *Grothendieck pretopology* is a collection of distinguished covering families (simply referred to as covering families for simplicity) that satisfy the following axioms.

1. If $x \in \mathcal{C}$, then for all $x_0 \rightarrow x$ in some covering family of x and all morphisms $y \rightarrow x$, the pullback $x_0 \times_x y$ exists.
2. For all $x \in \mathcal{C}$, all $y \rightarrow x$, and all covering families $\{x_\alpha \rightarrow x\}_{\alpha \in A}$, the family $\{x_\alpha \times_x y \rightarrow y\}_{\alpha \in A}$ is a covering family.
3. If $\{x_\alpha \rightarrow x\}_{\alpha \in A}$ is a covering family, and if $\{x_{\beta\alpha} \rightarrow x_\alpha\}_{\beta \in B_\alpha}$ is a covering family for all α , then the composites $\{X_{\beta\alpha} \rightarrow X_\alpha \rightarrow X\}_{\alpha \in A, \beta \in B_\alpha}$ is a covering family.
4. If $f : Y \cong X$ is an isomorphism, then $\{f\}$ is a covering family.

Given a pretopology for $\mathcal{C}$, the collection of all sieves that contain a covering family from the pretopology is always a Grothendieck topology.

Definition 3.5.4. A *presheaf* on a category $\mathcal{C}$ is a contravariant functor. A *sheaf* $\mathcal{F}$ on a site $(\mathcal{C}, J)$ is a presheaf F such that for all $x \in \mathcal{C}$ and all $S \in J(x)$, the map

$$i^* : \text{Nat}(\mathcal{C}(-, x), \mathcal{F}) \rightarrow \text{Nat}(S, \mathcal{F})$$

induced by the inclusion $i : S \subseteq \mathcal{C}(-, x)$ is an isomorphism. We generally refer to the image of a sheaf on a morphism f by f^* , and call it a *pullback map*.

Definition 3.5.5. Let X and $\{V_\alpha\}_{\alpha \in A}$ be spaces. A collection of maps $\{u_\alpha : V_\alpha \rightarrow X\}_{\alpha \in A}$ is *jointly surjective* if

$$\bigcup_{\alpha \in A} u_\alpha(V_\alpha) = X.$$

Define the pretopology on Man by taking covering families to be jointly surjective families whose members are all open immersions. Covering sieves S for X are then sieves such that

- For all manifolds Y and smooth maps $f : Y \rightarrow X$, there is a manifold V and an open immersion $g : V \rightarrow X$ such that $g \in S(V)$ and f factors through g (i.e. $f = g \circ \tilde{f}$).
- If W is the union of all sets $f(Y)$, where Y is a manifold and $f \in S(Y)$, then $W = X$.

We may now begin working towards the definition of a differentiable vector space.

Definition 3.5.6. Let C^∞ denote the sheaf of rings on Man that maps a manifold M to the commutative $\mathbb{R}$ -algebra $C^\infty(M)$. A C^∞ -module $\mathcal{F}$ is a sheaf of abelian groups on Man which is a module over C^∞ in the usual sense: for any manifold M , the abelian group $\mathcal{F}(M)$ is a $C^\infty(M)$ -module, and for any $f : M \rightarrow N$, the pullback map $f^* : \mathcal{F}(N) \rightarrow \mathcal{F}(M)$ is a morphism of $C^\infty(N)$ -modules.

For example, given a topological vector space V , there is a natural notion of a smooth map¹ $M \rightarrow V$. Thus we may define a C^∞ -module $C^\infty(-, V)$. Being even more specific, let V be the space of smooth maps $C^\infty(N)$ for some manifold N , equipped with the usual Fréchet topology. It is possible to show that

$$C^\infty(M, C^\infty(N)) \cong C^\infty(M \times N).$$

Sheaves of C^∞ -modules that arise from topological vector spaces are endowed with extra structure: we can always differentiate smooth maps $M \rightarrow V$. Differentiation can be viewed as an action of vector fields $\mathfrak{X}(M)$ on the space $C^\infty(M, V)$: in coordinates, $X \in \mathfrak{X}(M)$ is written as $\sum_i X^i \partial_i$, so we simply take partial derivatives in coordinates. Dually, we can view derivatives as coming from a connection

$$\nabla : C^\infty(M, V) \rightarrow \Omega^1(M, V),$$

where

$$\Omega^1(M, V) := \Omega^1(M) \otimes_{C^\infty(M)} C^\infty(M, V).$$

This tensor product is the usual algebraic tensor product, which is well-behaved since $\Omega^1(M)$ is a projective $C^\infty(M)$ -module of finite rank. I believe the definition is

$$\nabla f := df = \sum_i \frac{\partial f}{\partial x^i} \otimes dx^i =: \sum_i \frac{\partial f}{\partial x^i} dx^i$$

in coordinates. This would agree with what the book says next: $\nabla \circ \nabla = 0$, meaning the connection is flat. Note that evaluating this connection on a vector field yields

$$\nabla_X f = (df)(X) = \sum_i \frac{\partial f}{\partial x^i} (dx^i X) = \sum_i X^i \frac{\partial f}{\partial x^i},$$

since $dx^i X := X^i$. Flatness ensures that the action of vector fields is a Lie algebra action, so we get an algebra of *all* differential operators on M , not just vector fields on M .

Definition 3.5.7. Let Ω^1 denote the C^∞ -module $M \mapsto \Omega^1(M)$. For F any C^∞ -module, the C^∞ -module of k -forms valued in F , denoted $\Omega^k(F)$, is the (algebraic) tensor product

$$M \mapsto \Omega^k(M, F) := \Omega^k(M) \otimes_{C^\infty_M} F(M).$$

A *connection* on a C^∞ -module F is a map of sheaves on Man

$$\nabla : F \rightarrow \Omega^1(F),$$

that satisfies the Leibniz rule for any manifold M .

A *differentiable vector space* F is a C^∞ -module F equipped with a flat connection. A map $f : F \rightarrow G$ of differentiable vector spaces is a morphism of C^∞ -modules that intertwines with the flat connections. The category of differentiable vector spaces is denoted $\mathcal{DVS}$.

Note that a differentiable vector space $\mathcal{F}$ is a C^∞ -module. However, since $C^\infty(M)$ is an $\mathbb{R}$ -algebra, we see that $\mathcal{F}(M)$ is an $\mathbb{R}$ -vector space with $\mathbb{R}$ -action defined as follows. The multiplicative unit of $C^\infty(M)$ is the function $1 : M \rightarrow \mathbb{R}$ which is constant at $1 \in \mathbb{R}$. Since a module can be written as a ring homomorphism $C^\infty(M) \rightarrow \text{End}(\mathcal{F}(M))$, we see that 1 is a scalar multiplicative unit since ring homomorphisms preserve 1 . Now for $v \in \mathcal{F}(M)$ and $c \in \mathbb{R}$, define $c \cdot v := (c1)v$.

¹For example, let $\Omega \subseteq \mathbb{R}^n$, and use the topology on V to define a derivative of a map $\Omega \rightarrow V$ at some $h \in \mathbb{R}^n$ with the usual limit definition. Then raise this definition to manifolds using local coordinates.

Most of the differentiable vector spaces we will work with are concrete in nature. Let $\mathcal{F}$ be a sheaf of abelian groups on $\mathbf{Man}$, and postcompose with the forgetful functor $U : \mathbf{Ab} \rightarrow \mathbf{Set}$ to view $\mathcal{F}$ as taking values in $\mathbf{Set}$. Then there is a natural map $\mathcal{F}(M) \rightarrow \mathbf{Set}(M, \mathcal{F}(*))$. This map is defined as follows. Given $p \in M$, there is a smooth map $* \rightarrow M$ mapping $*$ to p . Write ε_p for this map. Since $\mathcal{F}$ is a sheaf, there is an induced pullback map $\varepsilon_p^* : \mathcal{F}(M) \rightarrow \mathcal{F}(*)$. Now define the map $f : \mathcal{F}(M) \rightarrow \mathbf{Set}(M, \mathcal{F}(*))$ by, writing f_x for the function $f(x) : M \rightarrow \mathcal{F}(*)$,

$$f_x(p) := \varepsilon_p^*(x).$$

We say a sheaf $\mathcal{F}$ is *concrete* if this map is injective. The idea of a concrete sheaf is that a section over M is a “smooth” map $M \rightarrow \mathcal{F}(*)$. For example, define the sheaf $\underline{X} = C^\infty(-, X)$ for a manifold X . It is quite clear that $\underline{X}$ is concrete: a section is, by definition, a function $M \rightarrow X$, and $\mathcal{F}(*) = C^\infty(*, X) \cong X$. Hence, this sheaf justifies which set-theoretic functions are smooth. Moreover, as noted earlier, $\mathcal{F}(*)$ is guaranteed to be a vector space, and the spaces $C^\infty(M, V)$ are also vector spaces, as expected.

We generally work with the differentiable vector space arising from a topological vector space V , which justifies which maps to V are smooth. For this reason, we will usually think of a differentiable vector space $\mathcal{F}$ as an ordinary vector space, given by its value $V = \mathcal{F}(*)$, and will think of sections $\mathcal{F}(M)$ as smooth maps $C^\infty(M, V)$. We often use the notation $C^\infty(M, V)$ for $\mathcal{F}(M)$. We abuse notation further and refer to maps of differentiable vector spaces as *smooth maps*.

Variant 3.5.8. There is a Grothendieck topology on the category $\mathbf{CMan}$ of complex manifolds and holomorphic maps, and a sheaf $\mathcal{O} : \mathbf{CMan} \rightarrow \mathbf{Ring}$ of holomorphic functions to $\mathbb{C}$. The analog for a differentiable vector space in the complex world is an $\mathcal{O}$ -module with a flat connection, which we will call a *holomorphic vector space*. Unfortunately, holomorphic vector spaces are poorly behaved in homological algebra. The main reason differentiable vector spaces are nice is because they are *fine*, meaning they admit a partition of unity in a particular sense, and the global sections functor $\Gamma(M, -)$ is *exact* for fine sheaves. Of course, you cannot define a partition of unity using holomorphic functions, so this fails. This is a big issue, since a local quasi-isomorphism no longer yields a global quasi-isomorphism in this world, and taking global sections over a manifold introduces large issues. We fix this issue by not working with holomorphic vector spaces. As we will see in chapter 5, the main trick for avoiding this issue is by passing to the Dolbeault complex.

3.5.2 Differentiable cochain complexes

The category $\mathcal{DVS}$ is much nicer than the category of topological vector spaces, so we usually use it, even if most of the objects we work with come from topological vector spaces. The key reason for this is that homological algebra is well-developed for sheaves, even those on a site, which solves many of the issues of doing homological algebra with infinite-dimensional vector spaces.

Definition 3.5.9. A *differentiable cochain complex* is a cochain complex in the category of differentiable vector spaces. A cochain map $f : V \rightarrow W$ of differentiable cochain complexes is a *quasi-isomorphism* if it is a quasi-isomorphism pointwise. This condition is equivalent to asking that the map be a quasi-isomorphism at the level of stalks (usually this is not an equivalence, just an implication).

The category of differentiable cochain complexes is denoted $\mathbf{Ch}_{\mathcal{DVS}}$, but may sometimes be denoted $\mathbf{Ch}_{\mathbb{R}}$ in abuse of notation. It is enriched over itself in the usual way.

Discussion 3.5.10. It is useful to further discuss quasi-isomorphisms in this category. Let $f : C^\bullet \rightarrow D^\bullet$ be a morphism of differentiable cochain complexes. The data of such a morphism can be expressed in two equivalent ways:

- For every $i \in \mathbb{Z}$, a morphism $f^i : C^i \rightarrow D^i$ of differentiable vector spaces;
- For every manifold M , a morphism $f_M : C^\bullet(M) \rightarrow D^\bullet(M)$ of cochain complexes of $\mathbb{R}$ -vector spaces (or $C^\infty(M)$ -modules, or abelian groups).

Note that the second condition actually preserves exactness of sequences. Note that although we think of $C^\bullet(*)$ as the vector space itself, a quasi-isomorphism $C^\bullet(*) \simeq D^\bullet(*)$ does not yield a quasi-isomorphism $C^\bullet \simeq D^\bullet$. One must show that the choices of smooth maps are also quasi-isomorphic.

3.5.3 Differentiable prefactorization algebras

To define a prefactorization algebra valued in $\mathcal{DVS}$ or $\mathcal{Ch}_{\mathcal{DVS}}$, we need a colored operad structure on $\mathcal{DVS}$. Since intuitively a differentiable vector space is just a vector space with a distinguished collection of smooth maps, it would make sense to define a tensor product using multilinearity.

Definition 3.5.11. Let $\{V_i\}_{1 \leq i \leq n}$ be differentiable vector spaces. A *smooth multilinear map*

$$\Phi : V_1 \times \cdots \times V_n \rightarrow W$$

is a C^∞ -multilinear map Φ of sheaves that satisfies the following Leibniz rule with respect to the connections on V_i and W . For every manifold M , and for every $v_i \in V_i(M)$, we require that

$$\nabla \Phi(v_1, \dots, v_n) = \sum_{i=1}^n \Phi(v_1, \dots, \nabla v_i, \dots, v_n) \in \Omega^1(M, W).$$

We let $\mathcal{DVS}(V_1, \dots, V_n; W)$ denote the space of smooth multilinear maps.

Discussion 3.5.12. I expand on the definition somewhat. First, recall that since sheaves are functors whose forgetful functor to the functor category preserves limits, by pointwise computation of limits in functor categories we see that

$$(V_1 \times \cdots \times V_n)(M) \cong V_1(M) \times \cdots \times V_n(M).$$

Note that Φ , presented as a map from the product, is not a map of sheaves: we would need Φ_M to be a morphism of $C^\infty(M)$ -modules, but it is $C^\infty(M)$ -multilinear, not linear. However, we factor it through the tensor product to redefine it as a map of sheaves

$$\Phi : V_1 \otimes_{C^\infty} \cdots \otimes_{C^\infty} V_n \rightarrow W.$$

Thus it is indeed a morphism of sheaves.

This yields a colored operad structure for $\mathcal{DVS}$, which promotes to a colored operad structure for $\mathcal{Ch}_{\mathcal{DVS}}$ by grading the tensor product over C^∞ . To be more precise, given differentiable cochain complexes $V_i^\bullet, W^\bullet$ for $1 \leq i \leq n$, a smooth multilinear map is simply a map of cochain complexes

$$\Phi^\bullet : V_1^\bullet \otimes_{C^\infty} \cdots \otimes_{C^\infty} V_n^\bullet \rightarrow W^\bullet$$

which is compatible with the connections pointwise, meaning for any k , the map

$$\Phi^k : \bigoplus_{a_1 + \cdots + a_n = k} (V_1^{a_1} \otimes_{C^\infty} \cdots \otimes_{C^\infty} V_n^{a_n}) \rightarrow W^k$$

is a smooth multilinear map of differentiable vector spaces.

Definition 3.5.13. A *differentiable prefactorization algebra* is a prefactorization algebra in $\mathcal{Ch}_{\mathcal{DVS}}$.

In other words, a differentiable prefactorization algebra $\mathcal{F}$ on a space X assigns a differentiable cochain complex $\mathcal{F}(U)$ to every open set $U \subseteq X$. We intuitively think of $\mathcal{F}(U)$ as the cochain complex of vector spaces $\mathcal{F}(U)(*)$, and simply note that the datum of a (quasi-iso)morphism carries the structure of the smooth maps on $\mathcal{F}(U)(*)$. This philosophy is not only well-supported by the theory, but is well-supported by our use case: the main reason for defining differentiable vector spaces is not for the added mobility of being able to choose smooth maps on a vector space, but rather for being able to work with a (usually canonical) smooth structure on fairly tame topological vector spaces.

3.5.4 Relationship with topological vector spaces

Recall that a topological vector space is just a vector space with a topology such that $+, \cdot$ are continuous. We have seen that we may view (sufficiently well-behaved) topological vector spaces as differentiable vector spaces in at least one way. This method is fairly canonical:

Theorem 3.5.14. Let $\mathbf{LCTVS}$ denote the category of locally-convex Hausdorff topological vector spaces, and continuous maps. Let $\mathbf{BVS}$ denote the category with the same objects, and morphisms given by bounded linear maps. The functor $\mathbf{LCTVS} \rightarrow \mathbf{DVS}$ embeds $\mathbf{BVS} \subseteq \mathbf{LCTVS}$ as a full sub-colored operad of $\mathbf{DVS}$.

This theorem tells us that viewing topological vector spaces as differentiable vector spaces does not lose much information. However, we should also say something about completeness for a topological vector space.

Definition 3.5.15. A topological vector space $V \in \mathbf{BVS}$ is c^∞ -complete, or *convenient*, if every smooth map $c : \mathbb{R} \rightarrow V$ has an antiderivative. The category of convenient vector spaces and bounded linear maps is denoted $\mathbf{CVS}$.

This definition is somewhat weaker than the usual notion of completeness for a topological vector space. Thus, all complete topological vector spaces are c^∞ -complete, including Banach and Hilbert spaces. The usefulness of this approach is it focuses only on differentiability.

Proposition 3.5.16. The full subcategory $\mathbf{CVS} \subseteq \mathbf{DVS}$ is closed under small limits, countable coproducts, and sequential colimits of closed embeddings.

We give $\mathbf{CVS}$ a colored operad structure inherited from $\mathbf{DVS}$ and $\mathbf{BVS}$. We will now recast this colored operad structure as a particular tensor product on the category.

Theorem 3.5.17. The colored operad structure on $\mathbf{CVS}$ is represented by a symmetric monoidal structure with tensor product denoted by $\widehat{\otimes}_\beta$.

This symmetric monoidal structure is called the *completed bornological tensor product*. If $E, F \in \mathbf{CVS}$, this completed bornological tensor product is written as $E \widehat{\otimes}_\beta F$. ?? means that smooth multilinear maps $f : E_1 \times E_2 \rightarrow F$ are the same as smooth/bounded linear maps $f : E_1 \widehat{\otimes}_\beta E_2 \rightarrow F$. When no confusion may arise, we often write $\otimes$ instead of $\widehat{\otimes}_\beta$.

3.5.5 Examples from differential geometry

The following examples will form the building blocks of most factorization algebras we will consider. Much of the theory reduces to that of usual vector spaces.

Let E be a vector bundle on a manifold X . Let $\mathcal{E}$ denote the sheaf of sections, and $\mathcal{E}_c$ the cosheaf of compactly supported sections. We construct DVS structures for the vector spaces of sections of these objects.

Let a smooth map $M \rightarrow \mathcal{E}$ be defined as a smooth section of the bundle $\pi_X^* E$ on $M \times X$. Thus the set of smooth maps $C^\infty(M, \mathcal{E}(X))$ is defined as

$$C^\infty(M, \mathcal{E}(U)) := \Gamma(M \times U, \pi_X^* E).$$

The map $U \mapsto C^\infty(-, \mathcal{E}(U))$ is a sheaf on X valued in C^∞ -modules $M \mapsto C^\infty(M, \mathcal{E}(U))$. Fixing U , define a flat connection on this sheaf as follows. Let s be a section of $\pi_X^* E$ over $M \times U$, and note that $(\pi_X^* E)_{(m,x)} = E_x$. Choose a point (m, x) , and choose coordinates $\{m^i, x^j\}$ for a neighborhood of this point. Choose a local frame $\{e^i\}$ for E , and thus $\pi_X^* E$, on these coordinates. Then s can be written as

$$s_{(m,x)} = \sum_i s_i(m, x) e_x^i$$

for some component functions s_i . Since the vector spaces do not change as we move along M , it makes sense to define the connection by differentiating along the coordinates for M . Indeed, define

$$\nabla s := \sum_j \left(\sum_i \frac{\partial s_i}{\partial m^j} \otimes e_x^i \right) \otimes dm^j.$$

Thus, $\mathcal{E}$ is a sheaf of differentiable vector spaces.

The cosheaf $\mathcal{E}_c$ is defined similarly. Smooth maps $M \rightarrow \mathcal{E}_c(U)$ are smooth sections of the same bundle whose support maps properly to M . Here, the word “properly” means proper in the sense of a proper map, meaning the projection $\pi_M|(M \times \text{Supp } s) : M \times \text{Supp } s \rightarrow M$ is proper (preimages of compact sets are compact). This yields a cosheaf on X valued in differentiable vector spaces (the connection is the same).

Theorem 3.5.18. *With this differentiable structure, $\mathcal{E}(X)$ and $\mathcal{E}_c(X)$ are in the full subcategory $\mathcal{CVS}$ of convenient vector spaces. Further, since any open subset $U \subseteq X$ is a submanifold in its own right, $\mathcal{E}$ and $\mathcal{E}_c$ promote to (co)sheaves valued in convenient vector spaces. Finally, these differentiable structures are the same as those arising from the natural topologies on sections of $\mathcal{E}, \mathcal{E}_c$.*

Let E be a vector bundle over X . Denote by $\bar{\mathcal{E}}$ the *distributional sections* of E , defined by

$$\bar{\mathcal{E}} := \mathcal{E} \otimes_{C_X^\infty} \mathcal{D},$$

where $\mathcal{D}(U)$ is the space of distributions on $U \subseteq X$. Similarly, let $\bar{\mathcal{E}}_c$ denote the compactly supported distributional sections of E on X . We define the inclusions

$$\begin{aligned} \mathcal{E}_c &\hookrightarrow \bar{\mathcal{E}}_c \hookrightarrow \bar{\mathcal{E}}, \\ \mathcal{E}_c &\hookrightarrow \mathcal{E} \hookrightarrow \bar{\mathcal{E}} \end{aligned}$$

by viewing smooth functions as distributions. More precisely, if 1 is the constant function $X \rightarrow \mathbb{R}$ at 1 , and $\iota : C_X^\infty \rightarrow \mathcal{D}$ is the standard inclusion, then the inclusion $\mathcal{E} \hookrightarrow \bar{\mathcal{E}}$ is the map $s \mapsto s \otimes \iota(1)$. Note that distributions on manifolds are the dual of top forms. This inclusion effectively explains all the other non-obvious inclusions as well.

Let $E^! := E^* \otimes \text{Dens}_X$, and note that there is a pairing $\text{ev} : E \otimes E^! \rightarrow \text{Dens}_X$. The differential structure on $\bar{\mathcal{E}}, \bar{\mathcal{E}}_c$ can now be described as follows.

Theorem 3.5.19. *For any manifold M , a smooth map $M \rightarrow \overline{\mathcal{E}}(U)$ is the same as a smooth linear map*

$$\mathcal{E}_c^!(U) \rightarrow C^\infty(M).$$

Similarly, a smooth map $M \rightarrow \overline{\mathcal{E}}_c(U)$ is the same as a smooth linear map

$$\mathcal{E}^! \rightarrow C^\infty(M).$$

3.5.6 Multilinear maps and enriched spaces of maps

The category $\mathcal{DVS}$ also has a natural tensor product and internal hom. If V, W are differentiable vector spaces, then since they are C^∞ -modules, we might define the tensor product to be $\otimes_{C^\infty}$. Indeed, it inherits a new flat connection $\nabla_V \otimes 1_W + 1_V \otimes \nabla_W$. However, this tensor product is poorly behaved in analysis. For example, if M, N are manifolds, then

$$C^\infty(M) \otimes_{C^\infty} C^\infty(N) \neq C^\infty(M \times N)$$

in general. This is an issue, since this is what we would expect from an appropriately completed tensor product. Thus, the colored operad structure is preferable to the tensor product. When restricting to convenient vector spaces, we may freely use the tensor product in $\mathcal{CVS}$, which coincides with the colored operad structure. Oftentimes we will still consider a tensor product of differentiable vector spaces in the case that they are *convenient* vector spaces. This is a fair use of notation by theorem 3.5.17.

There is an internal hom $\mathcal{H}om_{\mathcal{DVS}}(V, W)$ given by the sheaf hom. However, the sheaf hom does not really capture the notion of “smooth families of maps”, so this definition doesn’t make much sense. Instead, we use a different enrichment.

Definition 3.5.20. Let V be a differentiable vector space and M a manifold. Write $\mathbf{C}^\infty(M, V)$ for the differentiable vector space

$$N \mapsto C^\infty(N \times M, V)$$

with connection

$$\nabla_{N, \mathbf{C}^\infty(M, V)} : C^\infty(N, \mathbf{C}^\infty(M, V)) \rightarrow \Omega^1(N, \mathbf{C}^\infty(M, V))$$

given by the composition of the flat connection

$$\nabla_{N \times M} : C^\infty(M \times N, V) \rightarrow \Omega^1(M \times N, V) \cong \Omega^1(M, \mathbf{C}^\infty(N, V)) \oplus \Omega^1(N, \mathbf{C}^\infty(M, V))$$

with the projection onto $\Omega^1(N, \mathbf{C}^\infty(M, V))$.

Here we use $T_{M \times N}^* \cong \pi_M^* T^* M \oplus \pi_N^* T_N^*$. This makes $\mathcal{DVS}$ into a category cotensored over $\mathbf{Man}$. Note that $C^\infty(M)$ is a ring object in $\mathcal{DVS}$, so we refer to it as a *differentiable ring*.

This construction is fairly intuitive. We want $C^\infty(M, V)$ to be a differentiable vector space itself. We would want to think of a smooth map $N \rightarrow C^\infty(M, V)$ as the same as a smooth map $N \times M \rightarrow V$ in the following way. If $f : N \rightarrow C^\infty(M, V)$ maps $n \mapsto f_n : M \rightarrow V$, then we could identify f with a map $\hat{f} : N \times M \rightarrow V$ which maps $\hat{f}(n, m) = f_n(m)$. Hence we think of $\mathbf{C}^\infty$ as a sort of “almost” internal hom for manifolds and differentiable vector spaces. Hence $\mathcal{DVS}$ being cotensored over $\mathbf{Man}$.

This construction generalizes. If E is a vector bundle, define

$$\mathbf{C}^\infty(M, E \otimes V) := \mathcal{E}(M) \otimes_{C^\infty(M)} \mathbf{C}^\infty(M, V).$$

Although in general tensor products in $\mathcal{DVS}$ are ill-behaved analytically, problems do not arise here since $\mathcal{E}$ is a projective C_M^∞ -module of finite rank. We will denote

$$\Omega^1(M, E) := C^\infty(M, T^*M \otimes E).$$

We want to create an internal hom for $\mathcal{DVS}$ using C^∞ .

Definition 3.5.21. Let $\{V_i\}_{1 \leq i \leq n}$ and W be differentiable vector spaces. Given a manifold M , a smooth family of multilinear maps $V_1 \times \cdots \times V_n \rightarrow W$ *parameterized by* M is an element of

$$\mathcal{DVS}(V_1, \dots, V_n; C^\infty(M, W))$$

where we regard $C^\infty(M, W)$ as a differentiable vector space via definition 3.5.20.

The intuition for this definition is as follows. Let N be any test manifold. A smooth multilinear map

$$\Phi : V_1 \times \cdots \times V_n \rightarrow C^\infty(M, W)$$

yields a map

$$\Phi_N : V_1(N) \times \cdots \times V_n(N) \rightarrow C^\infty(N \times M, W) \cong C^\infty(M, C^\infty(N, W)) = C^\infty(M, W(N)).$$

The final isomorphism was explored informally earlier. Fix $m \in M$. Including $i_m : N \hookrightarrow N \times M$ along $m \in M$ yields a pullback map $i_m^* : W(N \times M) \rightarrow W(N)$. Thus we obtain a map

$$\Phi_{N,m} = \left(V_1(N) \times \cdots \times V_n(N) \xrightarrow{\Phi_N} W(N \times M) \xrightarrow{i_m^*} W(N) \right).$$

Naturality allows us to rewrite $\Phi_m : V_1 \times \cdots \times V_n \rightarrow W$ as a map of differentiable vector spaces, justifying the terminology.

It can then be shown that there is a differentiable vector space $\mathbb{H}\text{om}_{\mathcal{DVS}}(V_1, \dots, V_n; W)$ defined by

$$\mathbb{H}\text{om}_{\mathcal{DVS}}(V_1, \dots, V_n; W) : M \mapsto \mathcal{DVS}(V_1, \dots, V_n; C^\infty(M, W)).$$

The flat connection is the natural map

$$\nabla_{C^\infty(M, W)} : C^\infty(M, W) \rightarrow \Omega^1(M, W)$$

given by applying the projection in the M -direction to the connection $\nabla_{M \times N, W}$.

In the rest of the text, if we consider a smooth family of maps between differentiable vector spaces, we always mean an element of the internal hom in this sense.

3.5.7 Algebras of observables

The prefactorization algebras of observables we will see in this book are mostly built as functions on convenient vector spaces, which will mean symmetric algebras and their dual spaces. Most of these will be built on sections of graded vector bundles. Let $\widehat{\otimes}_\pi$ denote the completed projective tensor product in LCTVS.

Proposition 3.5.22. *Let E be a vector bundle on a manifold X , and let F be a vector bundle on a manifold Y . Let $\mathcal{E}$ denote the sheaf of convenient vector spaces of smooth sections of E on X , and likewise for $\mathcal{F}$. Then*

$$\mathcal{E}(X) \widehat{\otimes}_\beta \mathcal{F}(Y) \cong \Gamma(X \times Y, E \boxtimes F) \cong \mathcal{E}(X) \widehat{\otimes}_\pi \mathcal{F}(Y),$$

where $E \boxtimes F$ denotes the external tensor product of vector bundles and Γ denotes smooth sections.

The book then does some nonsense with symmetric algebras that I do not feel the need to review. What I will note is that the usual tensor product is dense in the bornological tensor product, meaning it suffices to still work with tensors as usual when considering elements.

Part II

First examples of field theories and their observables

Chapter 4

Free field theories

The beginning of this chapter is fully review. I am well-versed on the contents of sections 1 through 3, and am fairly confident in sections 4-6. Beyond that was not covered in class, meaning content I am unfamiliar with begins at section 7. For these reasons, section 1 is fully omitted, sections 2 and 3 are very sparse and only contain some details I felt worth copying down, and sections 4-6 should not be considered full notes either.

4.2 The prefactorization algebra of a free field theory

4.2.1 The classical observables of the free scalar field

For this section, let M be a Riemannian manifold with metric g , which provides a natural density vol_g for integrating against. Consider the free scalar field theory with naive fields C_M^∞ and action functional

$$S_U(\phi) = \int_U \phi \Delta \phi,$$

where Δ is the Laplacian on M , which we recall acts in local coordinates by $\Delta f = -\frac{1}{\sqrt{\det g}} \partial_j (g^{ij} \sqrt{\det g} \partial_i f)$, and more importantly, is self-adjoint. The Euler-Lagrange equations are $\Delta \phi = 0$, so we have a derived space of fields

$$\mathcal{E} = \left(C_M^\infty \xrightarrow{\Delta} C_M^\infty[-1] \right).$$

The sheaf $\mathcal{E}$ is a local dg Lie algebra with trivial bracket $[-, -] = 0$, and thus has a factorization envelope. The classical observables on this theory are

$$\text{Obs}^{cl} := \mathbb{U}\mathcal{E} = C_\bullet(\mathcal{E}_c) = \text{Sym}(C_c^\infty[1]).$$

Here C_c^∞ is the cosheaf of compactly supported smooth maps, $\mathcal{E} + c$ is compactly supported homogeneous elements of $\mathcal{E}$, and $C_\bullet$ is the Chevalley-Eilenberg chains for a dg Lie algebra, which is defined as the graded symmetric . The differential is given by $d_{\mathcal{E}} + d_{\text{CE}}$, but since $[-, -] = 0$ and $d_{\text{CE}} = [-, -]$ applied as a derivation, we see that the differential reduces simply to $d_{\mathcal{E}}$, extended via the Leibiz rule, thus forming a dg associative algebra.

4.2.2 Interpreting this construction

One can interpret this as polynomial polyvector fields on C_M^∞ in the following sense: since $\mathcal{E}[1]$ is concentrated in nonpositive degrees, we can compute

$$\text{Obs}^{cl} \cong \left(\cdots \longrightarrow \Lambda^2 C_c^\infty \otimes \text{Sym} C_c^\infty \longrightarrow C_c^\infty \otimes \text{Sym} C_c^\infty \longrightarrow \text{Sym} C_c^\infty \right).$$

All tensor products are completed tensor products. Since C_c^∞ is concentrated in degree 0, its symmetric algebra can be interpreted as an algebra of polynomial functions on C_M^∞ by viewing compactly supported functions as distributions, identifying $C_c^\infty \subseteq (C_M^\infty)^*$.

4.2.3 General free field theories

Definition 4.2.1. Let M be a manifold. A *free field theory* on M is the following data.

1. A graded vector bundle E on M , with sheaf of sections $\mathcal{E}$ and compactly supported sections $\mathcal{E}_c$.
2. A differential operator $d : \mathcal{E} \rightarrow \mathcal{E}[1]$ with square 0, making $\mathcal{E}$ into an elliptic complex.
3. An isomorphism $\varphi : E \rightarrow (E^\vee \otimes \text{Dens}_M)[-1]$ compatible with differentials. This is generally repackaged into the data of a graded anti-symmetric pairing $\langle -, - \rangle : \mathcal{E}_c \otimes_{C_M^\infty} \mathcal{E}_c \rightarrow \text{Dens}_M[-1]$ via mapping a pair (α, β) of compactly supported sections to evaluating the $\mathcal{E}^\vee$ component of $\varphi(\alpha)$ at β , and multiplying with the density. Integrating this gives a pairing $\mathcal{E}_c \otimes_{\mathbb{R}} \mathcal{E}_c \rightarrow \mathbb{R}[-1]$. On a Riemannian manifold, the metric g gives a canonical density vol_g .

The complex $\mathcal{E}$ is the derived version of the solutions to the equations of motion, and we refer to it as the derived space of fields or the derived solutions to the equations of motion.

For the free scalar field theory with nonzero mass m on a Riemannian manifold M , we have

$$\mathcal{E} = \left(C_M^\infty \xrightarrow{\Delta + m^2} C_M^\infty[-1] \right).$$

The pairing $\langle -, - \rangle : \mathcal{E}_c \otimes_{\mathbb{R}} \mathcal{E}_c \rightarrow \mathbb{R}[-1]$ is given by

$$\langle \phi_0, \phi_1 \rangle_U = \int_U \phi_0 \phi_1 \text{vol}_g.$$

Abelian Yang-Mills with gauge group $\mathbb{R}$ is described as follows. Let M be a 4-manifold, and $A \in \Omega_M^1(U)$ a connection on the trivial line $\mathbb{R}$ -bundle. Then the Yang-Mills action functional is

$$S_{YM}(A) = -\frac{1}{2} \int_U dA \wedge \star dA,$$

for $\star$ the Hodge star. If $d \star d$ is the Hodge Laplacian, then the equations of motion are $(d \star d)A = 0$. Gauge symmetry is given by $A \rightarrow A + dX$, for $X \in \Omega_M^0(U)$. The complex describing this is

$$\mathcal{E} = \left(\Omega_M^0[1] \xrightarrow{d} \Omega_M^1 \xrightarrow{d \star d} \Omega_M^3[-1] \xrightarrow{d} \Omega_M^4[-2] \right).$$

Definition 4.2.2. Let $\mathcal{E}$ be the derived space of fields for a free field theory. The prefactorization algebra of classical observables Obs^{cl} is defined as

$$\text{Obs}^{cl} := \text{Sym}(\mathcal{E}_c[1])$$

with differential $d_{\mathcal{E}}$ extended to a derivation.

To show that Obs^{cl} is a prefactorization algebra, note that it is a sheaf of differential graded commutative algebras with multiplication defined by the symmetric product (the differential is compatible by definition). If $U \subseteq V$ is an inclusion of open sets, then extension by 0 gives an

inclusion $C_c^\infty(U) \hookrightarrow C_c^\infty(V)$, which induces a map $\text{Obs}^{cl}(U) \hookrightarrow \text{Obs}^{cl}(V)$. If $\{U_i \subseteq V\}_{1 \leq i \leq n}$ are disjoint open sets, then the factorization product is the map

$$(\alpha_1, \dots, \alpha_n) \mapsto \prod_{i=1}^n i_V^{U_i}(\alpha_i),$$

where $i_V^{U_i} : \text{Obs}^{cl}(U_i) \hookrightarrow \text{Obs}^{cl}(V)$, and the product shown is the symmetric product. We evaluate this observable on a field ϕ by

$$\left(\prod_{i=1}^n i_V^{U_i} \alpha_i \right) (\phi) = \prod_{i=1}^n \alpha_i(\phi|_{U_i}),$$

where the product on the right is the product in $\mathbb{R}$. Here we note that any observable may be evaluated on a field in a similar way.

4.2.4 The one-dimensional case, in detail

Consider $M = \mathbb{R}$. Here the spacetime manifold has only one dimension, so we call it time, and write $t \in \mathbb{R}$. The classical field theory of a particle in 1 dimension is just classical mechanics. Given an open interval U , the cohomology of $\text{Obs}^{cl}(U)$ is $\mathbb{R}[p, q]$, concentrated in degree 0. To see this, if $m = 0$, let $\phi_q(x) = 1$ and $\phi_p(x) = x$. If $m > 0$, let

$$\phi_q(x) = \frac{1}{2} (e^{mx} + e^{-mx}), \quad \phi_p(x) = \frac{1}{2m} (e^{mx} - e^{-mx}).$$

The map $\Delta + m^2$ is simply $-\partial_x^2 + m^2$. Both functions are annihilated by this operator, and form a basis for its kernel. Further, $\phi_q(0) = 1$ and $\phi_p(0) = 0$, whereas $\phi'_q(0) = 0$ and $\phi'_p(0) = 1$. Finally, $\phi_q = \phi'_p$. Then there is a cochain map $\pi : \mathcal{E}_c(U) \rightarrow \mathbb{R}[p, q]$ by

$$g \mapsto q \int_U g(t) \phi_q dx + p \int_U g(t) \phi_p dt.$$

This map is a quasi-isomorphism.

4.2.5 The Poisson bracket

Let M be a smooth manifold with free field theory whose complex of fields is $\mathcal{E}$. Classical observables are described by $\text{Sym}(\mathcal{E}[1])$. Recall that there is an antisymmetric pairing $\langle -, - \rangle : \mathcal{E}_c \otimes_{\mathbb{R}} \mathcal{E}_c \rightarrow \mathbb{R}[-1]$ associated to this free field theory. This translates to a symmetric pairing on $\mathcal{E}_c[1]$. Proving this is a mess of dealing with degrees and I choose to omit it here.

Lemma 4.2.3. *There is a unique smooth Poisson bracket on Obs^{cl} of cohomological degree 1 such that*

$$\{\alpha, \beta\} = \langle \alpha, \beta \rangle$$

for any two linear observables $\alpha, \beta \in \mathcal{E}_c[1]$.

4.2.6 The quantum observables of a free field theory

In this section, we use the pairing $\langle -, - \rangle : \mathcal{E}_c \otimes_{\mathbb{R}} \mathcal{E}_c \rightarrow \mathbb{R}[-1]$. To construct Obs^q from Obs^{cl} , we start with a Heisenberg central extension $\widehat{\mathcal{E}}_c := \mathcal{E}_c \oplus \mathbb{R}\hbar$, where $\mathbb{R}\hbar$ is the 1-dimensional $\mathbb{R}$ -vector space spanned by $\hbar$. This central extension is given by the 2-cocycle $\omega(\alpha, \beta) = \langle \alpha, \beta \rangle$. The Lie bracket is given by mapping $\alpha, \beta \in \mathcal{E}_c$ to $[\alpha, \beta] = \hbar \langle \alpha, \beta \rangle$, and acting trivially on $\mathbb{R}\hbar$. The differential on the $\mathbb{R}\hbar$ component is trivial. Then $\text{Obs}^q := \mathbb{U}\widehat{\mathcal{E}}_c$:

$$\text{Obs}^q := \text{Sym}(\widehat{\mathcal{E}}_c[1]) \cong \text{Obs}^{cl}[\hbar],$$

where the latter is polynomials in $\hbar$ with coefficients in Obs^{cl} . The differential arises from the Lie bracket and the differential on $\widehat{\mathcal{E}}_c$. Note that the above isomorphism $\text{Obs}^q \cong \text{Obs}^{cl}[\hbar]$ is *only an isomorphism of graded vector spaces*, and does *not* respect differentials. The differential is $d_{\mathcal{E}} + d_{\text{CE}}$, for $d_{\text{CE}}(\alpha\beta) = [\alpha, \beta]$ the usual Chevalley-Eilenberg differential. Recall the full formula

$$d_{\text{CE}}(x_1 \cdots x_n) = \sum_{1 \leq i < j \leq n} \pm [x_i, x_j] x_1 \cdots \widehat{x_i} \widehat{x_j} \cdots x_n$$

with sign given by commuting the things all the way down.

The differential on Obs^q satisfies

$$d(ab) = (da)b + (-1)^{|a|} ab + (-1)^{|a|} \hbar \{a, b\}. \quad (4.1)$$

4.3 Quantum mechanics and the Weyl algebra

Lemma 4.3.1. *The prefactorization algebra Obs^q on $\mathbb{R}$ constructed from the free field theory is locally constant.*

Proof. There is a filtration

$$F^{\leq k} \text{Obs}^q := \text{Sym}^{\leq k}(\widehat{\mathcal{E}}[1]).$$

The associated graded is $\text{Obs}^{cl}[\hbar]$, so it suffices to show $H^\bullet(\text{Obs}^{cl})$ is locally constant. We have already seen $H^\bullet(\text{Obs}^{cl}(a, b)) = \mathbb{R}[p, q]$ for any interval (a, b) , and inclusion maps induce isomorphisms. ■

The associative algebra corresponding to this prefactorization algebra Obs^q will be denoted A_m . We will show this is the Weyl algebra, and is independent of the mass m . Of course we still want to keep track of the mass, so we will see that this is encoded in a *Hamiltonian operator* H .

The prefactorization algebra of quantum observables is the Chevalley-Eilenberg chains on the Heisenberg central extension of the complex $\Delta + m^2 : C_c^\infty \rightarrow C_c^\infty$. Note that ∂_x acts on C_c^∞ in the usual way, and more importantly, it commutes with the differential $\Delta + m^2 = -\partial_x^2 + m^2$ (you might think ∂_x annihilates m so this isn't true, but recall that the commutator is really the difference of the two *composites*, and you are composing ∂_x with the map $f \mapsto m^2 f$, so this is fine.). This means that $\partial_x : \mathcal{E}_c \rightarrow \mathcal{E}_c$ is a map of cosheaves. We will extend this to a map $\text{Obs}^q \rightarrow \text{Obs}^q$, not by requiring that it respect the factorization product, but rather that it act as a derivation on Obs^q .

Definition 4.3.2. Let $\mathcal{F}$ be a prefactorization algebra on X with values in Ch_R . A *degree k derivation* D of $\mathcal{F}$ is a collection of maps $\{D_U : \mathcal{F}(U) \rightarrow \mathcal{F}(U)\}_{U \in \text{Open}(X)}$ of cohomological degree k , which act as a derivation on the factorization product in the sense that if $\{U_i \subseteq V\}_{1 \leq i \leq n}$ are pairwise disjoint, $\{\alpha_i \in \mathcal{F}(U_i)\}_{1 \leq i \leq n}$ are homogeneous elements, and $m : \bigotimes_i \mathcal{F}(U_i) \rightarrow \mathcal{F}(V)$ is the factorization product, then

$$(D_V \circ m)(\alpha_1, \dots, \alpha_n) = \sum_{1 \leq i \leq n} (-1)^k \sum_{j=1}^{i-1} |\alpha_j| m(\alpha_1, \dots, D_{U_i} \alpha_i, \dots, \alpha_n).$$

Remark 4.3.3. If we apply definition 4.3.2 to the unary factorization product on an inclusion $U \subseteq V$, we obtain the relation

$$(D_V \circ m)(\alpha) = (m \circ D_U)(\alpha).$$

This is precisely the naturality condition for precosheaves, meaning any derivation of prefactorization algebras is automatically a morphism of precosheaves. However, such a morphism cannot be a morphism of prefactorization algebras (except possibly in some trivial edge cases which aren't worth considering).

Here, note that ∂_x is clearly a derivation on Obs^q , and it follows immediately that ∂_x is a derivation on A_m .

Definition 4.3.4. The *Hamiltonian* H is the derivation of the associative algebra A_m arising from the derivation $-\partial_x$ on the prefactorization algebra Obs^q for the free scalar field theory with mass m .

Proposition 4.3.5. *The associative algebra A_m is the Weyl algebra*

$$\mathcal{W} \cong \left\langle x^\alpha \partial_x^\beta \right\rangle_{\alpha, \beta}.$$

It is generated by $p, q, \hbar$ modulo the relations $[p, q] = \hbar$ and all other commutators vanishing. The Hamiltonian is

$$H(a) = \frac{1}{2\hbar} [p^2 - m^2 q^2, a].$$

Note that it is an inner derivation.

The proof is like 4 pages long so I went through it in the book but didn't copy it down.

4.4 Pushforward and canonical quantization

Given $\pi : X \rightarrow Y$ a continuous map between spaces, we can push forward precosheaves $\mathcal{PCshf}_X \rightarrow \mathcal{PCshf}_Y$ by defining $\pi_* \mathcal{F}(U) := \mathcal{F}(\pi^{-1}(U))$; the same as pushforwards of sheaves. We can do the same construction on the level of prefactorization algebras.

Definition 4.4.1. Let $\pi : X \rightarrow Y$ be a continuous map between spaces, and $\mathcal{F}$ a prefactorization algebra on X . Define the pushforward prefactorization algebra $\pi_* \mathcal{F}$ on Y as follows:

- If $U \subseteq Y$ is open, then $(\pi_* \mathcal{F})(U) := \mathcal{F}(\pi^{-1}(U))$.
- If $\{U_i \subseteq V\}_{1 \leq i \leq n}$ are pairwise disjoint opens, then note that $\pi^{-1}(U_i), \pi^{-1}(V)$ are open in X for all i , and taking preimages preserves disjointness, so $\mathcal{F}$ comes with the datum of a factorization product

$$\bigotimes_{1 \leq i \leq n} (\pi_* \mathcal{F})(U_i) \rightarrow (\pi_* \mathcal{F})(V).$$

Define this to be the relevant factorization product for $\pi_* \mathcal{F}$.

Warning 4.4.2. Although pushforwards always exist for prefactorization algebras, when we look at factorization algebras, not all pushforwards will always be a factorization algebra. However in the relevant cases, it usually will be. We will discuss this more in a few chapters.

Consider a manifold of the form $N \times \mathbb{R}$ with the product metric, and assume for simplicity that N is compact. Let $\pi : N \times \mathbb{R} \rightarrow \mathbb{R}$ be the canonical projection. Then $\pi_* \text{Obs}^q$ is a prefactorization algebra on $\mathbb{R}$. Let $\{e_i\}_{i \in I}$ be an orthonormal eigenbasis for $\Delta + m^2$ on $C_N^\infty(N)$, and associate eigenvalue λ_i to each e_i . Then by definition, $\bigoplus_i \mathbb{R} e_i$ is a dense subset of $C_N^\infty(N)$. For $m \in M$, let A_m denote the cohomology of the prefactorization algebra associated to the free 1-dimensional scalar field theory with mass m , which we previously showed is the Weyl algebra generated by $p, q, \hbar$ subject to $[p, q] = \hbar$ and all other brackets 0. Define A_N to be the infinite tensor product

$$A_N = A_{\sqrt{\lambda_1}} \otimes_{\mathbb{R}[\hbar]} A_{\sqrt{\lambda_2}} \otimes_{\mathbb{R}[\hbar]} \cdots,$$

which is defined as the relevant colimit.

Proposition 4.4.3. *There is a dense sub-prefactorization algebra of $\pi_* \text{Obs}^q$ that is locally constant, and whose cohomology prefactorization algebra corresponds to the associative algebra A_N .*

Remark 4.4.4. This proposition encodes the notion of canonical quantization in the sense of a physicist. The Weyl algebra $A_{\sqrt{\lambda_i}}$ corresponds to a quantum mechanical particle evolving in N with mass m and energy λ_i . The observables of the free scalar field $\pi_* \text{Obs}^q$ are essentially described by the combination of all of these algebras corresponding to each possible energy level; since the theory is free, they do not interact, meaning there are no relations to consider. Essentially, this allows us to describe the free scalar field theory on a manifold using the free scalar field theory on $\mathbb{R}$, which is already well-understood, in such a way that encodes the notion of canonical quantization.

Proof. The prefactorization algebra $\pi_* \text{Obs}^q$ maps open subsets $U \subseteq \mathbb{R}$ to the Chevalley-Eilenberg chains

$$\text{Sym} \left(C_c^\infty(U \times N)[-1] \xrightarrow{\Delta + m^2} C_c^\infty(U \times N) \right) [\hbar].$$

A dense subcomplex of these observables is

$$\bigoplus_{i \in I} \left(C_c^\infty(U) e_i \xrightarrow{\Delta + \lambda_i} C_c^\infty(U) e_i[-1] \right).$$

Define $\mathcal{F}_i$ to be the prefactorization algebra for the scalar free field theory on $\mathbb{R}$ with mass m . Note that $\mathcal{F}_i$ is a prefactorization algebra of $\mathbb{R}[\hbar]$ -modules. Using the tensor product of prefactorization algebras, define

$$\mathcal{F} := \mathcal{F}_1 \otimes_{\mathbb{R}[\hbar]} \mathcal{F}_2 \otimes_{\mathbb{R}[\hbar]} \cdots,$$

where the infinite tensor product is the appropriate colimit. We can equivalently view this tensor product as being the factorization envelope of the Heisenberg central extension of the dense subcomplex of $\pi_* \text{Obs}^q$ noted above. Because the complex is a dense subcomplex, the evident map $\mathcal{F} \rightarrow \pi_* \text{Obs}^q$ has a dense image, and passing to homology maintains that dense image. Note that $H^\bullet(\mathcal{F}_i)$ corresponds to the Weyl algebra $A_{\sqrt{\lambda_i}}$, and thus $H^\bullet \mathcal{F}$ corresponds to A_N . $\blacksquare$

4.5 Abelian Chern-Simons

We now consider Abelian Chern-Simons theory, which is a simple example of a gauge theory and a topological field theory. We will consider perturbative Chern-Simons with gauge group $U(1)$.

Let M be an orientable 3-manifold, with $\mathcal{E} = \Omega_M^\bullet[1]$ and differential given by the exterior derivative. The pairing $\mathcal{E}_c \otimes_{\mathbb{R}} \mathcal{E}_c \rightarrow \mathbb{R}[-1]$ is given by

$$\langle \alpha, \beta \rangle = (-1)^{|\beta|} \int_U \alpha \wedge \beta.$$

The action functional is $S(\alpha) = \langle \alpha, d\alpha \rangle$, and the equations of motion are $d\alpha = 0$. The action functional has a gauge symmetry $\alpha \rightarrow \alpha + df$, hence $\Omega_M^0[1]$ being in $\mathcal{E}$.

4.5.1 Observables

We have

$$\text{Obs}^{cl} = \text{Sym} \left(\Omega_c^0[2] \xrightarrow{d} \Omega_c^1[1] \xrightarrow{d} \Omega_c^2 \xrightarrow{d} \Omega_c^3 \right).$$

Writing H_c for the compactly supported de Rham cohomology, and recalling that cohomology commutes with taking the symmetric algebra, we see that $H^\bullet \text{Obs}^{cl} \cong \text{Sym}(H_c^\bullet[2])$. The dimensions of $H^\bullet \text{Obs}^{cl}(U)$ are thus always finite dimensional, which is much easier to understand than an arbitrary free scalar field theory, where the cohomology can be infinite dimensional.

This continues to quantum observables. Filter Obs^q by

$$F^k \text{Obs}^q(U) = \text{Sym}^{\leq k}(\Omega_c^\bullet(U)[2])[\hbar].$$

This filtration induces a spectral sequence with first page $H^\bullet \text{Obs}^{cl}(U)[\hbar]$. The differential on this first page is 0, but the differential on the second arises from the pairing of linear observables, which descends to cohomology:

$$\langle [\alpha], [\beta] \rangle_{H^\bullet} = [\langle \alpha, \beta \rangle].$$

The differential d_2 on the second page vanishes on Sym^0 and Sym^1 , and satisfies

$$d_2([\alpha][\beta]) = \hbar \langle [\alpha], [\beta] \rangle_{H^\bullet}.$$

Equation (4.1) determines d_2 on terms that are higher. Thus, $H^\bullet \text{Obs}^q$ simplifies to understanding the integration pairing between cohomology classes of compactly supported de Rham forms.

Lemma 4.5.1. *Let M be a closed, connected 3-manifold (I think it also has to be orientable). Then*

$$H^\bullet \text{Obs}^q(M) \cong \mathbb{R}[\hbar][1 - \dim H^2(M)].$$

Here the first bracket is polynomials, and the second is a shift.

Proof. Poincaré duality for closed, connected manifolds yields $H^n(M) \cong \mathbb{R}$, and $H^i(M) \cong H^{n-i}(M)$. Let $b_2 = \dim H^2(M)$. Choose a basis $\{\alpha_1, \dots, \alpha_{b_2}\}$ for $H^2(M)$ and a dual basis $\{\beta_1, \dots, \beta_{b_2}\}$ for $H^1(M)$:

$$\langle \alpha_j, \beta_k \rangle_{H^\bullet} = \delta_{jk}.$$

Let v denote the generator for $H^0(M)$, which we know is 1-dimensional by Poincaré duality, and $[M]$ the dual generator for $H^3(M)$. Thus, $d_2(\alpha_j \beta_k) = \hbar \delta_{jk}$ and $d_2(v[M]) = \hbar$, and d_2 vanishes on all other quadratic monomials. Explicit computation shows that $v\alpha_1 \cdots \alpha_{b_2}$ is closed but not exact, and it generates the cohomology. ■

4.5.2 Compactifying along a closed surface

Consider a 3-manifold of the form $M := \mathbb{R} \times \Sigma$, where Σ is a closed, orientable surface. Let $\pi : M \rightarrow \mathbb{R}$ be the natural projection. We will show that $\pi_* \text{Obs}^q$ is locally constant, so that it corresponds to an associative algebra. This will turn out to be the Weyl algebra for the graded symplectic vector space $H^\bullet(\Sigma)[1]$.

Let g be the genus of Σ . Pick a symplectic basis $\{\alpha_1, \dots, \alpha_g, \beta_1, \dots, \beta_g\}$ for $H^1(\Sigma)$ such that

$$\int_{\Sigma} \alpha_j \beta_k = \delta_{jk}.$$

Remark 4.5.2. It's time I become more familiar with symplectic geometry. A symplectic form ω is a nondegenerate skew-symmetric bilinear form. Clearly the pairing $\langle -, - \rangle$ is a symplectic form. A *symplectic basis* is a basis $\{e_i, f_i\}$ of the vector space such that $\omega(e_i, e_j) = 0$, $\omega(f_i, f_j) = 0$, and $\omega(e_i, f_j) = \delta_{ij}$. A symplectic manifold is a manifold with a symplectic form.

Let ν be the constant function 1, which yields the basis for $H^0(\Sigma)$. Let μ be a basis for $H^2(\Sigma)$ with the property that

$$\int_{\Sigma} \mu = 1.$$

Then $H^\bullet(\Sigma)[1]$ has a natural symplectic structure given by the integration pairing, and ν, μ are dual under this pairing.

Proposition 4.5.3. *The prefactorization algebra $H^\bullet \pi_* \text{Obs}^q$ is locally constant, and it corresponds to the Weyl algebra A_Σ for $H^\bullet(\Sigma)[1]$. In terms of our chosen generators, this Weyl algebra has commutators $[\nu, \mu] = \hbar$, $[\alpha_j, \beta_k] = \hbar \delta_{jk}$, and all others vanish. The Hamiltonian of this Weyl algebra is 0.*

The fact that the Hamiltonian is 0 makes Abelian Chern-Simons a topological field theory.

Proof. By Hodge theory, $\Omega_\Sigma^\bullet(\Sigma) \cong H^\bullet(\Sigma)$. We will use this fact in this proof. We start by showing local constancy. This is essentially the same proof as showing that $\mathbb{U}\mathfrak{g}^\mathbb{R} \cong U\mathfrak{g}$; our construction of the universal enveloping algebra via the factorization envelope.

Let $U \subseteq \mathbb{R}$ be open. Define $\mathcal{L} := \pi_* \Omega_M^\bullet[1]$. Then $\mathbb{U}\mathcal{L} \cong \pi_* \text{Obs}^{cl}$:

$$\mathbb{U}\mathcal{L}(U) = C_\bullet(\pi_* \Omega_{M,c}^\bullet(U)) = C_\bullet(\Omega_{M,c}^\bullet(\pi^{-1}(U))) = \text{Obs}^{cl}(\pi^{-1}(U)) = \pi_* \text{Obs}^{cl}(U).$$

We thus have that $\text{Obs}^q = \mathbb{U}\widehat{\mathcal{L}}$. Finally, note that

$$\Omega_M^\bullet(\pi^{-1}(U)) = \Omega_M^\bullet(U \times \Sigma) \cong \Omega_\mathbb{R}^\bullet(U) \otimes_\mathbb{R} \Omega_\Sigma^\bullet(\Sigma).$$

Thus we have

$$\mathcal{L} \cong \Omega_\mathbb{R}^\bullet[1] \otimes_\mathbb{R} \Omega_\Sigma^\bullet(\Sigma).$$

Using our result from Hodge theory,

$$\mathcal{L} \simeq \Omega_\mathbb{R}^\bullet[1] \otimes_\mathbb{R} H^\bullet(\Sigma).$$

Here we note that the completed tensor product is isomorphic to the usual tensor product when one of the entries is finite-dimensional. Since closed orientable surfaces have finite-dimensional cohomology, we can simply tensor over $\mathbb{R}$.

Now let $I \subseteq J \subseteq \mathbb{R}$ be an inclusion of open intervals. We will show local constancy of both classical and quantum observables. First, we note that since there is a homotopy equivalence $I \simeq J$, homotopicity of the de Rham functor $I \mapsto \Omega_\mathbb{R}^\bullet(I)[1]$ (not the de Rham sheaf, which is a sub-functor of the de Rham functor) yields a quasi-isomorphism $\Omega_\mathbb{R}^\bullet(I) \simeq \Omega_\mathbb{R}^\bullet(J)$. Since vector spaces are flat, $\otimes_\mathbb{R} H^\bullet(\Sigma)$ is exact, and thus preserves quasi-isomorphisms. We can thus tensor to obtain the quasi-isomorphisms

$$\mathcal{L}(I) \simeq \Omega_\mathbb{R}^\bullet(I) \otimes_\mathbb{R} H^\bullet(\Sigma) \simeq \Omega_\mathbb{R}^\bullet(J) \otimes_\mathbb{R} H^\bullet(\Sigma) \simeq \mathcal{L}(J).$$

Thus, $\mathcal{L}$ is locally constant up to homotopy. Note that passing to compactly supported sections preserves quasi-isomorphisms since it clearly preserves homotopy. Additionally, the shift functor

clearly preserves everything. Although Sym does not preserve quasi-isomorphisms, we recall that Sym commutes with cohomology, and thus

$$H^\bullet \text{Obs}^{cl} = H^\bullet(\text{Sym } \mathcal{L}_c[1]) \cong \text{Sym } H^\bullet(\mathcal{L}_c[1]).$$

Since $H^\bullet$ inverts quasi-isomorphisms, by functoriality of Sym we see that $H^\bullet \text{Obs}^{cl}$ is locally constant.

Now we pass to quantum observables. The same argument works; the only thing we must note is that if $\mathcal{L}(I) \simeq \mathcal{L}(J)$, then $\mathcal{L}(I) \oplus \mathbb{R}\hbar \simeq \mathcal{L}(J) \oplus \mathbb{R}\hbar$. It is trivial to check that $-\oplus \mathbb{R}\hbar$ is exact; hence it preserves quasi-isomorphisms. Therefore, $H^\bullet \text{Obs}^q$ is locally constant.

Write $\widehat{\mathfrak{g}} = \widehat{\mathcal{L}}_c$. We have previously seen that $H^\bullet(C_\bullet \widehat{\mathfrak{g}}) \cong \text{Sym}^\bullet(H^\bullet \widehat{\mathfrak{g}}[1]) \cong U\widehat{H^\bullet(\Sigma)}[1]$ in our construction of the universal enveloping algebra (note that $\mathfrak{g}$ is just $\mathfrak{g}^\Sigma$). Here, the central element is $\hbar$.

Now we consider the Hamiltonian. The conceptual reason that the Hamiltonian vanishes is since the de Rham complex is preserved up to homotopy under every diffeomorphism of $\mathbb{R}$, which includes infinitesimal translations.

Let x be a coordinate system on $\mathbb{R}$. Fix a function $f \in C_c^\infty((-\frac{1}{2}, \frac{1}{2}))$ such that $\int_{\mathbb{R}} f dx = 1$. Hence $f dx$ generates $H_c^1(\mathbb{R})$. Let $f_t(x) = f(x - t)$. Then there are linear observables in $C_\bullet \widehat{\mathfrak{g}}((t - \frac{1}{2}, t + \frac{1}{2}))$ given by

$$\begin{aligned} A_{t,j} &= f_t dx \otimes \alpha_j, \\ B_{t,k} &= f_t dx \otimes \beta_k, \\ \nu_t &= f_t dx \otimes \nu, \\ \mu_t &= f_t dx \otimes \mu. \end{aligned}$$

These elements are all cohomologically nontrivial, and their cohomology classes generate the Weyl algebra. Observe that, for instance,

$$\frac{\partial}{\partial x} A_{t,j} = f'_t dx \otimes \alpha_j = df_t \otimes \alpha_j,$$

so the infinitesimal translation is cohomologically trivial (df_t is exact, and is thus 0 in cohomology). This logic applies to any generator. Thus the derivation on the Weyl algebra given by ∂_x is trivial. ■

4.6 Another take on quantizing observables

We have given an abstract definition of Obs^q as the factorization envelope of a particular Heisenberg dg Lie algebra. This is mathematically useful, but physically it may not be fully clear how this is interesting. We provide another useful description here, which helps to answer that question.

As (precosheaves of) graded vector spaces, $\text{Obs}^q \cong \text{Obs}^{cl}[\hbar]$. Additionally, the factorization products are the same. However, the differentials are not compatible. Previously we saw that taking cohomology changes the differentials and leads to a Weyl algebra; in particular the change in differential modifies the structure maps at the level of cohomology. However, deformation quantization deforms the product of observables. It is possible to approach quantization in this sense: modifying the structure maps at the cochain level, rather than the differential.

Hence we will construct a different isomorphism of precosheaves $\text{Obs}^q \cong \text{Obs}^{cl}$ which compatible with differentials. This isomorphism is *not* compatible with the factorization product. Thus, this isomorphism induces a deformed factorization product on $\text{Obs}^{cl}[\hbar]$ corresponding to the factorization product on Obs^q .

In other words, instead of viewing Obs^q as being obtained from Obs^{cl} by keeping the factorization product and deforming the differential, we deform the factorization product and keep the differential.

One advantage of this is that correlation functions and vacua are easier to construct in this language. We introduce these ideas for the free scalar field.

4.6.1 Green's functions

Definition 4.6.1. A *Green's function* (for the Laplacian Δ) is a distribution G on $M \times M$, preserved under reflection across the diagonal (i.e., symmetric) and satisfying

$$(\Delta \otimes 1)G = \delta_\Delta,$$

where δ_Δ is the delta distribution¹ on the diagonal $M \hookrightarrow M \times M$. A Green's function for the Laplacian with mass satisfies

$$(\Delta \otimes 1)G + m^2 G = \delta_\Delta.$$

If M is compact, then there is no Green's function for the Laplacian. Instead, there is a unique $\tilde{G}$ satisfying $(\Delta \otimes 1)\tilde{G} = \delta_\Delta - 1$. However, if mass is nonzero, then $\Delta + m^2$ is an isomorphism on $C^\infty(M)$, yielding a unique Green's function.

If M is non-compact, then there can be a Green's function when there is no mass term. For example, a Green's function for Δ on $\mathbb{R}^n$ is

$$G(x, y) = \begin{cases} \frac{1}{4\pi^{d/2}} \Gamma(\frac{d}{2} - 1) |x - y|^{2-n}, & n \neq 2 \\ -\frac{1}{2\pi} \ln |x - y|, & n = 2 \end{cases}.$$

Here, I believe d is distance. The role of the Green's function is to produce an isomorphism. Intuitively, this is since it serves as an “inverse” for a differential operator. Different choices of Green's functions thus lead to different isomorphisms. These choices do not change the quantum observables themselves though, just the process of promoting a classical observable to a quantum one.

4.6.2 The isomorphism of graded vector spaces

Consider free field theory $\mathcal{E} = C_M^\infty \xrightarrow{\Delta + m^2} C_M^\infty[-1]$. The underlying graded vector space of Obs^q is

$$\text{Sym}(C_c^\infty(U)[1] \oplus C_c^\infty(U))[\hbar].$$

Let V a vector space and $P \in (V^\vee)^{\otimes 2}$. Then P defines a differential operator ∂_P on $\text{Sym } V$ by $\partial_P| \text{Sym}^{\leq 1} V = 0$, and ∂_P on quadratic terms by contraction (symmetrize and evaluate). In the same way, if P is a distribution on $U \times U$, we can define a continuous, second-order differential operator on $\text{Sym}(C_c^\infty(U)[1] \oplus C_c^\infty(U))$ that vanishes on linear and lower terms, vanishes elements of negative cohomological degree in Sym^2 , and

$$\partial_P(\phi\psi) = \int_{U \times U} P(x, y) \phi(x) \psi(y)$$

for any $\phi, \psi \in C_c^\infty(U)$ of cohomological degree 0 (hence degree 1 after the shift, landing $\phi\psi$ in Sym^2).

The mild abuse of notation here tripped me up, so I will make this more precise. Define $(\phi \boxtimes \psi) : M \times M \rightarrow \mathbb{R}$ by $(\phi \boxtimes \psi)(x, y) = \phi(x)\psi(y)$. Then $\partial_P(\phi\psi) := \langle P, \phi \boxtimes \psi \rangle$, where of course the pairing here is just contraction, which is given by symmetrizing P and evaluating. We abuse

¹Note that δ -sub- Δ does not mean “ δ -sub-Laplacian, even though the Δ on the LHS is the Laplacian. There are two different uses of Δ in this equation: Laplacian on the left and diagonal on the right.

notation and write $P^{sym}(\phi \boxtimes \psi)$ as an integral against a test function $P(x, y)$ since almost all useful distributions arise in this way, or at least up to arbitrary precision. Especially since ϕ, ψ are compactly supported, so the test function P need not be compactly supported.

Choose a Green's function G for Δ on M . Thus, G restricts to a Green's function for the Laplacian on any open subset $U \subseteq M$. Thus, we define a second-order differential operator ∂_G on the algebra $\text{Sym}(C_c^\infty(U)[1] \oplus C_c^\infty(U))$, and extend by $\mathbb{R}[\hbar]$ -linearity to an operator on $\text{Obs}^q(U)$.

The differential on $\text{Obs}^q(U)$ can be written as $d = d_1 + d_2$, for $d_1 = d_\varepsilon$ a first-order differential operator arising from the space of fields, and d_2 a second-order $\mathbb{R}[\hbar]$ -linear differential operator given by the BV Laplacian. In the scalar field theory, this means

$$d_2(\phi_{-1}\phi_0) = \hbar \int_U \phi_{-1}\phi_0,$$

for ϕ_k in the copy of $C_c^\infty(U)$ sitting in degree k , sitting in Obs^q .

Remark 4.6.2. Let $a \in \text{Sym}(V^\bullet)$ for some complex $V^\bullet$, and write μ_a for the map $x \mapsto ax$. A differential operator of order $\leq n$ is an operator D such that $[D, \mu_a]$ is a differential operator of order $\leq n - 1$. A differential operator of order 0 is an operator given by right-multiplication by some element $D(1)$. Most differential operators annihilate some lower degree of elements in the symmetric algebra.

Since

$$((\Delta + m^2) \otimes 1)G = \delta_\Delta,$$

we can show that

$$[\hbar\partial_G, d_1] = d_2.$$

Indeed, both sides of the equation are second-order differential operators, so it suffices to check it on a quadratic term. If $\phi_k \in C_c^\infty(U)$ has cohomological degree k , then

$$\begin{aligned} \hbar\partial_G d_1(\phi_0\phi_{-1}) &= \hbar\partial_G(\phi_0(\Delta + m^2)\phi_{-1}) = \hbar \int_{U \times U} G(x, y)\phi_0(x)((\Delta + m^2)\phi_{-1})(y)\text{vol}_g \\ &= \int_{U \times U} ((\Delta_y + m^2)G(x, y))\phi_0(x)\phi_{-1}(y)\text{vol}_g = \int_U \phi_0(x)\phi_{-1}(y)\text{vol}_g = d_2(\phi_0\phi_{-1}). \end{aligned}$$

Remark 4.6.3. This calculation confused me, so I provide further commentary. First, note that

$$[\hbar\partial_G, d_1] = \hbar\partial_G d_1 - (-1)^{|\partial_G||d_1|} d_1 \hbar\partial_G.$$

On quadratic terms, ∂_G is second order, thus mapping quadratic terms to constants, i.e. Sym^0 . Note that d_1 kills Sym^0 , so $d_1 \hbar\partial_G = 0$. Thus $[\hbar\partial_G, d_1] = \hbar\partial_G d_1 = d_2$. Recall that G is symmetric, so the choice of variable assignment in the integral doesn't matter. This is also due to symmetrization. The Laplacian commutes past G since it is self-adjoint.

It is also immediate that ∂_G commutes with d_2 : both are just derivatives, and derivatives commute. Thus if W is a map assigning a classical observable α to a quantum observable $W(\alpha) = e^{\hbar\partial_G(\alpha)}$, then

$$(d_1 + d_2)W(\alpha) = W(d_1\alpha).$$

The book does not explain why this is true. Google's Gemini suggested that I recall Hadamard's lemma. The following is my own work. Recall Hadamard's lemma

$$e^A B e^{-A} = B + [A, B] + \frac{1}{2!} [A, [A, B]] + \dots$$

This is a non-graded statement, and must be adjusted to account for grading. Substitute $A = \hbar \partial_G$ and $B = d_1 + d_2$. Then note that since ∂_G has cohomological degree -2, the above identity holds in the graded setting. Now use $[\hbar \partial_G, d_1] = d_2$ and $[\hbar \partial_G, d_2] = 0$ to simplify the expression to only the first two terms

$$e^{\hbar \partial_G} d_1 e^{-\hbar \partial_G} = d_1 + d_2.$$

Multiplying on the left by $e^{\hbar \partial_G}$ yields the result

$$e^{\hbar \partial_G} d_1 = (d_1 + d_2) e^{\hbar \partial_G},$$

recovering the book's statement.

It is important to note here that this is exactly the statement that $W = e^{\hbar \partial_G}$ commutes with the differentials of Obs^{cl} and Obs^q . Moreover, it is $\mathbb{R}[\hbar]$ -linear: if $c \in \mathbb{R}$ and $\alpha \in \text{Obs}^{cl}$,

$$W(c\hbar\alpha) = e^{\hbar \partial_G}(c\hbar\alpha) = \sum_{n \geq 0} \frac{1}{n!} \hbar \partial_G^n (c\hbar\alpha) = \sum_{n \geq 0} \frac{1}{n!} \hbar (c\hbar) \partial_G^n \alpha = c\hbar \sum_{n \geq 0} \hbar \partial_G^n \alpha = c\hbar e^{\hbar \partial_G} \alpha = c\hbar W(\alpha).$$

Thus, W is an $\mathbb{R}[\hbar]$ -linear cochain map $\text{Obs}^{cl}[\hbar] \rightarrow \text{Obs}^q$. Since it has an inverse given by $e^{-\hbar \partial_G}$, it is an isomorphism.

We can promote this to a map of prefactorization algebras by defining a new factorization product for $\text{Obs}^{cl}[\hbar]$. If $U, V \subseteq M$ are disjoint and $\alpha \in \text{Obs}^{cl}(U)$, $\beta \in \text{Obs}^{cl}(V)$ are observables, then define the factorization product $\star_h$ to be

$$\alpha \star_h \beta := e^{-\hbar \partial_G} \left(\left(e^{\hbar \partial_G} \alpha \right) \left(e^{\hbar \partial_G} \beta \right) \right) = W^{-1}(W(\alpha)W(\beta)).$$

We can easily show that this is the correct definition. Let $\star_h$ denote the factorization product on $\text{Obs}^{cl}[\hbar]$, and recall that the factorization product on Obs^q is simply the $\mathbb{R}[\hbar]$ -multilinear symmetric algebra product. Let $\alpha, \beta \in \text{Obs}^{cl}(U)[\hbar]$ be homogeneous elements. For W to be a morphism of prefactorization algebras, we must have $W(\alpha \star_h \beta) = W(\alpha)W(\beta)$. Using its inverse W^{-1} , we obtain

$$\alpha \star_h \beta = W^{-1}(W(\alpha \star_h \beta)) = W^{-1}(W(\alpha)W(\beta)).$$

Thus, we have our definition.

4.6.3 Beyond the free scalar field

This construction can be readily generalized. Let M be a manifold with space of fields $\mathcal{E}$ and differential d . Instead of a Green's function, let $G : (\mathcal{E}(M)[1])^{\otimes 2} \rightarrow \mathbb{R}$ be a symmetric continuous linear operator such that

$$Gd(e_1 \otimes e_2) = \langle e_1, e_2 \rangle$$

for $\langle -, - \rangle$ the BV pairing. The analog of the operator $e^{\hbar \partial_G}$ is the "renormalization group flow operator from scale zero to scale ∞ ". See [Cos11] for more on whatever that means.

4.7 Correlation functions

Let $\mathcal{E}$ be the space of fields of some free field theory on a *compact* manifold M (compactness is important!). Additionally, suppose that $H^\bullet(\mathcal{E}(M)) = 0$. For example, the free scalar field theory on M where

$$\mathcal{E} = \left(C_M^\infty \xrightarrow{\Delta+m^2} C_M^\infty[-1] \right).$$

The book claims that since our conventions are such that Δ has non-negative eigenvalues, adding the m^2 yields a term with no zero eigenvalues, meaning the complex has no cohomology by Hodge theory (this doesn't hold on non-compact manifolds).

In this situation, the dg Lie algebra $\widehat{\mathcal{E}}(M)$ has cohomology spanned by the element $\hbar$ in degree 1, and cohomology still vanishes in the lowest degree. Since cohomology commutes with taking Sym , there is an isomorphism of $\mathbb{R}[\hbar]$ -modules

$$H^\bullet(\text{Obs}^q(M)) \cong \text{Sym} \left(H^1(\widehat{\mathcal{E}}(M)[-1][1]) \right) \cong \text{Sym}(\langle \hbar \rangle) \cong \mathbb{R}[\hbar].$$

Let us normalize this isomorphism by requiring that the element

$$1 \in \text{Sym}^0(\widehat{\mathcal{E}}[1]) = \mathbb{R}$$

is mapped to $1 \in \mathbb{R}[\hbar]$.

Definition 4.7.1. In this situation, we define the *correlation functions* of the free field theory as follows. For a finite collection $\{U_i \subseteq M\}_{1 \leq i \leq n}$ of disjoint open sets and a closed element $O_i \in \text{Obs}^q(U_i)$ for each open, we define

$$\langle O_1 \cdots O_n \rangle := [O_1 \cdots O_n] \in H^\bullet(\text{Obs}^q(M)) \cong \mathbb{R}[\hbar].$$

Here $O_1 \cdots O_n \in \text{Obs}^q(M)$ denotes the image under the factorization product

$$\bigotimes_{1 \leq i \leq n} \text{Obs}^q(U_i) \rightarrow \text{Obs}^q(M)$$

of the tensor $O_1 \otimes \cdots \otimes O_n$.

Essentially, our choice of a normalization for this isomorphism gives a real value in $\mathbb{R}[\hbar]$ for each cohomology class, and we identify this with the “expected value” of the observable. The map $W : \text{Obs}^{cl}[\hbar] \cong \text{Obs}^q$ simplifies this computation somewhat. Suppose we have the free scalar field theory with a nonzero mass term on a compact manifold, yielding a unique Green's function for $\Delta + m^2$.

Lemma 4.7.2 (Wick). *Let $\{U_i \subseteq M\}_{1 \leq i \leq n}$ be pairwise disjoint open sets. Let*

$$\alpha_i \in \text{Obs}^{cl}(U_i)$$

be classical observables of the free scalar field theory. Let

$$W(\alpha_i) = e^{\hbar \partial_G} \alpha_i \in \text{Obs}^q(U_i)$$

be the corresponding quantum observables under the isomorphism W of cochain complexes. Then

$$\langle W(\alpha_i) \cdots W(\alpha_n) \rangle = W^{-1}(W(\alpha_1) \cdots W(\alpha_n))(0) \in \mathbb{R}[\hbar].$$

On the right hand side, the product indicates the symmetric product in Sym , and evaluation at 0 indicates evaluating the function at $0 \in \text{Sym}^0$.

Proof. The map W is an isomorphism $\text{Obs}^q(U) \cong \text{Obs}^{cl}(U)[\hbar]$ for every open $U \subseteq M$. Let $\star_\hbar$ denote the deformed factorization product on $\text{Obs}^{cl}[\hbar]$. It is easily seen that

$$\alpha_1 \star_\hbar \cdots \star_\hbar \alpha_n = e^{-\hbar \partial_G} \left(\left(e^{\hbar \partial_G} \alpha_1 \right) \cdots \left(e^{\hbar \partial_G} \alpha_n \right) \right).$$

Since W is an isomorphism, it preserves cohomology classes, meaning the correlation function for the observables $W(\alpha_i)$ is the cohomology class of $\alpha_1 \star_\hbar \cdots \star_\hbar \alpha_n$. The map $\text{Obs}^q(M) \rightarrow \mathbb{R}[\hbar]$ is an $\mathbb{R}[\hbar]$ -linear quasi-isomorphism sending $1 \mapsto 1$. Such a map can be shown to be unique up to chain homotopy. It suffices to show that the map $\alpha \mapsto \alpha(0)$ suffices as the dotted arrow in the diagram

$$\begin{array}{ccc} \text{Obs}^{cl}(M)[\hbar] & & \\ \uparrow W^{-1} & \searrow & \\ \text{Obs}^q(M) & \xrightarrow{1 \mapsto 1} & \mathbb{R}[\hbar]. \end{array}$$

We will thus have that evaluation at 0 is the unique map sending $1 \mapsto 1$ on the level of cohomology (since cohomology identifies homotopies). Indeed, all we must show is that $W^{-1}(1)(0) = 1$, where 1 is the constant observable, and indeed

$$e^{\hbar \partial_G}(1) = \sum_{n \geq 0} \frac{1}{n!} \partial_G^n(1) = \frac{1}{0!} 1 = 1$$

is still the constant observable, and $1(0) = 1$. ■

This statement does not resemble the usual statement of Wick's lemma in a physics text, so we explore it further. Suppose that we have two linear observables α_1, α_2 of cohomological degree 0 over open sets U, V . By the isomorphism of graded vector spaces, we may view these as either quantum or classical observables, and we will change our viewpoint as is convenient. Since these are linear observables, they are killed by ∂_G , so we have that $e^{\hbar \partial_G}(\alpha_k) = \alpha_k$. But their product is quadratic, so we obtain

$$W^{-1}(\alpha_1 \alpha_2) = \alpha_1 \alpha_2 - \hbar \partial_G(\alpha_1 \alpha_2).$$

Recall the formula

$$\partial_G = \int_{M \times M} G(x, y) \alpha_1(x) \alpha_2(y).$$

Applying lemma 4.7.2 and these equations, we can compute the expectation value as

$$\begin{aligned} \langle W(\alpha_1) W(\alpha_2) \rangle &= W^{-1}(W(\alpha_1) W(\alpha_2)) \\ &= W(\alpha_1) W(\alpha_2) - \hbar \partial_G(W(\alpha_1) W(\alpha_2)) \\ &= -\hbar \partial_G(W(\alpha_1) W(\alpha_2)) \\ &= -\hbar \int_{M \times M} G(x, y) \alpha_1(x) \alpha_2(y). \end{aligned}$$

The second to last step follows since W kills Sym^1 . This is what a physicist would write down as the expectation value of two linear observables. This is also what follows from our lemma – note that since ∂_G is a differential operator of degree 2, it drops the quadratic term to a constant term in Sym^0 , meaning evaluation at 0 is trivial.

We have now given a precise definition to the earlier heuristic of integrating against a "Lebesgue" measure $e^{S/\hbar} d\mu$ over the space of field configurations.

4.8 Translation-invariant prefactorization algebras

Most physical theories possess some notion of translation invariance. In this section, we will analyze this notion for prefactorization algebras on $\mathbb{R}^n$. For one-dimensional theories, one often uses the phrase “time-independent Hamiltonian” for this property. We will show how the Hamiltonian formalism of quantum mechanics can be related to our approach.

4.8.1 The definition

Definition 4.8.1. For any $U \subseteq \mathbb{R}^n$ and $x \in \mathbb{R}^n$, let $\tau_x U := U + x$ denote the translation of U by x . A prefactorization algebra $\mathcal{F}$ on $\mathbb{R}^n$ is *discretely translation-invariant* if we have isomorphisms

$$\tau_x : \mathcal{F}(U) \cong \mathcal{F}(\tau_x U)$$

for all $x \in \mathbb{R}^n$ and all open subsets $U \subseteq \mathbb{R}^n$, which satisfy the following conditions:

1. We have $\tau_x \circ \tau_y = \tau_{x+y}$ for all $x, y \in \mathbb{R}^n$.
2. For all pairwise disjoint collections $\{U_i \subseteq V\}_{1 \leq i \leq k}$ of open sets, the diagram

$$\begin{array}{ccc} \bigotimes_{1 \leq i \leq k} \mathcal{F}(U_i) & \xrightarrow{\tau_k} & \bigotimes_{1 \leq i \leq k} \mathcal{F}(\tau_x U_i) \\ \downarrow & & \downarrow \\ \mathcal{F}(V) & \xrightarrow{\tau_x} & \mathcal{F}(\tau_x V) \end{array}$$

commutes. Here the vertical arrows are the factorization product.

Most of the theories we have seen so far are discretely translation-invariant. For example, any locally constant theory is (homotopy) discretely translation invariant. However, this notion turns out to be a bit too general for physical interest. We are more interested in the case where the factorization product depends smoothly on the position of the open sets, in a particular sense which we now make precise. Recall definition 4.3.2 which says the following:

Definition 4.8.2. Let $\mathcal{F}$ be a prefactorization algebra on M with values in Ch_R . A *degree k derivation* D of $\mathcal{F}$ is a collection of maps $\{D_U : \mathcal{F}(U) \rightarrow \mathcal{F}(U)\}_{U \in \mathcal{O}_{\text{pen}}(M)}$ of cohomological degree k , which act as a derivation on the factorization product in the sense that if $\{U_i \subseteq V\}_{1 \leq i \leq n}$ are pairwise disjoint, $\{\alpha_i \in \mathcal{F}(U_i)\}_{1 \leq i \leq n}$ are homogeneous elements, and $m : \bigotimes_i \mathcal{F}(U_i) \rightarrow \mathcal{F}(V)$ is the factorization product, then

$$(D_V \circ m)(\alpha_1, \dots, \alpha_n) = \sum_{1 \leq i \leq n} (-1)^{k \sum_{j=1}^{i-1} |\alpha_j|} m(\alpha_1, \dots, D_{U_i} \alpha_i, \dots, \alpha_n).$$

Recall that remark 4.3.3 proves that a derivation is necessarily a natural transformation. Given a prefactorization algebra $\mathcal{F}$ valued in cochain complexes, we define a dg Lie algebra $\text{Der}^\bullet(\mathcal{F})$ of derivations on $\mathcal{F}$ by letting $\text{Der}^k(\mathcal{F})$ to be the degree k derivations of $\mathcal{F}$. The differential is defined by $(dD)_U := [d_U, D_U]$ for d_U the differential on $\mathcal{F}(U)$. The Lie bracket is defined by $[D, D']_U := [D_U, D'_U]$. This concept allows us to introduce the notion of an action of a dg Lie algebra on a prefactorization algebra.

Definition 4.8.3. Let $\mathcal{F}$ be a prefactorization algebra on M valued in $\mathcal{Ch}_R$, and let $\mathfrak{g}$ be a differential graded Lie R -algebra. An *action* of $\mathfrak{g}$ on $\mathcal{F}$ is a morphism of differential graded Lie algebras $\mathfrak{g} \rightarrow \text{Der}^\bullet(\mathcal{F})$.

Now we introduce the requisite machinery to discuss the smoothness condition for a discretely translation-invariant prefactorization algebra. Define the operad Disj to have colors given by finite tuples $(U_1, \dots, U_k)$ of disjoint open subsets. If $U_i \subseteq V$ for all i , define the k -ary multimorphisms $\text{Disj}(U_1, \dots, U_k; V)$ to be all k -tuples $(x_1, \dots, x_k) \in (\mathbb{R}^n)^k$ such that $\tau_{x_1}(U_1), \dots, \tau_{x_k}(U_k)$ are still disjoint and are contained in V . Hence $\text{Disj}(U_1, \dots, U_k; V)$ parameterizes the way we can move the open sets without causing overlaps. If the closure of each U_i are disjoint and contained in V , then $\text{Disj}(U_1, \dots, U_k; V)$ has nonempty interior.

Let $\mathcal{F}$ be any discretely translation-invariant prefactorization algebra. For each $(x_1, \dots, x_k) \in \text{Disj}(U_1, \dots, U_k; V)$, we have a multilinear map

$$\prod_i \mathcal{F}(U)_i \longrightarrow \prod_i \mathcal{F}(\tau_{x_i} U_i) \longrightarrow \mathcal{F}(V),$$

where the second map is the factorization product. Write $m_{x_1, \dots, x_k}$ for this composite.

Definition 4.8.4. A discretely translation-invariant prefactorization algebra $\mathcal{F}$ is *smoothly translation-invariant* if the following hold.

1. The map $m_{x_1, \dots, x_k}$ depends smoothly on $(x_1, \dots, x_k)$.
2. The prefactorization algebra $\mathcal{F}$ is equipped with an action of the abelian Lie algebra $\mathbb{R}^n$ of translations. If $v \in \mathbb{R}^n$, we denote the corresponding action maps by

$$\frac{d}{dv} : \mathcal{F}(U) \rightarrow \mathcal{F}(U).$$

We view this Lie algebra action as an infinitesimal version of the global translation invariance.

3. The infinitesimal action is compatible with the global translation invariance in the following sense. For $v \in \mathbb{R}^n$, let $v_i \in (\mathbb{R}^n)^k$ be the vector $(0, \dots, v, \dots, 0)$ containing v in the i th slot. If $\alpha_i \in \mathcal{F}(U_i)$, then we require that

$$\frac{d}{dv_i} m_{x_1, \dots, x_k}(\alpha_1, \dots, \alpha_k) = m_{x_1, \dots, x_k} \left(\alpha_1, \dots, \frac{d}{dv} \alpha_i, \dots, \alpha_k \right).$$

Whenever we refer to a translation-invariant prefactorization algebra without further qualification, we always mean a smoothly translation-invariant one.

Example 4.8.5. Let $\mathcal{F}$ be the cohomology prefactorization algebra $H^\bullet \text{Obs}^q$ on the free scalar field theory on $\mathbb{R}$ with mass m . Recall that this is locally constant, and corresponds to the Weyl algebra A , generated by $p, q, \hbar$ subject to the relation $[p, q] = \hbar$. Clearly this is discretely translation invariant, so we must check that it is smoothly translation invariant. We start with the action of $\mathbb{R}$ on $H^\bullet \text{Obs}^q$. Choose some $x \in \mathbb{R}$; we want to define an action of x on $\mathcal{F}$, denoted $\frac{d}{dx}$. The natural choice here is to simply use the derivative as the derivation: given $\phi \in C_c^\infty(U)$, define $\frac{d}{dx} \phi(t) := -x \frac{d\phi}{dt}$. Since the derivative is supported within the original support, this returns an element of $C_c^\infty(U)$. Now we simply demand that $\frac{d}{dx}$ act as a derivation on the symmetric algebra. Since the factorization product is precisely the symmetric product, we obtain a derivation on $\mathcal{F}$. It is immediate to check compatibility with global translation invariance. Note that this derivation induces the Hamiltonian derivation $H(a) = \frac{1}{2\hbar} [p^2 - m^2 q^2, a]$ on the Weyl algebra.

4.8.2 Operadic reformulation

Definition 4.8.6. Let Disks_n be the $\mathbb{R}_{>0}$ -colored operad in the category of smooth manifolds whose space of multimorphisms is defined as follows. Given $r_1, \dots, r_k, s \in \mathbb{R}_{>0}$, define

$$\text{Disks}_n(r_1, \dots, r_k; s) \subseteq (\mathbb{R}^n)^k$$

to be the (possibly empty) open subset of k -tuples $x_1, \dots, x_k \in \mathbb{R}^n$ such that the closures of the balls $B_{r_i}(x_i)$ are all disjoint and contained in $B_s(0)$. We refer to this as a *disk configuration*. Composition is defined by inserting balls into bigger balls.

An important map to consider is as follows. Given $r_1, \dots, r_k, t_1, \dots, t_m, s \in \mathbb{R}_{>0}$, define

$$\circ_i : \text{Disks}_n(r_1, \dots, r_k; t_i) \times \text{Disks}_n(t_1, \dots, t_m; s) \rightarrow \text{Disks}_n(t_1, \dots, t_{i-1}, r_1, \dots, r_k, t_{i+1}, \dots, t_m; s)$$

by only inserting in the r -balls at t_i .

Let $\mathcal{F}$ be a translation-invariant prefactorization algebra on $\mathbb{R}^n$. Let $\mathcal{F}_r := \mathcal{F}(B_r(0))$; by translation invariance we can replace 0 with any $x \in \mathbb{R}^n$ up to isomorphism. The structure maps yield, for any configuration of disks $p \in \text{Disks}(r_1, \dots, r_k; s)$, a multiplication operation

$$m[p] : \mathcal{F}_{r_1} \times \dots \times \mathcal{F}_{r_k} \rightarrow \mathcal{F}_s.$$

Here we must take $\mathcal{F}_{r_i}$ to be $\mathcal{F}(B_{r_i}(p_i))$ such that all the balls are disjoint, but of course once again we can shift these to $B_{r_i}(0)$ up to isomorphism. This map is a smooth multilinear map of differentiable spaces, and depends smoothly on p .

These operations make the complexes $\{\mathcal{F}_r\}_{r>0}$ into an algebra over the operad Disks_n valued in (differentiable) chain complexes. Additionally, there is an action of the abelian Lie algebra $\mathbb{R}^n$ on the Disks_n -algebra $\mathcal{F}$ given by the action on the prefactorization algebra $\mathcal{F}$.

Now we will explain what it means to be a Disks_n -algebra in $\text{Ch}_{\mathbb{R}}$. Let $\mathcal{F}$ be such an algebra, and denote the operations by $m[p]$, as above. Given a configuration $p \in \text{Disks}_n(r_1, \dots, r_k; s)$, the map $m[p]$ is a multilinear map of cohomological degree 0, and is compatible with the differentials.

Second, if N is a manifold and $f_i : N \rightarrow \mathcal{F}_{r_i}^{d_i}$ be smooth maps into the space of homogeneous elements with degree d_i . The smoothness of $m[p]$ means that

$$N \times \text{Disks}_n(r_1, \dots, r_k; s) \rightarrow \mathcal{F}_s$$

$$(x, p) \mapsto m[p](f_1(x), \dots, f_k(x))$$

is smooth. Next, let $\sigma \in S_k$ be a permutation. There is an isomorphism

$$\sigma : \text{Disks}_n(r_1, \dots, r_k; s) \cong \text{Disks}_n(r_{\sigma(1)}, \dots, r_{\sigma(k)}; s).$$

We require that $m[\sigma(p)] = m[p]$. Finally, multiplication is functorial in the sense that if $p \in \text{Disks}_n(r_1, \dots, r_k; t_i)$ and $q \in \text{Disks}_n(t_1, \dots, t_l; s)$, then for any collections $\alpha_i \in \mathcal{F}_{r_i}$ and $\beta_j \in \mathcal{F}_{t_j}$,

$$m[q](\beta_1, \dots, \beta_{i-1}, m[p](\alpha_1, \dots, \alpha_k), \beta_{i+1}, \dots, \beta_l) = m[q \circ_i p](\beta_1, \dots, \beta_{i-1}, \alpha_1, \dots, \alpha_k, \beta_{i+1}, \dots, \beta_l).$$

Essentially, $m[q] \circ_i m[p] = m[q \circ_i p]$ for an appropriate notion of $\circ_i$ in the category $\text{Ch}_{\mathbb{R}}$. In addition, the action of $\mathbb{R}^n$ on each $\mathcal{F}_r$ is compatible with these multiplication maps.

4.8.3 A co-operadic reformulation

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4.8.4 The free scalar field

Let us write down explicit formulas for the multiplication maps in the working example of the free scalar field theory. We will utilize the isomorphism $W : \text{Obs}^{cl}[\hbar] \rightarrow \text{Obs}^q$ given by a Green's function for Δ . Recall that

$$\alpha \star_{\hbar} \beta := W_{U \sqcup V}^{-1}(W_U \alpha \cdot W_V \beta),$$

with notation as is evident. The multiplication map $m[p]$ varies smoothly with p , and thus m is a map

$$m : \mathcal{F}_{r_1} \otimes \cdots \otimes \mathcal{F}_{r_k} \rightarrow C^\infty(\text{Disks}_n(r_1, \dots, r_k; s), \mathcal{F}_s)$$

via $(\alpha_1, \dots, \alpha_k) \mapsto m[-](\alpha_1, \dots, \alpha_k)$. We will write this map m as $\mu_{r_1, \dots, r_k}^s$. Using $\star_{\hbar}$ and W , we can fully describe this map in the case of the free scalar field theory.

We use $\mathbb{R}^2$ as an example on how to do this. Let α_1, α_2 be compactly supported smooth functions on disks $B_{r_i}(0)$. Then we can view α_i as homogeneous elements of $\text{Obs}^{cl}(B_{r_i}(0))[\hbar]$ of degree 0. Denote by $\tau_x \alpha_i$ the image of α_i in $C_c^\infty(B_{r_i}(x))$ under the translation τ_x .

For $x_1, x_2 \in \mathbb{R}^2$ such that $B_{r_i}(x_i)$ are disjoint and contained within $B_s(0)$ for some s , we have

$$\mu_{r_1, r_2}^s(\alpha_1, \alpha_2) = \tau_{x_1} \alpha_1 \star_{\hbar} \tau_{x_2} \alpha_2 = (\tau_{x_1} \alpha_1) \cdot (\tau_{x_2} \alpha_2) - \hbar \int_{u_1, u_2 \in \mathbb{R}^2} \alpha_1(u_1 + x_1) \alpha_2(u_2 + x_2) \ln |u_1 - u_2|.$$

Here we used the Green's function

$$G(x, y) = -\frac{1}{2\pi} \ln |x - y|.$$

4.9 States and vacua for translation invariant theories

We now develop the notions of *states* and *vacuum* in this language for translation-invariant theories. Note that prefactorization algebras only encode the local relationships between observables. Long-range and global aspects of a physical situation appear in a different way, and the notion of state provides one way to introduce them.

Definition 4.9.1. Let $\mathcal{F}$ be a smoothly translation-invariant prefactorization algebra on $\mathbb{R}^n$, over a ring R . (In practice, R is $\mathbb{R}, \mathbb{C}, \mathbb{R}[[\hbar]], \mathbb{C}[[\hbar]]$). A *state* for $\mathcal{F}$ is a smooth linear map $\langle - \rangle : H^\bullet(\mathcal{F}(\mathbb{R}^n)) \rightarrow R$. The state is *translation invariant* if it commutes with both the action of the group $\mathbb{R}^n$, and the infinitesimal action of the Lie algebra $\mathbb{R}^n$, where the actions on R are trivial.

Previously, we only defined correlation functions for the free scalar field theory when the manifold was compact. Now, we can define correlation functions more generally using a state. Given observables $O_i \in \mathcal{F}(B_{r_i}(0))$ and a configuration $(x_1, \dots, x_k)$ such that the disks $B_{r_i}(x_i)$ are disjoint, then we define the *correlation function*

$$\langle O_1(x_1) \cdots O_n(x_n) \rangle$$

by applying the state $\langle - \rangle$ to the product observable $\tau_{x_1} O_1 \cdots \tau_{x_n} O_n \in \mathcal{F}(\mathbb{R}^n)$. Since the factorization algebra is *smoothly* translation invariant, we obtain that the correlation function depends smoothly on the configuration $(x_1, \dots, x_n)$; i.e. it forms a smooth map $\text{Disks}(r_1, \dots, r_k; \infty) \rightarrow R$. This function is invariant under simultaneous translation of all the points x_i .

Definition 4.9.2. A translation-invariant state $\langle - \rangle$ is called a *vacuum* if it satisfies the cluster decomposition principle: in the situation above, for any observables $O_1 \in \mathcal{F}(B_{r_1}(0))$ and $O_2 \in \mathcal{F}(B_{r_2}(0))$, we have

$$\langle O_1(0)O_2(x) \rangle - \langle O_1(0) \rangle \langle O_2(0) \rangle \xrightarrow{x \rightarrow \infty} 0.$$

A vacuum is *massive* if the above equation tends to 0 exponentially fast.

Consider the free scalar field with mass m . If $m > 0$, then there is a unique Green's function of the form $G(x, y) = f(x - y)$ for a distribution f on $\mathbb{R}^n$ that is smooth away from the origin and tends to 0 exponentially fast at ∞ . For example, if $n = 1$, then $f(x) = \frac{1}{2m}e^{-m|x|}$. If $m \neq 0$, then the canonically defined Green's function is

$$G(x, y) = \begin{cases} \frac{1}{4\pi^{d/2}}\Gamma(d/2 - 1)|x - y|^{2-n}, & n \neq 2 \\ -\frac{1}{2\pi} \ln|x - y|, & n = 2 \end{cases}.$$

Consider the map $\text{Obs}^{cl}(\mathbb{R}^n) \rightarrow \mathbb{R}$ which is the identity on Sym^0 and vanishes elsewhere. Since Sym^0 lives in degree 0, this is a cochain map of cohomological degree 0. It extends to an $\mathbb{R}[\hbar]$ -linear map

$$\langle - \rangle : \text{Obs}^{cl}(\mathbb{R}^n)[\hbar] \rightarrow \mathbb{R}[\hbar].$$

This map is clearly translation-invariant and smooth, since constants are invariant under translation. The isomorphism $\text{Obs}^{cl}[\hbar] \cong \text{Obs}^q$ yields a translation-invariant quantum state.

Lemma 4.9.3. *For $m > 0$, this state is a massive vacuum. For $m = 0$, this state is a vacuum if and only if $n > 2$.*

Proof. Let $F_1 \in C_c^\infty(B_{r_1}(0))^{\otimes k_1}$ and $F_2 \in C_c^\infty(B_{r_2}(0))^{\otimes k_2}$, viewed as observables in $\text{Obs}^{cl}(B_{r_i}(0))$ using the natural map $C_c^\infty(U)^{\otimes k} \rightarrow \text{Sym}^k C_c^\infty(U)$. Let c be sufficiently large such that $B_{r_1}(c)$ and $B_{r_2}(0)$ are disjoint, and let $\tau_c F_2$ denote the translation of F_2 by c . Explicitly, $(\tau_c F_2)(y_1, \dots, y_{k_2}) = F_2(y_1 - c, \dots, y_{k_2} - c)$. We want to compute the expected value $\langle F_1(\tau_c F_2) \rangle$. Note that these states are sufficiently arbitrary for full generality due to translation invariance.

The factorization product in this higher-dimensional example is the messy expression

$$F_1 \star_h \tau_c F_2 = \sum_{r=0}^{\min(k_1, k_2)} \hbar^r \sum_{1 \leq i_1 < \dots < i_r \leq k_1} \sum_{1 \leq j_1 < \dots < j_r \leq k_2} \int_{\mathbf{x} \in \mathbb{R}^{nk_1}} \int_{\mathbf{y} \in \mathbb{R}^{nk_2}} F_1(\mathbf{x}) F_2(\tau_c \mathbf{y}) \prod_{k=1}^r G(x_{i_k}, y_{j_k}).$$

Note that after performing the integrals, we are left with a function of the $k_1 + k_2 - 2r$ copies of $\mathbb{R}^n$ that we have not integrated over. Some remarks to make this confusing line make sense: this is the Taylor expansion of the exponential $e^{\hbar \partial G}$. The integrals are written wrong I think. Each one is only supposed to integrate over $k_i - r$ copies of r , not k_i copies. This is reflected more accurately in the Green's function product (im psure the integral is just a nice shorthand for taking the distributional Green's function anyways). Hence we are indeed left with that many copies of $\mathbb{R}^n$. We view this as an observable in $\text{Sym}^{k_1+k_2+2r} C_c^\infty(\mathbb{R}^n)$.

If $k_1, k_2 > 0$, then $\langle F_1 \rangle = 0$ and $\langle F_2 \rangle = 0$. Furthermore, $\langle F_1(\tau_c F_2) \rangle$ selects the constant term in the factorization product $F_1(\tau_c F_2)$. Passing to the deformed factorization product, this is the constant term in $F_1 \star_h \tau_c F_2$. As seen above, there is only a nonzero constant term if $k_1 = k_2 = k$, otherwise r never ranges high enough that $k_1 + k_2 - 2r = 0$ (it ranges to the min over k_1, k_2). If this happens, then the constant term is simply the $r = k$ term in the series:

$$\langle F_1(\tau_c F_2) \rangle = \hbar^k \int_{\mathbf{x}, \mathbf{y} \in \mathbb{R}^{kn}} F_1(\mathbf{x}) F_2(\mathbf{y} - c^{\otimes k}) \prod_{i=1}^k G(x_i, y_i).$$

We now must check the cluster decomposition principle. If $k_1 = k_2 = 0$, then the sum has a single term, which can be shown to be $F_1 \langle \tau_c F_2 \rangle$. Using $\langle F_i \rangle = F_i$, we conclude the statement. Now assume $k_1, k_2 > 0$, and as seen before we immediately have that the cluster decomposition equation vanishes unless $k_1 = k_2$. In this case, it reduces to

$$\langle F_1(\tau_c F_2) \rangle - \langle F_1 \rangle \langle F_2 \rangle = \langle F_1(\tau_c F_2) \rangle.$$

We must take $c \rightarrow \infty$ and see that this tends to 0. If $m > 0$, then G tends to 0 exponentially fast as $x - y \rightarrow \infty$, which can be seen in our definition. Thus we have a massive vacuum in this case. If $m = 0$, then $G(x, y)$ tends to 0 like the inverse of a polynomial as long as $n > 2$. Otherwise, we do not see a vacuum at all, since G does not tend to 0. $\blacksquare$

We can give a more abstract construction for the vacuum associated to a massive scalar field theory on $\mathbb{R}^n$, which illuminates how boundary conditions relate to observables. Let $\mathcal{E}$ be the complex of fields, and $\widehat{\mathcal{E}}_c = \mathcal{E}_c \oplus \mathbb{R}\hbar$ be the Heisenberg central extension, where $\hbar$ has degree 1.

A function $\mathbb{R}^n \rightarrow \mathbb{R}$ is *Schwartz* if it tends to zero at ∞ faster than the reciprocal of any polynomial. Let $\mathcal{S}(\mathbb{R}^n)$ be the collection of Schwartz functions, and define $\mathcal{E}_s(\mathbb{R}^n)$ to be the complex

$$\mathcal{S}(\mathbb{R}^n) \xrightarrow{\Delta+m^2} \mathcal{S}(\mathbb{R}^n).$$

We once again define a Heisenberg central extension

$$\widehat{\mathcal{E}}_s(\mathbb{R}^n) := \mathcal{E}_s(\mathbb{R}^n) \oplus (\mathbb{R}\hbar[-1]), \quad [\phi^0, \phi^1] := \hbar \int_U \phi^0 \phi^1.$$

Here ϕ^k is a Schwartz function of cohomological degree k . We remark here that the product of two Schwartz functions is Schwartz, and Schwartz functions are integrable.

Some Grothendieck magic shows us that the completed tensor product of Schwartz spaces can be defined, and coincides with

$$\mathcal{S}(\mathbb{R}^n) \widehat{\otimes} \mathcal{S}(\mathbb{R}^m) \cong \mathcal{S}(\mathbb{R}^{n+m}).$$

This allows us to define the Chevalley-Eilenberg complex of the Heisenberg algebra as usual:

$$\text{Obs}_s^q(\mathbb{R}^n) := C_\bullet(\widehat{\mathcal{E}}_s(\mathbb{R}^n)).$$

Since compactly supported functions are Schwartz, there is a map $\text{Obs}^q(\mathbb{R}^n) \rightarrow \text{Obs}_s^q(\mathbb{R}^n)$.

Lemma 4.9.4. *The cohomology of $\text{Obs}_s^q(\mathbb{R}^n)$ is $\mathbb{R}[\hbar]$ concentrated in degree 0.*

Proof. The complex $\mathcal{E}_s$ has no cohomology. I don't know why; I've never seen Schwartz functions before. I'll trust the authors here. It immediately follows that the cohomology of $\text{Obs}_s^q(\mathbb{R}^n)$ has the same cohomology of $C_\bullet(\mathbb{R}\hbar[-1])$. Since $\mathbb{R}\hbar$ is abelian, the Chevalley-Eilenberg chains have no differential, and thus are their own cohomology. We have

$$C_\bullet(\mathbb{R}\hbar[-1]) := \text{Sym}^\bullet(\mathbb{R}\hbar[-1][1]) = \text{Sym}^\bullet(\mathbb{R}\hbar) \cong \mathbb{R}[\hbar].$$

Thus we have a translation-invariant state

$$\langle - \rangle : H^\bullet(\text{Obs}^q(\mathbb{R}^n)) \rightarrow H^\bullet(\text{Obs}_s^q(\mathbb{R}^n)) = \mathbb{R}[\hbar].$$

This state is uniquely characterized by the fact that it is defined on Schwartz observables, and is a massive vacuum. $\blacksquare$

Chapter 5

Holomorphic field theories and vertex algebras

5.0.1 Vertex algebras and holomorphic prefactorization algebras on $\mathbb{C}$

Vertex algebras are the standard formalism of a two-dimensional chiral field theory. We will define a vertex algebra such that we can compare it to our own approach in this chapter.

Definition 5.0.1. Let V be a vector space. An element $a(z) = \sum_{n \in \mathbb{Z}} a_n z^{-n}$ in $\text{End } V[[z, z^{-1}]]$ is a *field* if, for each $v \in V$, there is some N such that $a_j v = 0$ for all $j > N$.

Warning 5.0.2. The usage of the word “field” in definition 5.0.1 achieves maximum possible confusion. Even though vertex algebras show up in field theory, a field in the sense of definition 5.0.1 is not a field in the sense of field theory, nor is it a field in the sense of algebra. I have no idea why this terminology was chosen, and hope that there is a good reason for it.

Definition 5.0.3. A *vertex algebra* is the following data:

- A $\mathbb{C}$ -vector space V , referred to as the *state space*;
- A nonzero vector $|0\rangle \in V$, referred to as the *vacuum vector*;
- A linear map $T : V \rightarrow V$, referred to as the *shift operator*;
- A linear map $Y(-, z) : V \rightarrow \text{End } V[[z, z^{-1}]]$ that sends every vector v to a field, referred to as the *vertex operation*;

subject to the following axioms:

1. (Vacuum axiom) $Y(|0\rangle, z) = 1_V$ and $Y(v, z)|0\rangle = v + zV[[z]]$ for all $v \in V$;
2. (Translation axiom) $[T, Y(v, z)] = \partial_z Y(v, z)$ for all $v \in V$ and $T|0\rangle = 0$;
3. (Locality axiom) For any pair of vectors $v, v' \in V$, there exists a nonnegative integer N such that $(z - w)^N [Y(v, z), Y(v', w)] = 0$ as an element of $\text{End } V[[z^{\pm 1}, w^{\pm 1}]]$.

This is a complex definition, so we consider the following intuition. We can think of V as the set of pointwise measurements one can make of the fields. Essentially, given a field ϕ , a measurement made at a single point $z \in \mathbb{C}$ would live in V . Thus we imagine attaching a copy of V to each $z \in \mathbb{C}$, which we denote V_z . Given a disk D containing z , the measurements at z are a dense subset of the measurements one can make on D . We denote the observables on D by V_D . The vertex operation is a way of combining the pointwise measurements. Given a disk D centered at the origin and $z \neq 0$, we can multiply observables at 0 and z to get a map

$$Y_z : V_0 \otimes V_z \rightarrow V_D.$$

This *vertex operator* should vary holomorphically in $z \in D \setminus \{0\}$. In other words, we should get something with properties like the formal definition $Y(-, z)$ above.

The language of factorization algebras has the benefit of making this intuition rigorous. We can formalize this picture using the structure maps of a factorization algebra, and in particular we can do so in a coordinate free way. This relationship is given by the main theorem for the chapter. Let $\mathcal{F}$ be a prefactorization algebra on $\mathbb{C}$. We want $\mathcal{F}$ to be *holomorphically translation-invariant*. A rough description of that is as follows. We saw earlier that if $\mathcal{F}$ is smoothly translation-invariant, then multiplication $p \mapsto m[p](\alpha_1, \dots, \alpha_k)$ yields a map

$$\mu : \mathcal{F}_{r_1} \otimes \dots \otimes \mathcal{F}_{r_n} \rightarrow C^\infty(\text{Disks}(r_1, \dots, r_k; s), \mathcal{F}_s).$$

Here, $\text{Disks}(r_1, \dots, r_k; s)$ is the open subset of $\mathbb{C}^k$ consisting of valid configurations of disks in $\mathbb{C}$, in the sense made precise in definition 4.8.6. Note that since this is now an open subset of complex space, it is a complex manifold. If $\mathcal{F}$ is *holomorphically translation-invariant*, then μ lifts to a cochain map

$$\mu : \mathcal{F}_{r_1} \otimes \dots \otimes \mathcal{F}_{r_n} \rightarrow \Omega^{0,\bullet}(\text{Disks}(r_1, \dots, r_k; s), \mathcal{F}_s).$$

Here, $\Omega^{0,\bullet}$ is the Dolbeault complex of $(0, \bullet)$ -forms for the complex manifold $\text{Disks}(r_1, \dots, r_k; s)$ with differential $\bar{\partial}$. There are also some compatibility conditions on these lifts, which we consider later.

In other words, holomorphic translation-invariance means that the operator product map is holomorphic up to homotopy in the location of the disks.

Theorem 5.0.4. *Let $\mathcal{F}$ be a holomorphically translation invariant prefactorization algebra on $\mathbb{C}$. Let $\mathcal{F}$ be equivariant under the action of S^1 on $\mathbb{C}$ by rotation, and let $\mathcal{F}_r^k$ denote the weight k eigenspace of the S^1 action on the complex $\mathcal{F}_r$. Assume that for ever $r < s$, the extension map $\mathcal{F}_r^k \rightarrow \mathcal{F}_s^k$ associated to the inclusion $D_r(0) \subseteq D_0(s)$ is a quasi-isomorphism. Finally, assume that the S^1 action on each $\mathcal{F}_r$ satisfies a certain technical “tameness” condition. Then the vector space*

$$V_{\mathcal{F}} := \bigoplus_{k \in \mathbb{Z}} H^\bullet(\mathcal{F}_r^k)$$

has the structure of a vertex algebra. The vertex algebra structure map

$$Y_{\mathcal{F}} : V_{\mathcal{F}} \otimes V_{\mathcal{F}} \rightarrow V_{\mathcal{F}}[[z, z^{-1}]]$$

is the Laurent expansion of the operator product map

$$H^\bullet \mu : H^\bullet(\mathcal{F}_{r_1}^{k_1}) \otimes H^\bullet(\mathcal{F}_r^{k_2}) \rightarrow \mathcal{O}(\text{Disks}(r_1, r_2; s), H^\bullet(\mathcal{F}_s))$$

for $\mathcal{O}$ the space of holomorphic maps.

In other words, in this situation, the intuition that the vertex map is our factorization product is made precise here. The nice thing about holomorphically translation invariant prefactorization algebras is that they make sense in complex dimensions other than 1, yielding a sort of higher vertex algebra as the algebraic structure attached to those theories.

5.1 Holomorphically translation-invariant prefactorization algebras

5.1.1 The definition

Let $\mathcal{F}$ be a smoothly translation-invariant prefactorization algebra. We want to describe what it means for $\mathcal{F}$ to be *holomorphically translation-invariant*. Of course, this requires $\mathcal{F}$ be defined over $\mathbb{C}$, so we now assume that $\mathcal{F}$ take values in $\mathcal{Ch}_{\mathbb{C}}$.

Recall that the datum of a smoothly translation-invariant prefactorization algebra includes an action of the (real) Lie algebra $\mathbb{R}^{2n} \cong \mathbb{C}^n$ by derivations. Since $\mathcal{F}$ is defined over $\mathbb{C}$, this action extends to a complexified action by the Lie algebra $\mathbb{R}^{2n} \otimes_{\mathbb{R}} \mathbb{C}$. The action of $z_i = x_i + iy_i$ and $\bar{z}_j = x_j - iy_j$ are denoted

$$\frac{\partial}{\partial z_i}, \frac{\partial}{\partial \bar{z}_j} : \mathcal{F}(U) \rightarrow \mathcal{F}(U),$$

and are generated using the standard Wirtinger derivatives (i.e. $\partial_{z_i} = \frac{1}{2}(\partial_{x_i} - i\partial_{y_i})$.)

Definition 5.1.1. A translation-invariant prefactorization algebra $\mathcal{F}$ on $\mathbb{C}^n$ is *holomorphically translation-invariant* if it is equipped with derivations $\eta_i : \mathcal{F} \rightarrow \mathcal{F}[-1]$ of cohomological degree -1 for $i = 1, \dots, n$, such that

$$\begin{aligned} d\eta_i &= \frac{\partial}{\partial \bar{z}_i} \in \text{Der}(\mathcal{F}), \\ [\eta_i, \eta_j] &= 0, \\ \left[\eta_i, \frac{\partial}{\partial \bar{z}_i} \right] &= 0. \end{aligned}$$

Here, d is the differential on the dg Lie algebra $\text{Der}(\mathcal{F})$.

Intuition 5.1.2. Recall that the differential on $\text{Der}(\mathcal{F})$ is $d_{\text{Der}}D = [d_{\mathcal{F}}, D]$ for the bracket given by the commutator. The idea is that for the field theory to be holomorphic, the antiholomorphic derivatives should be strictly 0. However, the derivatives are acting on the level of $\mathcal{F}$, and the physically meaningful information is on the level of cohomology, so we should not require that $\partial_{\bar{z}_i} = 0$ on the level of $\mathcal{F}$. Rather, we require that the derivatives vanish up to homotopy. This is the role of the maps η_i .

By guaranteeing that $\partial_{\bar{z}_i} = d\eta_i$, we guarantee that antiholomorphic derivatives are exact, and thus vanish in cohomology. The other two conditions guarantee that certain pathologies don't arise. For example, if we expect the antiholomorphic action to be homotopy trivial, then they should probably commute with each other, and they should be invariant under antiholomorphic translation.

5.1.2 An operadic reformulation

Let $\Delta_r(z) = PD_r(z)$ be the polydisk of radius r about $z \in \mathbb{C}^n$. Define the $\mathbb{R}_{>0}$ -colored operad enriched over complex manifolds to have multimorphisms

$$\text{PDisks}_n(r_1, \dots, r_k; s) \subseteq (\mathbb{C}^n)^k$$

given by all configurations $(z_1, \dots, z_k)$ of polydisks of radii r_i such that their closures are disjoint and fit inside of $\Delta_s(0)$. We will see that holomorphically translation-invariant prefactorization algebras form algebras for this colored operad.

As before, if $\mathcal{F}$ is a holomorphically translation-invariant prefactorization algebra on $\mathbb{C}^n$, denote by $\mathcal{F}_r$ the cochain complex $\mathcal{F}(\Delta_r(0))$. For any polydisk configuration $p \in \text{PDisks}(r_1, \dots, r_k; s)$, translation-invariance yields a map given by the factorization product

$$m[p] : \mathcal{F}_{r_1} \otimes \dots \otimes \mathcal{F}_{r_k} \rightarrow \mathcal{F}_s.$$

This map is a smooth map of cochain complexes, and varies smoothly with p . To show that $\mathcal{F}$ is an algebra for $\text{PDisks}(r_1, \dots, r_k; s)$, we must show that these maps are holomorphic in a particular sense.

We must use the extra structure of the derivations η_i which make $\partial_{\bar{z}_i}$ holomorphically trivial. Let $\{z_{ij}, \bar{z}_{ij}\}_{1 \leq i \leq k}^{1 \leq j \leq n}$ be coordinates on $(\mathbb{C}^n)^k$. This yields coordinates on the open subset

$$\text{PDisks}_n(r_1, \dots, r_k; s) \subseteq (\mathbb{C}^n)^k.$$

Thus we can define operations

$$\frac{\partial}{\partial \bar{z}_{ij}} m[p] : \mathcal{F}_{r_1} \otimes \dots \otimes \mathcal{F}_{r_k} \rightarrow \mathcal{F}_s$$

by differentiating $m[p]$, which depends on p , in the $\bar{z}_{ij}$ -direction. To show $m[p]$ depends holomorphically on p (up to homotopy), it suffices to show that this operation annihilates $m[p]$ in cohomology.

Let $m[p] \circ_i \eta_j$ denote the operation

$$(\alpha_1, \dots, \alpha_k) \mapsto (-1)^{\sum_{\ell=1}^{i-1} |\alpha_\ell|} m[p](\alpha_1, \dots, \eta_j \alpha_i, \dots, \alpha_k)$$

where the sign arises from Koszul's sign rule. The axioms of a smoothly translation-invariant prefactorization algebra (definition 4.8.4 condition 3) tell us that

$$\frac{\partial}{\partial \bar{z}_{ij}} m[p] = m[p] \circ_i \frac{\partial}{\partial \bar{z}_j}.$$

Here, $\partial_{\bar{z}_j}$ is the derivation of the prefactorization algebra $\mathcal{F}$. Note that since $d_{\mathcal{F}}$ and η_j have cohomological degree 1 and -1 , respectively,

$$[d_{\mathcal{F}}, \eta_j] = d_{\mathcal{F}} \eta_j - (-1)^{(1)(-1)} \eta_j d_{\mathcal{F}} = d_{\mathcal{F}} \eta_j + \eta_j d_{\mathcal{F}}.$$

Additionally, recall that the differential of a derivation $d\eta_j$ is defined as the commutator with the differential $[d_{\mathcal{F}}, \eta_j]$. Thus the first equation of definition 5.1.1 can be rewritten as

$$[d_{\mathcal{F}}, \eta_j] = \frac{\partial}{\partial \bar{z}_j}.$$

Applying this fact and doing some messy algebra yields

$$[d_{\mathcal{F}}, m[p] \circ_i \eta_j] = \frac{\partial}{\partial \bar{z}_{ij}} m[p].$$

The algebra is not hard, just use the fact that m is a chain map and commutes with differentials, and chase elements under the action of the tensor product differential. The commutator once again simplifies to a homotopy witnessing the fact that the right hand side is homotopic to 0. Thus, $m[p]$ depends holomorphically on p up to homotopy.

5.1.3 A cooperadic reformulation

We will form a colored cooperad in the category of convenient cochain complexes using the Dolbeault complex. We will utilize the fact that if $\widehat{\otimes}_\beta$ is the symmetric monoidal product for convenient vector spaces, then

$$\Omega^{0,\bullet}(X \times Y) \cong \Omega^{0,\bullet}(X) \widehat{\otimes}_\beta \Omega^{0,\bullet}(Y).$$

I don't actually know what a cooperad is, but the notation for the maps is useful, so I have copied down the definitions from this section.

Let $r_1, \dots, r_k, t_1, \dots, t_m, s \in \mathbb{R}_{>0}$. Define

$$\circ_i^* : \Omega^{0,\bullet}(\text{PDisks}_n(t_1, \dots, t_{i-1}, r_1, \dots, r_k, t_{i+1}, \dots, t_m; s)) \rightarrow \Omega^{0,\bullet}(\text{PDisks}_n(r_1, \dots, r_k; t_i) \times \text{PDisks}(t_1, \dots, t_m; s))$$

be the pullback of $\circ_i$ on Dolbeault complexes. Given a differentiable vector space V , define

$$\Omega^{0,\bullet}(X, V) := \Omega^{0,\bullet}(X) \otimes_{C^\infty(X)} C^\infty(X, V).$$

Since differential operators on X act on $C^\infty(X, V)$, we can define $\bar{\partial}$ on $\Omega^{0,\bullet}(X, V)$. Anyways the cooperad structure on $\mathcal{F}$ is given by lifting $m[p]$ to a map

$$\mu^{\bar{\partial}}(r_1, \dots, r_k; s) : \mathcal{F}_{r_1} \otimes \dots \otimes \mathcal{F}_{r_k} \rightarrow \Omega^{0,\bullet}(\text{PDisks}_n(r_1, \dots, r_k; s), \mathcal{F}_s)$$

which are compatible with differentials and satisfy the relevant conditions to be a cooperad. See the book for the specifics, and a nice remark on comparisons with other theories.

5.2 A general method for constructing vertex algebras

In this section, we show that the cohomology of a holomorphically translation-invariant prefactorization algebra on $\mathbb{C}$ with a compatible S^1 -action yields a vertex algebra. Using some results from the second volume, this will allow the construction of vertex algebras using compatible Lagrangians.

5.2.1 The circle action

Definition 5.2.1. A *holomorphically translation-invariant prefactorization algebra $\mathcal{F}$ with compatible S^1 action* is a smoothly $S^1 \ltimes \mathbb{R}^2$ -invariant prefactorization algebra $\mathcal{F}$ on $\mathbb{C}$ with values in $\text{Ch}_{\mathbb{C}}$, together with an extension of the action of the complex Lie algebra

$$\text{Lie}_{\mathbb{C}}(S^1 \ltimes \mathbb{R}^2) = \mathbb{C} \{ \partial_\theta, \partial_z, \partial_{\bar{z}} \},$$

where ∂_θ is a basis of $\text{Lie}_{\mathbb{C}}(S^1)$, to an action of the dg Lie algebra

$$\mathbb{C} \{ \eta \} [1] \xrightarrow{\text{d}\eta = \partial_{\bar{z}}} \mathbb{C} \{ \partial_\theta, \partial_z, \partial_{\bar{z}} \}.$$

In this dg Lie algebra, the only nonvanishing commutator involving η is

$$[\partial_\theta, \eta] = -\eta.$$

I take some time to explore this definition, since it is somewhat opaque. The intuition is that this definition encodes all the data of definition 5.1.1, but adds on an action by S^1 in a compatible way via the semidirect product. Write

$$\text{Lie}_{\mathbb{C}}(S^1 \ltimes \mathbb{R}^2) = (S^1 \ltimes \mathbb{R}^2) \otimes_{\mathbb{R}} \mathbb{C}.$$

Note that S^1 acts on $\mathbb{C}$ via rotation in the usual way. We add this into the action of $\mathbb{C} \cong \mathbb{R}^2$ via the semi-direct product, which is used since rotations don't commute with translations. Thus the fact that $\mathcal{F}$ is smoothly $(S^1 \ltimes \mathbb{R}^2)$ -invariant means that it is discretely translation and rotation invariant, and has infinitesimal actions

$$\frac{\partial}{\partial z}, \frac{\partial}{\partial \bar{z}}, \frac{\partial}{\partial \theta}$$

for translation in z and $\bar{z}$, and rotation, which act as derivations for the factorization product, and are compatible with global $(S^1 \ltimes \mathbb{R}^2)$ -invariance in the sense of definition 4.8.4. Note that the brackets for the complexified Lie algebra can be derived to be

$$[\partial_\theta, \partial_z] = \partial_z, \quad [\partial_\theta, \partial_{\bar{z}}] = -\partial_{\bar{z}}.$$

Note that part of the datum of this definition is that $\mathcal{F}$ be translation-invariant. We have just added the fact that it is also rotation-invariant.

We now want to consider the extension to the dg Lie algebra. The reason for this extension is that we want $\mathcal{F}$ to be *holomorphically invariant*, not just smoothly, so we need to define η like we did in definition 4.8.4 to witness the fact that the infinitesimal generators of translation are null-homotopic. We do not require that the infinitesimal generator of rotation is null-homotopic, since the prefactorization algebra is *holomorphically translation-invariant*, not holomorphically rotation-invariant; it only comes with a compatible smooth rotation action. Write $\mathfrak{g}$ for the dg Lie algebra. This is done simply by adding in η such that $d\eta = \partial_{\bar{z}}$. Since the action $\mathfrak{g} \rightarrow \text{Der}(\mathcal{F})$ is a chain homomorphism, we see that this relationship holds in the action, meaning the action of $\partial_{\bar{z}}$ vanishes in cohomology.

The definition of the bracket is forced by the definition of d : recall that d acts as a derivation for the bracket in the sense that

$$d[x, y] = [dx, y] + (-1)^{|x|} [x, dy].$$

Note that $d\mathfrak{g}^0 = 0$, meaning that for any $\partial_x \in \mathfrak{g}^0$, we have

$$d[\eta, \partial_x] = [d\eta, \partial_x] + [\eta, 0] = [\partial_{\bar{z}}, \partial_x].$$

We now simply read off the fact that $d[\eta, \partial_\theta] = -\partial_{\bar{z}}$ and 0 otherwise from the definition of the bracket in $\mathfrak{g}^0$ from earlier.

Discussion 5.2.2. It is finally time to make precise some technicalities about what we mean when we say $\mathcal{F}$ takes values in $\text{Ch}_{\mathbb{C}}$. Our prefactorization algebras all take values in *differentiable vector spaces*, which need only be $\mathbb{R}$ -vector spaces. Thus, a *complex* differentiable vector space will refer to a differentiable vector space E such that for all manifolds M , the $C^\infty(M)$ -module $E(M)$ has a $\mathbb{C}$ -vector space structure. Additionally, pullback morphisms should be $\mathbb{C}$ -linear.

There is a way of using the flat connection ∇ for E to obtain a notion of the $\bar{\partial}$ -operator, which will be our preferred way of describing when a complex differentiable vector space is holomorphic (since working with $\mathcal{O}$ -modules on $\mathcal{C}\text{Man}$ is homologically poorly-behaved). First, recall that ∇ is a map

$$\nabla : C^\infty(M, E) \rightarrow \Omega^1(M, E) =: \Omega^1(M) \otimes_{C^\infty(M)} C^\infty(M, E),$$

natural in M , where $C^\infty(M, E) = E(M)$. We can obtain a decomposition into $\Omega^{1,0}(M, E) \oplus \Omega^{0,1}(M, E)$ for all complex manifolds M . Due to the use of C^∞ -multilinear maps, some extra algebraic machinery is required to construct this decomposition, which we now detail.

Recall the extension of scalars map: given a commutative ring R , an R -algebra A , an R -module M , and an A -module N , there is a canonical isomorphism of A -modules

$$M \otimes_R N \cong (M \otimes_R A) \otimes_A N.$$

Setting $R = C^\infty(M)$ for M a complex manifold, let $A = C^\infty(M, \mathbb{C}) \cong C^\infty(M) \otimes_{\mathbb{R}} \mathbb{C}$, let $M = \Gamma(M, T^*M) = \Omega^1(M)$, and let $N = C^\infty(M, E)$. Note that $C^\infty(M, E)$ is indeed an $C^\infty(M, \mathbb{C})$ -module

since it is a complex vector space, and thus for any $f \in C^\infty(M, E)$ and smooth $u + iv : M \rightarrow \mathbb{C}$, we see that $(u + iv)f = uf + ivf$, which is defined. Then we obtain

$$\Gamma(M, T^*M) \otimes_{C^\infty(M)} C^\infty(M, E) \cong (\Gamma(M, T^*M) \otimes_{C^\infty(M)} \mathbb{C}) \otimes_{C^\infty(M, \mathbb{C})} C^\infty(M, E).$$

Using the fact that global sections is a monoidal functor, and thus commutes with tensor products, we have

$$(\Gamma(M, T^*M) \otimes_{C^\infty(M)} \mathbb{C}) \otimes_{C^\infty(M, \mathbb{C})} C^\infty(M, E) \cong \Gamma(M, T_{\mathbb{C}}^*M) \otimes_{C^\infty(M, \mathbb{C})} C^\infty(M, E).$$

Decompose $T_{\mathbb{C}}^*M \cong T^{*1,0}M \oplus T^{*0,1}M$, and using the fact that $\oplus$ is a product and Γ is right-adjoint, we obtain

$$\Gamma(M, T_{\mathbb{C}}^*M) \otimes_{C^\infty(M, \mathbb{C})} C^\infty(M, E) \cong (\Gamma(M, T^{*1,0}M) \oplus \Gamma(M, T^{*0,1}M)) \otimes_{C^\infty(M, \mathbb{C})} C^\infty(M, E).$$

Using the fact that $- \otimes_{C^\infty(M, \mathbb{C})} C^\infty(M, E)$ is left-adjoint, and thus commutes with the coproduct $\oplus$, we have

$$\begin{aligned} & \cong (\Gamma(M, T^{*1,0}M) \otimes_{C^\infty(M, \mathbb{C})} C^\infty(M, E)) \oplus (\Gamma(M, T^{*0,1}M) \otimes_{C^\infty(M, \mathbb{C})} C^\infty(M, E)) \\ & = (\Omega^{1,0}(M) \otimes_{C^\infty(M, \mathbb{C})} C^\infty(M, E)) \oplus (\Omega^{0,1}(M) \otimes_{C^\infty(M, \mathbb{C})} C^\infty(M, E)) =: \Omega^{1,0}(M, E) \oplus \Omega^{0,1}(M, E). \end{aligned}$$

We thus obtain a projection

$$\pi^{0,1} : \Omega^1(M, E) \rightarrow \Omega^{0,1}(M, E).$$

Define $\bar{\partial} := \pi^{0,1} \circ \nabla$. Define ∂ in the same way. Thus we obtain a morphism

$$\bar{\partial} : C^\infty(M, E) \rightarrow \Omega^{0,1}(M, E)$$

natural in M . Note that this in particular gives us both a morphism of sheaves on $\mathcal{C}\text{Man}$

$$\bar{\partial} : E \rightarrow \Omega^{0,1}(-, E),$$

and a morphism of sheaves on M

$$\bar{\partial} : C_M^\infty(-, E) \rightarrow \Omega_M^{0,1}(-, E).$$

Now letting $E^\bullet$ be a differentiable complex cochain complex, these constructions promote to cochain complexes (degreewise), yielding a map $\bar{\partial}^\bullet$ of sheaves of complex differentiable cochain complexes. We will focus on this being a map of sheaves on M . Since $E^\bullet$ is a cochain complex of complex differentiable vector spaces, we can take its cohomology $H^\bullet(E^\bullet)$ as a cochain complex. This yields a sheaf $H^\bullet(E^\bullet)$ of graded differentiable vector spaces. Note that since the sections functor does *not* commute with cohomology, $H^\bullet(E^\bullet)(U) \neq H^\bullet(E^\bullet(U))$. Rather, we denote $\Gamma(-, H^\bullet(E^\bullet))$ by $C^\infty(-, H^\bullet(E^\bullet))$ as usual. Regardless, we obtain two graded complex differentiable vector spaces

$$C^\infty(-, H^\bullet(E^\bullet)), \quad \Omega^{0,1}(-, H^\bullet(E^\bullet)).$$

Restricting along a complex manifold yields sheaves of graded complex (not differentiable) vector spaces

$$C_M^\infty(-, H^\bullet(E^\bullet)), \quad \Omega_M^{0,1}(-, H^\bullet(E^\bullet)).$$

The $\bar{\partial}^\bullet$ -operator is a cochain map, and by functoriality of cohomology descends to a morphism of graded complex differentiable vector spaces

$$H^\bullet(\bar{\partial}^\bullet) : C^\infty(-, H^\bullet(E^\bullet)) \rightarrow \Omega^{0,1}(-, H^\bullet(E^\bullet)).$$

This restricts along the complex manifold M to a map of sheaves of complex (not differentiable) vector spaces

$$H^\bullet(\bar{\partial}_M^\bullet) = H^\bullet(\bar{\partial}^\bullet)_M : C_M^\infty(-, H^\bullet(E^\bullet)) \rightarrow \Omega_M^{0,1}(-, H^\bullet(E^\bullet)).$$

The equality noted holds since restriction from the site $\mathcal{M}an$ to a single manifold is an exact functor. Thus, we will not be extremely careful about the placements of the subscripts, since it can be inferred from context without dealing with technicalities.

We have now finally introduced enough machinery for a proper notion of holomorphicity of maps $M \rightarrow E$. The kernel $\text{Ker } H^\bullet(\bar{\partial}_M^\bullet)$ is itself a sheaf on M , and since kernels are computed pointwise in sheaf categories, we can describe the kernel very well. We view $H^\bullet(\bar{\partial}_M^\bullet)$ as a map of sheaves of graded vector spaces (we can equivalently view it as a map of graded sheaves of vector spaces!), so we start by passing to sections over U . Thus we have a map

$$H^\bullet(\bar{\partial}_M^\bullet)_U : C_M^\infty(U, H^\bullet(E^\bullet)) \rightarrow \Omega_M^{0,1}(U, H^\bullet(E^\bullet)).$$

We now pass to homogeneous elements. For any $i \in \mathbb{Z}$, we have a map of $\mathbb{C}$ -vector spaces

$$H^i(\bar{\partial}_M^\bullet)_U : C_M^\infty(U, H^i(E^\bullet)) \rightarrow \Omega_M^{0,1}(U, H^i(E^\bullet)).$$

Suppose that $f \in \text{Ker } H^i(\bar{\partial}_M^\bullet)_U$. Then $H^i(\bar{\partial}_M^\bullet)_U(f) = 0$. We would like $H^i(\bar{\partial}_M^\bullet)_U(f)$ to lift to a closed section $\bar{\partial}_M^i(f)$ of $C_M^\infty(-, E^i)$, which would express the statement that f is holomorphic up to homotopy. However, this would require commuting sections with cohomology, which need not follow. Rather, we appeal to the explicit construction of limits in $\mathcal{S}hf$ and $\mathcal{P}shf$.

The cohomology is the quotient of the kernel by the image. Kernel and image are limits, and by continuity of the forgetful functor $U : \mathcal{S}hf \rightarrow \mathcal{P}shf$ and $[?]$, we can compute them pointwise. By cocontinuity of sheafification, the quotient in $\mathcal{S}hf$, which is a colimit, is the sheafification of the quotient in $\mathcal{P}shf$, which is once again computed pointwise in $\mathcal{P}shf$. Because (co)limits in $\mathcal{P}shf$ are computed pointwise, cohomology commutes with sections of presheaves, since cohomology is a composite of limits and colimits, and sections are just evaluation functors. Thus, if $(-)^+$ is sheafification, then

$$H^i(E^\bullet) \cong \left(\frac{\text{Ker}(E^i \rightarrow E^{i+1})}{\text{Im}(E^{i-1} \rightarrow E^i)} \right)^+,$$

where all (co)limits other than sheafification are computed in $\mathcal{P}shf$. Denote the unsheafified cohomology (aka the cohomology computed in $\mathcal{P}shf$) by $H_{\mathcal{P}shf}^\bullet$. Recall that the sheafification of a presheaf is given by collections of all compatible germs of the presheaf. Thus, a section f of $H^i(E^\bullet)(U)$ is a collection $\{f_p \in H_{\mathcal{P}shf}^i(E^\bullet)_p\}_{p \in U}$ of compatible germs on U . Germs are compatible if they are locally germs of some section, meaning that there is a collection $\{f_\alpha \in H_{\mathcal{P}shf}^i(E^\bullet)(U_\alpha)\}_{\alpha \in A}$ of sections of the presheaf cohomology, for some open cover $\{U_\alpha\}_{\alpha \in A}$ of U . Since every section gives rise to a collection of compatible germs, we obtain that $f|_{U_\alpha} = f_\alpha$. Thus, f can locally be written as a section f_α of $H_{\mathcal{P}shf}^i(E^\bullet)(U_\alpha)$. But since presheaf cohomology commutes with sections, we can view each f_α as a cohomology class

$$[f_\alpha] \in H_{\mathcal{P}shf}^i(E^\bullet(U_\alpha)), \quad \forall \alpha \in A.$$

Note that we use the notation $[-]$ only when considering elements as a cohomology class of presheaf cohomology. We will sometimes abuse notation and refer to f_α as a representative of this class, but we will avoid this where possible.

Now we act $H_{\mathcal{P}shf}^i(\bar{\partial}_M^\bullet)_U$ on each cohomology class f_α . It's important to note here that by universality of presheaf, $H^i(\bar{\partial}_M)$ restricted along the map $\iota : H_{\mathcal{P}shf}^i(E^\bullet) \rightarrow H^i(E^\bullet)$ acts the same as

$H_{\mathcal{P}\text{shf}}^i(\bar{\partial}_M)$. The morphism ι maps a section s over U to its collection $\{s_p\}_{p \in U}$ of compatible germs, which we view as being equivalent to the section s from the presheaf. However, multiple sections in a presheaf can have the same germs, so ι identifies all these sections). Thus, if $H^i(\bar{\partial}_M^\bullet)_U(f_\alpha) = 0$, then we may immediately conclude that $H_{\mathcal{P}\text{shf}}^i(\bar{\partial}_M^\bullet)_U(f_\alpha)$ has vanishing germs. Since $H^i(\bar{\partial}_M^\bullet)$ is a sheaf morphism, it commutes with restriction morphisms, and since $f|_{U_\alpha} = f_\alpha$, we have

$$H^i(\bar{\partial}_M^\bullet)_U(f_\alpha) = H^i(\bar{\partial}_M^\bullet)_U(f|_{U_\alpha}) = H^i(\bar{\partial}_M^\bullet)_U(f)|_{U_\alpha} = 0|_{U_\alpha} = 0.$$

The final step follows since restriction morphisms must be abelian group homomorphisms.

Now, we need to rewrite this in the language of presheaf cohomology. This tells us that for any $p \in U_\alpha$, the germ at p , denoted $H_{\mathcal{P}\text{shf}}^i(\bar{\partial}_M^\bullet)_U(f_\alpha)_p$, is the zero germ. This implies that for any point $p \in U_\alpha$, there is an open set $p \in V \subseteq U_\alpha$ with

$$[\bar{\partial}_M^i f_\alpha]|_V = [\bar{\partial}_M^i f|_V] = 0.$$

Choose an open cover $\{U_{\alpha\beta}\}_{\beta \in B_\alpha}$ of U_α by such subsets, and assemble this into a cover $\{U_\beta\}_{\beta \in B}$ of U . Since this is finer than $\{U_\alpha\}_{\alpha \in A}$, we still have $f|_{U_\beta} = f_\beta$. However, we now have that each cohomology class $[f_\beta]$ is strictly killed by $H_{\mathcal{P}\text{shf}}^i(\bar{\partial}_M^\bullet)$:

$$H_{\mathcal{P}\text{shf}}^i(\bar{\partial}_M^\bullet)[f_\beta] = 0.$$

Thus we can find an exact representative $d\eta_\beta \in \Omega_M^{0,1}(U_\beta, E^i)$ of $[\bar{\partial}_M^i(f_\beta)]$. Note that although cohomology for sheaves does not commute with taking sections, we still get a map of sheaves

$$\text{Ker } d_{E^\bullet} \rightarrow H^\bullet(E^\bullet).$$

Recall that $\Omega^{0,1}(-, E^\bullet) := \Omega_M^{0,1} \otimes_{C_M^\infty} E^\bullet$, where this tensor product is applied *pointwise*, and is *not* the graded tensor product. Thus applying $\Omega_M^{0,1} \otimes -$ to this map of sheaves yields a map

$$\Omega_M^{0,1}(-, \text{Ker } d_E) \rightarrow \Omega^{0,1}(-, H^\bullet(E^\bullet)).$$

Note that $\text{Ker } d_E \otimes \Omega_M^{0,1} = \text{Ker } d_\Omega$, since the differential on $\Omega_M^{0,1}(-, E^\bullet)$ is induced from $E^\bullet$. Thus, by taking the component of this map of sheaves at U_β , we see that $d\eta_\beta \in \Omega_M^{0,1}(U_\beta, E^i)$ maps to an element of $\Omega_M^{0,1}(U_\beta, H^\bullet(E^\bullet))$. I believe naturality of everything forces this image to be both $H^\bullet(\bar{\partial}_M^\bullet)_U(f)|_{U_\beta}$ and 0, expressing that f is locally d_E -exact.

To summarize, given any $i \in \mathbb{Z}$, any $U \subseteq M$, and any section $f \in \text{Ker } H^i(\bar{\partial}_M^\bullet)_U$, we can find an open cover $\{U_\beta\}_{\beta \in B}$ of U such that $H^\bullet(\bar{\partial}_M^\bullet)_U(f|_{U_\beta})$ lifts to a $(1 \otimes d_E)$ -exact element of $\Omega^{0,1}(U, E^i)$. Thus, $H^\bullet(\bar{\partial}_M^\bullet)_U(f)$ is locally exact.

Thus, we denote $\text{Ker}(H^\bullet(\bar{\partial}_M^\bullet))$ by $\mathcal{O}_M(-, H^\bullet(E^\bullet))$, and call its elements *holomorphic sections* of $H^\bullet(E^\bullet)$ on M . We think of holomorphic sections on $H^\bullet(E^\bullet)$ as locally given by smooth maps $M \rightarrow E^\bullet$ which are holomorphic up to homotopy.

The theorem on vertex algebras will require an extra hypothesis regarding the S^1 -action. Given any compact Lie group G , we can formulate a notion of *tameness* for a G -action on a differentiable vector space. The S^1 -action on $\mathcal{F}_r$ will then be required to be tame.

Given a Lie group G , the space $\mathcal{D}(G)$ of distributions on G is an algebra under convolution. Since distributions are naturally differentiable vector spaces, and the convolution

$$*: \mathcal{D}(G) \times \mathcal{D}(G) \rightarrow \mathcal{D}(G)$$

is smooth, $\mathcal{D}(G)$ forms an algebra in $\mathcal{DVS}$. Here, an action just means a smooth bilinear map

$$\mathcal{D}(G) \times E \rightarrow E$$

which is associative in the relevant sense. There is a map $\delta : G \rightarrow \mathcal{D}(G)^\times$ sending g to the δ -distribution at g . It is smooth and a homomorphism of monoids.

Definition 5.2.3. Let E be a differentiable vector space and let G be a compact Lie group. A *tame* action of G on E is a smooth action of the algebra $\mathcal{D}(G)$ on E . Note that this is, in particular, an action of G on E via the smooth map of groups $G \rightarrow \mathcal{D}(G)^\times$ sending $g \mapsto \delta_g$. If E is a differentiable cochain complex, a tame action is an action of $\mathcal{D}(G)$ on E which commutes with the differential.

Let us now specialize $G = S^1$. Let k be an integer. Then there is an irreducible representation ρ_k of S^1 on S^1 given by the function $\lambda \mapsto \lambda^k$, where $\lambda \in S^1 \subseteq \mathbb{C}$. We view ρ_k as a map of sheaves on Man from the representable sheaf $C^\infty(-, S^1)$ to the representable differentiable vector space $C^\infty(-, \mathbb{C})$. This lifts naturally to a representation of the algebra $\mathcal{D}(S^1)$ defined by

$$\tilde{\rho}_k(\phi) = \langle \phi, \rho_k \rangle$$

for $\langle -, - \rangle$ the pairing on distributions and functions. For simplicity, we will abuse notation and write ρ_k for this representation as well.

If E is a differentiable vector space equipped with such a smooth action of $\mathcal{D}(S^1)$, we use $E_k \subseteq E$ to denote the subspace on which $\mathcal{D}(S^1)$ acts by ρ_k . We call this the *weight k eigenspace* for the S^1 -action on E .

The map $\rho_k : S^1 \rightarrow S^1$ can be viewed as a distribution by integrating against it. If we denote the action of $\mathcal{D}(S^1)$ on E by $*$, then the map $\rho_k * - : E \rightarrow E$ defines a projection π_k from E onto E_k : since $\rho_k * \rho_k = \rho_k$ by idempotence,

$$\pi_k(\pi_k(v)) = \rho_k * (\rho_k * v) = (\rho_k * \rho_k) * v = \rho_k * v = \pi_k(v).$$

The statment now follows by some theorem.

5.2.2 The statement of the main theorem

Theorem 5.2.4. *Let $\mathcal{F}$ be a unital S^1 -equivariant holomorphically translation invariant prefactorization algebra on $\mathbb{C}$ valued in differentiable vector spaces. Assume that for each disk $\Delta_0(r)$ around the origin, the action of S^1 on $\mathcal{F}(\Delta_0(r))$ is tame. Let $\mathcal{F}_k(\Delta_0(r))$ be the weight k eigenspace of the S^1 -action on $\mathcal{F}(\Delta_0(r))$. We make the following additional assumptions.*

1. *Assume that, for $r < r'$, the structure map*

$$\mathcal{F}_k(\Delta_0(r)) \rightarrow \mathcal{F}_k(\Delta_0(r'))$$

is a quasi-isomorphism.

2. *For $k \gg 0$, the vector space $H^\bullet(\mathcal{F}_k(\Delta_0(r))) \cong 0$.*
3. *For each k and r , we require that $H^\bullet(\mathcal{F}_k(\Delta_0(r)))$ is isomorphic, as a sheaf on the site of smooth manifolds, to a countable sequential colimit of finite-dimensional graded vector spaces.*

Let $V_k = H^\bullet(\mathcal{F}_k(\Delta_0(r)))$, and let $V = \bigoplus_{k \in \mathbb{Z}} V_k$. This space is independent of r by assumption. Then V has the structure of a vertex algebra, determined by the structure maps of $\mathcal{F}$.

The construction is functorial: maps of prefactorization algebras that respect the equivariance conditions produce maps of vertex algebras. The book proves this later.

5.2.3 Using the theorem

Part III

Factorization algebras

Chapter 6

Factorization algebras: definitions and constructions

A prefactorization algebra is similar to a precosheaf or a presheaf. In the same way that it is usually beneficial to refine pre(co)sheaves to (co)sheaves, there is a “descent” axiom which refines a prefactorization algebra into a factorization algebra. We will see that the examples we have constructed so far satisfy this axiom later in this chapter.

This axiom is not always essential for field theory, but is still beneficial, and sometimes simplifies the arguments mathematically.

6.1 Factorization algebras

A factorization algebra is a prefactorization algebra that satisfies a particular local to global axiom. This axiom is similar to the cosheaf axiom, but defined for *Weiss covers*. We review the important notions.

6.1.1 Sheaves and cosheaves

Definition 6.1.1. Let M be a topological space and $\mathcal{C}$ any category. A *presheaf on M valued in $\mathcal{C}$* is a functor $\mathcal{A} : \text{Open}(M)^{\text{op}} \rightarrow \mathcal{C}$. Given an open cover $\{U_i\}_{i \in I}$ of some open set $U \subseteq M$, there is a canonical map

$$\prod_{i \in I} (-)|_{U_i} : \mathcal{A}(U) \rightarrow \prod_{i \in I} \mathcal{A}(U_i),$$

provided the product exists. We say $\mathcal{A}$ is a *sheaf* if for every open $U \subseteq M$ and every open cover $\{U_i\}_{i \in I}$ of U , this map is an equalizer

$$\mathcal{A}(U) \longrightarrow \prod_{i \in I} \mathcal{A}(U_i) \rightrightarrows \prod_{i, j \in I} \mathcal{A}(U_i \cap U_j)$$

of the parallel arrows given by the inclusions $U_i \cap U_j \rightarrow U_i$ and $U_i \cap U_j \rightarrow U_j$.

If $\mathcal{C}$ is a concrete category, then definition 6.1.1 says that given a collection $\{s_i \in \mathcal{A}(U_i)\}_{i \in I}$ of sections in the product $\prod_i \mathcal{A}(U_i)$, we can pull this back to a *unique* section $s \in \mathcal{A}(U)$ with $s|_{U_i} = s_i$ *if and only if* the property

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$$

holds. Note that this is precisely the sheaf axiom. Framing this definition allows it to be easily dualized.

Definition 6.1.2. A *precosheaf* Φ on M with values in $\mathcal{C}$ is a functor $\Phi : \text{Open}(M) \rightarrow \mathcal{C}$. It is a cosheaf if for every open cover $\{U_k\}_{k \in K}$ of an open set $U \subseteq M$, the canonical map

$$\coprod_{k \in K} \Phi(U_k) \rightarrow \Phi(U)$$

is a coequalizer

$$\coprod_{i,j \in K} \Phi(U_i \cap U_j) \rightrightarrows \coprod_{k \in K} \Phi(U_k) \longrightarrow \Phi(U)$$

of the parallel arrows given by the inclusions $U_i \cap U_j \rightarrow U_i$ and $U_i \cap U_j \rightarrow U_j$. Equivalently, Φ is a sheaf with values in $\mathcal{C}^{\text{op}}$.

Remark 6.1.3. In these notes, we focus on categories such as cochain complexes valued in an abelian category, dg Lie R -algebras, graded vector spaces, etc. These categories are linear/abelian in nature, so the cosheaf axiom can be reformulated as the exactness of the sequence

$$\bigoplus_{i,j \in K} \Phi(U_i \cap U_j) \longrightarrow \bigoplus_{k \in K} \Phi(U_k) \longrightarrow \Phi(U) \longrightarrow 0,$$

where the first map is the difference of the parallel arrows in definition 6.1.2. We will usually phrase our axioms in this style, since it is generally easier to work with.

6.1.2 Weiss covers

We will require our covers for factorization algebras to be finer than those for cosheaves. In particular, we want our open covers to capture the structure of the factorization product in the underlying prefactorization algebra. The proper notion will be a *Weiss cover*.

Definition 6.1.4. Let U be an open set of some space M . A *Weiss cover* is a collection of open sets $\mathfrak{U} = \{U_i \subseteq U\}_{i \in I}$ such that for any finite collection of points $S = \{x_1, \dots, x_k\} \subseteq U$, there is an open set $U_i \in \mathfrak{U}$ such that $S \subseteq U_i$.

The Weiss covers define a Grothendieck pretopology, and thus a Grothendieck topology, on $\text{Open}(M)$, called the *Weiss topology*. A Weiss cover is much larger than an open cover in most cases, and is better suited to studying all possible configuration spaces of finitely many points in U .

Example 6.1.5. Let M be a smooth n -manifold. One Weiss cover of M is the collection of all open sets diffeomorphic to a finite disjoint union of n -balls. Alternatively, the open sets $\{M \setminus \{p\} \mid p \in M\}$ is a Weiss cover. One can even form a Weiss cover with countably many elements by fixing a Riemannian metric g . Let d be the corresponding distance function. Choose points $\{p_n \in M\}_{n \in \mathbb{N}}$ such that the union of the g -balls

$$B_1(p_n) = \{q \in M \mid d(p_n, q) < 1\}$$

covers M . Consider the collection of all balls

$$\mathfrak{D} = \{D_{1/m}(p_n) \mid m \in \mathbb{N}, n \in \mathbb{N}\}.$$

The collection of all finite, pairwise disjoint unions of elements of $\mathfrak{D}$ is a countable Weiss cover.

These examples suggest the following definition.

Definition 6.1.6. Let $\mathfrak{B}$ be a Weiss cover. Then a cover $\mathfrak{U} = \{U_\alpha\}_{\alpha \in A}$ *generates* the Weiss cover $\mathfrak{B}$ if every $V \in \mathfrak{B}$ is a finite disjoint union of open sets in $\mathfrak{U}$.

6.1.3 Strict factorization algebras

The value of a factorization algebra on U will be determined by its behavior on a Weiss cover of U in the same way a cosheaf is determined by open covers.

Definition 6.1.7. Let $\mathcal{F}$ be a prefactorization algebra on M valued in a linear colored operad $\mathcal{C}$ (in particular, a linear symmetric monoidal category). Then $\mathcal{F}$ is a *factorization algebra* if it satisfies the following property: for every open subset $U \subseteq M$ and every Weiss cover $\{U_i \subseteq U\}_{i \in I}$ of U , the sequence

$$\bigoplus_{i,j \in K} \Phi(U_i \cap U_j) \longrightarrow \bigoplus_{k \in K} \Phi(U_k) \longrightarrow \Phi(U) \longrightarrow 0,$$

that was defined in remark 6.1.3 is exact. That is, $\mathcal{F}$ is a factorization algebra if it is a cosheaf with respect to the Weiss topology.

A factorization algebra is *multiplicative* if for every pair of disjoint open sets $U, V \subseteq M$, the factorization product

$$m_{U \cup V}^{U,V} : \mathcal{F}(U) \otimes \mathcal{F}(V) \rightarrow \mathcal{F}(U \cup V)$$

is an isomorphism.

The multiplicative axiom says that observables of finite collections of disjoint opens can be factored into tensor products of its value on each disjoint open.

Direct verification that a prefactorization algebra satisfies the descent axiom is typically not easy. We will later discuss results that exhibit many examples, allowing us to build more complex factorization algebras from simple examples.

6.1.4 The Čech complex and homotopy factorization algebras

If the target symmetric monoidal category has a notion of a weak equivalence, then there is a natural modification of the notion of a cosheaf. One should require not that $\Phi(U)$ be a colimit, but that it be a homotopy colimit. Applying this modification to a cosheaf yields the notion of a homotopy cosheaf. We will begin by describing these, since they are simpler.

In this book, we always take the target category to be unbounded cochain complexes with values in some additive category $\mathcal{A}$, with weak equivalences given by quasi-isomorphisms. Usually, we take $\mathcal{A} = \mathcal{DVS}$. In this case, $\mathcal{DVS}$ is a *Grothendieck abelian category*.

Digression on the derived ∞ -category $\mathcal{D}^\infty(\mathcal{A}) \cong \mathcal{N}_\bullet^{\text{dg}}(\mathcal{Ch}_\mathcal{A})$

Definition 6.1.8. A *Grothendieck abelian category* is an abelian category $\mathcal{A}$ which satisfies Grothendieck's AB5 axiom, and has a generator. More explicitly,

1. AB3: $\mathcal{A}$ has all small coproducts. This implies that $\mathcal{A}$ is cocomplete.
2. AB5: Filtered colimits are exact in $\mathcal{A}$. More precisely, given a filtered category $\mathcal{J}$ (see [?]), the functor $\text{colim} : \mathcal{A}^{\mathcal{J}} \rightarrow \mathcal{A}$ is an exact functor.
3. There exists an object $g \in \mathcal{A}$ called the *generator* such that $\mathcal{A}(g, -)$ is faithful.

There is a useful theorem of Lurie's. Let $\mathcal{A}$ be any Grothendieck abelian category, and let $\mathcal{Ch}_\mathcal{A}$ denote the 1-category of (unbounded) cochain complexes in $\mathcal{A}$. Let W denote the quasi-isomorphisms in $\mathcal{Ch}_\mathcal{A}$.

Proposition 6.1.9 (Lurie). *There is a left proper combinatorial model structure on $\mathcal{Ch}_{\mathcal{A}}$. Given a map $f : c \rightarrow d$,*

- *f is a weak equivalence if and only if it is a quasi-isomorphism;*
- *f is a cofibration if and only if $f^k : c^k \rightarrow d^k$ is a monomorphism in $\mathcal{A}$;*
- *f is a fibration if and only if it satisfies the right lifting property with respect to every trivial cofibration (also called acyclic cofibrations).*

Note that this is the same as the usual cofibrantly generated model structure for $\mathcal{Ch}_{\mathcal{A}}$; the new results are that it is left proper and combinatorial, which are terms I will gloss over.

Note that all cochains are cofibrant in this model structure. We then define the ∞ -categorical version of $\mathcal{Ch}_{\mathcal{A}}$, also called the derived ∞ -category and denoted $\mathcal{D}^{\infty}(\mathcal{A})$, to be the differential graded nerve $N_{\bullet}^{\text{dg}}(\mathcal{Ch}_{\mathcal{A}}^f)$ of the *full* subcategory $\mathcal{Ch}_{\mathcal{A}}^f$ of fibrant objects of $\mathcal{Ch}_{\mathcal{A}}$. Here we must note that $\mathcal{Ch}_{\mathcal{A}}$ is self-enriched via the usual internal hom

$$\underline{\mathcal{Ch}}_{\mathcal{A}}(c, d)^k \cong \prod_{n \in \mathbb{Z}} \mathcal{A}(c^n, d^{n+k}) \cong \mathcal{A}^{\mathbb{Z}}(c, d[k]),$$

for $\mathcal{A}^{\mathbb{Z}}$ the (functor) category of $\mathbb{Z}$ -graded objects of $\mathcal{A}$. A differential graded category only requires $\mathcal{Ch}_{\mathbb{Z}}$ -enrichment, and many of the differential graded categories we use are $\mathcal{Ch}_R$, which are self-enriched and come with a natural $\mathbb{Z}$ -action, which is a stronger result. Using this, we may now define the differential graded nerve on $\mathcal{Ch}_{\mathcal{A}}^f$. Here I follow [Lur26]’s approach in Tag 00PK.

Definition 6.1.10 ([Lur26] 2.5.3.1, 2.5.3.5). Let $\mathcal{C}$ be a differential graded category with $\mathcal{Ch}_{\mathbb{Z}}$ -hom $\underline{\mathcal{C}}$. Define the simplicial set $N_{\bullet}^{\text{dg}}(\mathcal{C})$ as follows.

- An n -simplex $\sigma \in N_n^{\text{dg}}(\mathcal{C})$ is a pair of tuples

$$\sigma = \left(\{x_i\}_{1 \leq i \leq n}, \{f_I \mid I = \{i_0 > i_1 > \dots > i_k\} \subseteq [n]\} \right),$$

where each x_i is an object of $\mathcal{C}$, and f_I is a homogeneous element of $\underline{\mathcal{C}}(x_{i_k}, x_{i_0})^{1-k}$ of degree $1 - k$, satisfying the identity

$$df_I = \sum_{a=1}^{k-1} (-1)^a (f_{\{i_0 > \dots > i_a\}} \circ f_{\{i_a > \dots > i_k\}} - f_{I \setminus \{i_a\}}).$$

Here d is the differential of $\underline{\mathcal{C}}$. Note that this requires I have at least two elements.

- Given a morphism $\alpha : [n] \rightarrow [m]$ in the simplex category, the map $\alpha^* : N_m^{\text{dg}}(\mathcal{C}) \rightarrow N_n^{\text{dg}}(\mathcal{C})$ is given by

$$\left(\{x_i\}_{1 \leq i \leq m}, \{f_I\} \right) \mapsto \left(\{x_{\alpha(j)}\}_{1 \leq j \leq n}, \{g_J\} \right)$$

for

$$g_J := \begin{cases} f_{\alpha(J)}, & \alpha|J \text{ is injective} \\ 1_{x_i}, & J = \{j_0 > j_1\} \text{ with } \alpha(j_0) = \alpha(j_1) = i. \\ 0, & \text{otherwise} \end{cases}$$

It is shown in [Lur26] 2.5.3.10 that $N_{\bullet}^{\text{dg}}(\mathcal{C})$ is a quasi-category. Note that the definition Lurie presents works with homological grading conventions, whereas we work with cohomological grading conventions. Thus, a morphism of degree k for Lurie has degree $-k$ for us.

This definition is a mess. We take a minute to look at the intuition behind the differential graded nerve. We start by looking at 0-simplicies. Since the set I must have at least two elements, a 0-simplex is a pair $(\{x\}, \emptyset)$. We identify this with the object $x \in \mathcal{C}$, yielding $N_0^{\text{dg}}(\mathcal{C}) \cong \text{Obj}(\mathcal{C})$.

Now consider a 1-simplex. Note that $[1]$ has two elements, so there is exactly one linearly ordered subset $I \subseteq [1]$, which is simply $I = [1]$. Thus, a 1-simplex σ is a pair $(\{x_0, x_1\}, \{f_{10}\})$ (here we recall $i_0 > i_1$, thus $i_0 = 1$). We simply write f for f_{10} . Writing $i_0 = 1$ and $i_1 = 0$ yields that $f \in \underline{\mathcal{C}}(x_0, x_1)^0$; i.e. f is a morphism $x_0 \rightarrow x_1$ of cohomological degree 0. Now we have that

$$df = \sum_{a=1}^{k-1} (-1)^a (f_{\{i_0 > \dots > i_a\}} \circ f_{\{i_a > \dots > i_k\}} - f_{I \setminus \{i_a\}}).$$

However, note that $k = 1$, so this sum is from 1 to 0. Hence, this is the empty sum, which is 0. Therefore we have $df = 0$, meaning a 1-simplex is a closed cochain map $f : x_0 \rightarrow x_1$.

In the setting $\text{Ch}_{\mathcal{A}}$, recall that if $f \in \underline{\text{Ch}}_{\mathcal{A}}(X, Y)^k$, then f is a map of graded $\mathcal{A}$ -objects $f : X \rightarrow Y[k]$ of cohomological degree k . Then $df : X \rightarrow Y[k+1]$ is also a map of graded $\mathcal{A}$ -objects, and is defined by

$$df = d_Y \circ f - (-1)^{|f|=k} f \circ d_X.$$

Applying our condition that (X, Y, f) is a 1-simplex of the differential graded nerve, this simplifies to

$$df = d_Y \circ f - f \circ d_X = 0 \iff d_Y \circ f = f \circ d_X.$$

This is exactly the condition that f is a map of cochains. Thus, 1-simplicies of $N_{\bullet}^{\text{dg}}(\text{Ch}_{\mathcal{A}})$ are precisely cochain maps, as we would expect.

We now explore 2-simplices of the differential graded category $\mathcal{C}$. First, note that $[2]$ admits four subsets with at least two elements: $\{1, 0\}, \{2, 1\}, \{2, 0\}, \{2, 1, 0\}$. By definition 6.1.10, a 2-simplex σ is the data $(x_0, x_1, x_2, f_{10}, f_{20}, f_{21}, f_{210})$, where

- $x_0, x_1, x_2 \in \mathcal{C}$ are objects.
- The maps f_{i_0, i_1} are elements of $\underline{\mathcal{C}}(x_{i_1}, x_{i_0})^0$. More precisely,

$$f_{10} \in \underline{\mathcal{C}}(x_0, x_1)^0, \quad f_{20} \in \underline{\mathcal{C}}(x_0, x_2)^0, \quad f_{21} \in \underline{\mathcal{C}}(x_1, x_2)^0.$$

- The map f_{210} is an element of $\underline{\mathcal{C}}(x_0, x_2)^{-1}$ satisfying the identity

$$df_{210} = \sum_{a=1}^{2-1} (-1)^a (f_{\{i_0 > \dots > i_a\}} \circ f_{\{i_a > \dots > i_k\}} - f_{I \setminus \{i_a\}}) = -(f_{21} \circ f_{10} - f_{20}) = f_{20} - f_{21} \circ f_{10}.$$

- Additionally, by the same logic applied to 1-simplicies, we have $df_{ij} = 0$ for all $i, j \in [2]$.

We now apply this in the case that $\mathcal{C} = \text{Ch}_{\mathcal{A}}$. We have that $f_{ij} : x_j \rightarrow x_i$ is a chain map for all $i, j \in [2]$. Additionally, $f_{210} : x_0 \rightarrow x_2[1]$ has cohomological degree 1, and yields the statement

$$df_{210} = f_{20} - f_{21} \circ f_{10} \iff df_{210} + f_{21} \circ f_{10} = f_{20}.$$

Additionally, note that $f_{21} \circ f_{10}$ is also closed: since f_{21}, f_{10} are closed, we have $f_{21} \circ d_1 = d_2 \circ f_{21}$, and $f_{10} \circ d_0 = d_1 \circ f_{10}$. Thus,

$$\begin{aligned}
d(f_{21} \circ f_{10}) &= d_2 \circ f_{21} \circ f_{10} - f_{21} \circ f_{10} \circ d_0 \\
&= (d_2 \circ f_{21}) \circ f_{10} - f_{21} \circ (f_{10} \circ d_0) \\
&= f_{21} \circ (d_1 \circ f_{10}) - f_{21} \circ (f_{10} \circ d_0) \\
&= f_{21} \circ (d_1 \circ f_{10} - f_{01} \circ d_0) \\
&= f_{21} \circ df \\
&= f_{21} \circ 0 \\
&= 0.
\end{aligned}$$

Therefore, the cochain maps $f_{21} \circ f_{10}$ and f_{20} differ by the exact element df_{210} , which means that they define the same cohomology class. Therefore, the datum of the arrow f_{210} can be interpreted as the statement that f_{20} and $f_{21} \circ f_{10}$ are cohomologous, which we depict as the diagram

$$\begin{array}{ccc}
& x_1 & \\
f_{10} \nearrow & \Downarrow f_{210} & \nwarrow f_{21} \\
x_0 & \xrightarrow{f_{20}} & x_2.
\end{array}$$

Thus, composition of maps is well-defined up to homology, and since $N_{\bullet}^{\text{dg}}(\text{Ch}_{\mathcal{A}})$ is a quasi-category, composites always exist. Thus we have the expected intuition.

This suggests a notion of faces and degeneracies. We will now show that the given definition of simplicial operators indeed yields the proper notion of faces and degeneracies. We follow the convention that d_i is the i th face operator, and d^i is the i th coface map in the simplex category Δ . We similarly write s_i and s^i for degeneracies and codegeneracies, respectively. Both of these operators are fairly simple to deal with.

Let $d^k : [n] \rightarrow [n+1]$ be a coface, and let $\sigma = (\{x_i\}_{1 \leq i \leq n+1}, \{f_I\})$ be an $(n+1)$ -simplex of $N_{\bullet}^{\text{dg}}(\mathcal{C})$. We work with the action of d_k on the objects $\{x_i\}_{1 \leq i \leq n+1}$ and morphisms $\{f_I\}$ independently. First, by definition, we have

$$d_k \{x_i\}_{1 \leq i \leq n+1} = \{x_0, \dots, x_{k-1}, x_{k+1}, \dots, x_{n+1}\} = \{x_0, \dots, \widehat{x_k}, \dots, x_{n+1}\},$$

where the hat denotes the omission of an argument. Next, note that d^k is injective, and thus

$$d_k \{f_I\} = \left\{ g_J = f_{d^k(J)} \mid J \subseteq [n] \right\}.$$

We explore a low-dimensional example in $\text{Ch}_{\mathcal{A}}$. Let $\sigma = (x_0, x_1, f)$ be a 1-simplex in $N_{\bullet}^{\text{dg}}(\text{Ch}_{\mathcal{A}})$. We expect that $d_1 \sigma = x_0$ and $d_0 \sigma = x_1$, which would allow us to say that f is a morphism $x_0 \rightarrow x_1$ in the ∞ -category $N_{\bullet}^{\text{dg}}(\text{Ch}_{\mathcal{A}})$. First, note that $d_i \sigma$ is a 0-simplex for any i , and thus has no morphisms. Thus we need only look at the action of d_i on objects, and indeed,

$$d_0 \sigma = \{\widehat{x_0}, x_1\} = x_1, \quad d_1 \sigma = \{x_0, \widehat{x_1}\} = x_0.$$

Thus, σ is a morphism $x_0 \rightarrow x_1$. We prefer to write $f : x_0 \rightarrow x_1$ for obvious reasons.

We also quickly check the faces of a 2-simplex. Let $\sigma = (x_0, x_1, x_2, f_{10}, f_{20}, f_{21}, f_{210})$ be a 2-simplex of $N_{\bullet}^{\text{dg}}(\mathcal{C})$. We start with d_0 . From the above diagram, we would expect that $d_0 \sigma = (x_1, x_2, f_{21})$.

We immediately see that the objects work out, so now we appeal to the morphisms. Recall that 1-simplices have a single morphism, determined by the subset $[1] \subseteq [1]$, which we will write as g_{10} . By definition,

$$g_{10} = f_{d^0(1), d^0(0)} = f_{21}.$$

Thus, $d_0\sigma = (x_1, x_2, f_{21})$, as expected. We similarly compute that the morphisms for $d_1\sigma$ and $d_2\sigma$ are, respectively,

$$f_{d^1(1), d^1(0)} = f_{20}, \quad f_{d^2(1), d^2(0)} = f_{10}.$$

These are precisely what we would expect, given the above intuition.

Finally, we check degeneracies. Recall that codegeneracies $s^i : [n] \rightarrow [n-1]$ are injective everywhere except the subset $\{i, i+1\} \subseteq [n]$, which both map to $i \in [n-1]$. Thus, combining rules 1 and 2 for simplicial operators yields that if $\sigma = (\{x_i\}_{1 \leq i \leq n-1}, \{f_I\})$ is an $(n-1)$ -simplex, then

$$s_k\sigma = (\{x_0, \dots, x_i, x_i, \dots, x_{n-1}\}, \{f_{s^i(J)} \mid J \subseteq [n]\})$$

is an n -simplex, where $f_{s^i(J)}$ simply inserts 1 at i , between the parts of J where s^i is injective (on those parts, it comes from the definition of σ , like before). Thus, it is trivial to see that for $x \in N_0^{\text{dg}}(\mathcal{C})$, we have $s^0x = (x, x, 1_x)$ is the identity morphism.

We now return to the notation $\mathcal{D}^\infty(\mathcal{A})$ for the simplicial set $N_\bullet^{\text{dg}}(\mathcal{Ch}_{\mathcal{A}})$. It is possible to state that $\mathcal{D}^\infty(\mathcal{A})$ is the ∞ -categorical localization of $\mathcal{Ch}_{\mathcal{A}}$ at weak equivalences in a more precise sense: there is an ∞ -functor $N(\mathcal{Ch}_{\mathcal{A}}) \rightarrow \mathcal{D}^\infty(\mathcal{A})$ which is universal amongst ∞ -categories with respect to inverting quasi-isomorphisms. The ∞ -category $\mathcal{D}^\infty(\mathcal{A})$ is a stable ∞ -category.

Finally, note that $\mathcal{D}^\infty(\mathcal{Ch}_{\mathcal{A}})$ is presented by $\mathcal{Ch}_{\mathcal{A}}$ as a combinatorial model category. This makes intuitive sense, since standard model category theory yields an equivalence $\mathcal{Ch}_{\mathcal{A}} \simeq \mathcal{Ch}_{\mathcal{A}}^f$.

Back to the normal content of the chapter

Since $\mathcal{Ch}_{\mathcal{DVS}}$ is Grothendieck abelian, all of this applies to our most common example of factorization algebras. We now have a notion of fibration and cofibration in $\mathcal{Ch}_{\mathcal{DVS}}$, as well as a simple way of constructing homotopy (co)limits using quasi-categories.

Let Φ be a precosheaf on a topological space M with values in cochain complexes in a sufficiently nice additive category $\mathcal{A}$. Let $\mathfrak{U} = \{U_i\}_{i \in I}$ be an open cover of some open subset $U \subseteq M$. Consider the cochain complex

$$C^k := \bigoplus_{j_0, \dots, j_k \in I; j_i \text{ all distinct}} \Phi(U_{j_0} \cap \dots \cap U_{j_k}).$$

There are two differentials here, defined as follows. Given a multi-index J , let U_J denote the intersection over all U_j , $j \in J$. Since $\Phi(U_J)$ is a cochain complex, it has a differential d_J . We can rewrite

$$C^{k,p} = \bigoplus_{|J|=k+1} \Phi(U_J)^p,$$

making C clearly bigraded. We then define the differential $d : C^{k,p} \rightarrow C^{k,p+1}$ as mapping some $x \in \Phi(U_J)^p \mapsto d_J x$. Thus, $d = \bigoplus_{|J|=k+1} d_J$. This is called the *internal differential*.

There is also an *external differential*, which acts via the extension maps arising from the precosheaf Φ . This is the dual of the usual Čech differential, meaning it makes $C^{\bullet,p}$ into a chain complex. Write ι_a for the pushforward map $\Phi(U_J) \rightarrow \Phi(U_{J \setminus \{j_a\}})$, where $j_a \in J = \{j_0 < \dots < j_a < \dots < j_k\}$

(here we assume tuples are increasing only). Let $x \in C^{k,p}$. Then x is a tuple, which has a value $x_J \in \Phi(U_J)$ for each $J \subseteq I$ with $k+1$ elements. We define

$$d(x)_K = \sum_{|J|=k+1, J \setminus \{j_a\} = K} (-1)^a \iota_a(x_J).$$

The *Čech complex of $\mathfrak{u}$ with coefficients in Φ* is defined as the *totalization* of this double complex, which is the complex

$$\check{C}(\mathfrak{U}, \Phi) := \bigoplus_{k=0}^{\infty} \left(\bigoplus_{j_0, \dots, j_k \in I; j_i \text{ all distinct}} \Phi(U_{j_0} \cap \dots \cap U_{j_k})[k] \right).$$

In particular,

$$\check{C}(\mathfrak{U}, \Phi)^n = \bigoplus_{k \geq 0} \bigoplus_{|J|=k+1} \Phi(U_J)^{n+k}.$$

Note that $C^{p,k}$ falls into degree $n = p - k$. Recall that the differential of the shifted complex $\Phi(U_J)[k]$ picks up a sign of $(-1)^k$. The differential of the totalization uses both differentials on the bigrading, so we denote the internal differential by d_i , and the external differential by d_e . Then we define the differential D as

$$D|C^{k,p} = (-1)^k d_i + d_e.$$

Note that this is a map $C^{k,p} \rightarrow C^{k-1,p} \oplus C^{k,p+1}$ which lives inside degree $p - (k-1) = p+1-k = n+1$. It is easy to show $D^2 = 0$, yielding a cochain complex.

There is a natural map $\check{C}(\mathfrak{U}, \Phi) \rightarrow \Phi(U)$ given as follows. Since Φ yields maps $\Phi(U_J) \rightarrow \Phi(U)$, we can define a map on the n th degree via the composite

$$\bigoplus_{k \geq 0} C^{k,n+k} \xrightarrow{\pi_0} C^{0,n} = \bigoplus_{|J|=n+1} \Phi(U_J)^n \xrightarrow{\bigoplus_J \iota_J} \Phi(U)^n,$$

where π_0 is the projection of the product onto $k = 0$, and $\bigoplus_J \iota_J$ is the coproduct of the pushforwards $\iota_J : \Phi(U_J) \rightarrow \Phi(U)$. These maps assemble into a cochain map.

Definition 6.1.11. A precosheaf Φ is a *homotopy cosheaf* if the natural map $\check{C}(\mathfrak{U}, \Phi)^0 \rightarrow \Phi(U)$ is a quasi-isomorphism for every open U , and every open cover $\mathfrak{U}$ of U .

This definition has a lot hiding behind it that I don't feel like going through. The idea is that there is a Čech nerve N which maps open covers $\mathfrak{U}$ of $U \subseteq X$ to a particular simplicial object in $(\mathcal{J}op \downarrow U)$. Then Kan extending a cosheaf to this simplicial object yields a particular diagram indexed by Δ^{op} . This diagram is 1-coskeletal, and its 1-truncation is a coequalizer diagram (the existence of the diagram posits the existence of a section for the parallel arrows, but given such a datum, the colimit is isomorphic to the coequalizer). When passing to ∞ -categories by taking the nerve of this 1-category, gluing laws have to hold up to coherent homotopy, and the coherence requires taking into account n -fold intersections. Hence, we take the homotopy colimit of the full Čech nerve.

We now suggest the notion of a homotopy factorization algebra.

Definition 6.1.12. A *homotopy factorization algebra* (valued in cochain complexes) is a prefactorization algebra $\mathcal{F}$ on X valued in $\mathcal{Ch}_{\mathcal{A}}$ with the property that for every open $U \subseteq X$, and every Weiss cover $\mathfrak{U}$ of U , the natural map from the Čech complex is a quasi-isomorphism

$$\check{C}(\mathfrak{U}, \mathcal{F}) \simeq \mathcal{F}(U).$$

If $\mathcal{A}$ is symmetric monoidal, then so is $\mathcal{Ch}_{\mathcal{A}}$ via the Day convolution. The homotopy factorization algebra $\mathcal{F}$ is *multiplicative* if for every disjoint pair of opens $U, V \subseteq X$, the factorization product

$$\mathcal{F}(U) \otimes \mathcal{F}(V) \rightarrow \mathcal{F}(U \cup V)$$

is a quasi-isomorphism.

Remark 6.1.13. Due to our applications having observables living in cohomology, it is almost always most natural to consider homotopy factorization algebras. Thus, the term “factorization algebra” will almost always denote a homotopy factorization algebra.

Definition 6.1.14. A *differentiable factorization algebra* is a factorization algebra valued in differentiable cochain complexes.

Due to differentiable vector spaces not having a tensor product, we have to modify the multiplicative axiom to a statement about the factorization product, viewed as a bilinear map. However, most vector spaces we work with are convenient vector spaces, which we recall *do* have a tensor product. Thus, this issue can usually be ignored. Regardless, we are more interested in the descent axiom than multiplicity.

6.1.5 Weak equivalences

One can promote the notion of weak equivalences of factorization algebras valued in an ∞ -category to the level of factorization algebras themselves.

Definition 6.1.15. Let $\mathcal{F}, \mathcal{G}$ be homotopy factorization algebras. Then a map $\phi : \mathcal{F} \rightarrow \mathcal{G}$ is a *weak equivalence* if each component $\phi_U : \mathcal{F}(U) \simeq \mathcal{G}(U)$ is a weak equivalence.

Definition 6.1.16. A *factorizing basis* $\mathfrak{U} = \{U_i\}_{i \in I}$ for a space X is a basis for the topology on X with the following properties:

1. For every finite set $\{x_i\}_{1 \leq i \leq n} \subseteq X$, there exists $i \in I$ such that $\{x_i\}_{1 \leq i \leq n} \subseteq U_i$. Equivalently, $\mathfrak{U}$ is a Weiss cover.
2. If U_i, U_j are disjoint, then $U_i \cup U_j \in \mathfrak{U}$.
3. $U_i \cap U_j \in \mathfrak{U}$ for all $U_i, U_j \in \mathfrak{U}$.

Lemma 6.1.17. A map $\phi : \mathcal{F} \rightarrow \mathcal{G}$ of differentiable factorization algebras on a Hausdorff space X is a weak equivalence if and only if, for every factorizing basis $\mathfrak{U}$ of X and every $U \in \mathfrak{U}$, the map $\phi_U : \mathcal{F}(U) \simeq \mathcal{G}(U)$ is a weak equivalence. More succinctly, it suffices to check weak equivalences on a factorizing basis.

Proof. Fix a factorizing basis $\mathfrak{U}$. Let $V \subseteq X$ be open, and let $\mathfrak{U}_V$ denote the Weiss cover of V generated by all open subsets in $\mathfrak{U}$ that lie in V . We must show that this is indeed a Weiss cover. This is trivial: let $\{V_i \subseteq V\}_{i \in J} \subseteq \mathfrak{U}$ be an open cover of V by $\mathfrak{U}$, which exists since $\mathfrak{U}$ is a basis. Now let $S \subseteq V$ be a finite set, and for $p, q \in S$, let W_p, W_q be disjoint open neighborhoods of p, q , respectively, which exist since X is Hausdorff. We may without loss of generality take $W_p, W_q \subseteq V$, since if they aren't subsets, we can intersect them with V to produce subsets. Now cover W_p, W_q with elements $\{U_i^p\}_{i \in J_p \subseteq I} \subseteq \mathfrak{U}$ and $\{U_j^q\}_{j \in J_q \subseteq I} \subseteq \mathfrak{U}$ of the factorizing basis, which can be done since it is in particular a basis. Then $U_i^p \cap U_j^q = \emptyset$ for all i, j . There exists at least one $i \in J_p$ and $j \in J_q$ such that $p \in U_i^p$ and $q \in U_j^q$. Thus, we have disjoint elements of $\mathfrak{U}$ which are subsets of V ,

and contain 2 of the points of S . Continuing this argument inductively yields a collection of pairwise disjoint opens whose union covers S , and is a subset of V . Since this belongs to $\mathfrak{U}_V$, we are done.

Now we continue with the proof. By the descent axiom, there is a weak equivalence $\check{C}(\mathfrak{U}_V, \mathcal{F}) \simeq \mathcal{F}(V)$, and similarly for $\mathcal{G}$. Thus, it suffices to check that the map

$$\check{\phi} : \check{C}(\mathfrak{U}_V, \mathcal{F}) \rightarrow \check{C}(\mathfrak{U}_V, \mathcal{G})$$

is a weak equivalence. Recall that

$$\check{C}^\bullet(\mathfrak{U}_V, \mathcal{F}) \cong \bigoplus_{k \geq 0} \bigoplus_{|J|=k+1} \mathcal{F}(U_J)[k]^\bullet, \quad \check{C}^\bullet(\mathfrak{U}_V, \mathcal{G}) \cong \bigoplus_{k \geq 0} \bigoplus_{|J|=k+1} \mathcal{G}(U_J)[k]^\bullet.$$

We must use the fact that ϕ is a quasi-isomorphism on all elements of the factorizing basis. Note that since $\mathfrak{U}_V$ is given by finite unions of pairwise disjoint elements of $\mathfrak{U}$, and since $\mathfrak{U}$ is closed under finite union of pairwise disjoint elements, we have $\mathfrak{U}_V \subseteq \mathfrak{U}$. Since U_J is a union of pairwise disjoint elements of $\mathfrak{U}_V$, and thus $\mathfrak{U}$, and since factorizing bases are closed under finite intersection, we have $U_J \in \mathfrak{U}$ for all J . Therefore, ϕ_{U_J} is a quasi-isomorphism.

Filter the complex by k . The differential on the E_0 page is the internal differential d_i , since the external differential lowers the degree of k , and thus is killed in $k = 0$. We recall that the internal differential is simply the direct sum of $d_{\mathcal{F}(U_J)}$. Since $\oplus : \text{Ch}_{\mathcal{D}\text{VS}}^2 \rightarrow \text{Ch}_{\mathcal{D}\text{VS}}$ is exact (easy to prove for arbitrary additive categories), it suffices to show each ϕ_J is a quasi-isomorphism. We have shown this, so $\check{\phi}$ is a quasi-isomorphism on the page E_0 .

The E_1 page is given by taking homology with respect to d_i . Since ϕ_{U_J} is a quasi-isomorphism, the induced map is an isomorphism on the E_1 page. The spectral sequence comparison theorem yields an isomorphism on E_∞ , meaning $\check{\phi}$ is a quasi-isomorphism. $\blacksquare$

6.1.6 The (multi)category structure on factorization algebras

The category $\mathcal{Fa}(X, \mathbb{C})$ is the full subcategory of $\mathcal{P}\mathcal{Fa}(X, \mathbb{C})$ spanned by factorization algebras. The tensor product of prefactorization algebras was defined as

$$(\mathcal{F} \otimes \mathcal{G})(U) := \mathcal{F}(U) \otimes \mathcal{G}(U).$$

This does not immediately promote, but if the tensor product commutes (separately in each variable) with geometric realizations of simplicial objects, then $\mathcal{Fa}(X, \mathbb{C})$ has a symmetric monoidal structure. This is to ensure $F \otimes G$ satisfies descent. We will not use this structure on factorization algebras, so we do not expand on this.

6.2 Factorization algebras in quantum field theory

So far, the definition of a factorization algebra is well-motivated by sheaf theory. It allows us to better study factorization algebras, and thus their quantum field theories. But one must ask what kinds of quantum field theories produce factorization algebras, and what the descent axiom means physically.

Consider the Weiss cover $\mathfrak{U}_\varepsilon$. The descent axiom roughly says that an observable O can be written as a sum of observables of the form $O_1 \cdots O_k$, for $O_i \in \text{Obs}(B_{\delta_i}(x_i))$, $x_1, \dots, x_k \in M$. Now by taking ε small, we can write any observable as a sum of products of observables supported arbitrarily close to a point.

To be more precise, we can (up to chain homotopy) write any $O \in \mathcal{F}(U)$ as a sum

$$O = \sum_{\alpha} \iota_{V_{\alpha}}(O_{\alpha}), \quad V_{\alpha} \in \mathfrak{U}_{\varepsilon}.$$

But since elements of $\mathfrak{U}$ are just pairwise disjoint unions of balls, we can decompose V_{α} into disjoint unions $B_{\delta_1}(x_1) \sqcup \cdots \sqcup B_{\delta_k}(x_k)$ of balls. If we also assume the factorization algebra is multiplicative, then up to chain homotopy, we can write any $O_{\alpha} \in V_{\alpha}$ as a factorization product $O_1 \cdots O_k$ of observables $O_i \in B_{\delta_i}(x_i)$.

By taking ε small, we can write observables as being supported arbitrarily close to a point. In particular, all observables can be written as sums of products of observables supported arbitrarily close to a point. This is a reasonable assumption, since most physical observables in QFT have pointlike support.

However, QFT also considers many global observables. The answer to how these fit into the factorization picture reveals that factorization algebras only work for *perturbative quantum field theories*. An arbitrary field theory need not yield a factorization algebra, only a prefactorization algebra.

Indeed, in a perturbative gauge theory, the gauge field (eg. a connection) is taken to be a small perturbation $A_0 + \delta A$ of a fixed and well-known connection A_0 . Consider the holonomy over some loop L . It is easy to work with $A_0 + \delta A$ as a power series of integrals over L^k , which shows that $A_0 + \delta A$ can be built from observables supported at points on L .

However, in a non-perturbative setting, we cannot do this. In general, the prefactorization algebra of observables need not satisfy descent.

6.3 Variant definitions of factorization algebras

Didn't feel like writing this bit down.

6.4 Locally constant factorization algebras

Recall that the little n -disks operad E_n is the topological (∞) -operad with 1 color whose k -ary operations are embeddings $\coprod_{i=1}^k B_{r_i}(x_i) \hookrightarrow B_1(0)$. Essentially, the datum of a k -ary operation is a k -tuple of pairs (x_i, r_i) , where $x_i \in \mathbb{R}^n$ and $r_i \in \mathbb{R}_{>0}$, such that the relevant balls are all disjoint, and fit into the unit ball. We thus identify $E_n(k) \subseteq (\mathbb{R}^n \times \mathbb{R}_{>0})^k$, and give it the subspace topology. Composition is defined by inserting a disk configuration into a single disk. The S_k -action is given by permuting the labels $1, \dots, k$ on the tuple.

An E_n -algebra A is thus an object A of a symmetric monoidal category $\mathcal{C}$ which is enriched, tensored, and cotensored over $\mathcal{T}\text{op}$, or a related category like $\mathbf{sSet}$; together with maps $A^{\otimes k} \rightarrow A$ for each disk configuration, compatible with the substitution composition and S_k -action. Recall that the S_k -action on a symmetric monoidal category is given by the symmetry isomorphism, appropriately applied.

Although fundamentally different objects, factorization algebras bear a resemblance to E_n -algebras in the sense that they describe maps between disjoint opens sitting inside of larger opens. In fact, E_n -algebras form a full subcategory of factorization algebras on $\mathbb{R}^n$, which we discuss in this section.

Definition 6.4.1. A factorization algebra $\mathcal{F}$ on an n -manifold M is *locally constant* if for each inclusion of open disks $D \subseteq D'$, the map $\mathcal{F}(D) \rightarrow \mathcal{F}(D')$ is a quasi-isomorphism.

For example, given an associative algebra A , the prefactorization algebra A^{fact} on $\mathbb{R}$ is locally constant. Lurie has extended this example as follows. First, we must note that the homotopy coherent nerve promotes E_n to an ∞ -operad, allowing us to consider E_n -algebras in any symmetric monoidal ∞ -category.

Theorem 6.4.2. *There is an equivalence of ∞ -categories between E_n -algebras and locally constant factorization algebras on $\mathbb{R}^n$, valued in an ∞ -category.*

This theorem provides a fascinating bridge between topology and physics. Topological field theories produce locally constant factorization algebras, which can then be viewed as E_n -algebras, and analyzed using homotopy theory. Conversely, intuition from the physical behavior of topological field theories can motivate new insights and definitions in topology.

6.4.1 Motivation from topology

Many examples of E_n -algebras arise naturally from topology. As such, we begin our motivational work by considering algebras and factorization algebras valued in a nice category of spaces, which we denote $\mathcal{Top}$. Cocontinuity of Hom yields an isomorphism

$$\mathcal{Top}(U \amalg V, X) \cong \mathcal{Top}(U, X) \times \mathcal{Top}(V, X).$$

If we use the cartesian monoidal structure (which would yield a cartesian closed category), then we might think that a representable functor could yield a prefactorization algebra of spaces. Indeed, our original intuition of observables was some type of functions $\text{Map}(\mathbb{P}^\bullet(M), \mathbb{R})$, so this is not unexpected. The issue is, Hom is contravariant in its first argument, whereas we require prefactorization algebras to be covariant. The solution is to pass to compactly supported sections.

To make sense of compactly supported sections for an arbitrary space, consider a pointed space (X, p) , and define a compactly supported map $U \rightarrow X$ to be a map such that the closure of $f^{-1}(X \setminus \{p\})$ is compact. The usual case of a compactly supported map is recovered by taking $(X, p) = (\mathbb{R}, 0)$.

This defines a prefactorization algebra on pointed spaces that is also valued in pointed spaces, given by taking the subfunctor of $\mathcal{Top}_*(-, (X, p))$ given by compactly supported maps. This is topologized via the subspace topology for the compact-open topology. The basepoint is simply the constant map. One recovers a prefactorization algebra on spaces by identifying compactly supported maps $U \rightarrow (X, p)$ with basepoint-preserving compactly supported maps from the one point compactification of U to (X, p) . Restricting this along the open sets of a space M yields a prefactorization algebra on M . Write F for this prefactorization algebra.

Consider $M = \mathbb{R}^n$, and let $D \subseteq \mathbb{R}^n$ be an open disk. There is a weak homotopy equivalence to the n -loop space based at p :

$$F(D) \simeq \Omega_p^n X.$$

This is easy to see: a compactly supported map $f : D \rightarrow X$ must extend uniquely to a map on $\overline{D}$, mapping $\partial \overline{D} \mapsto p$. Then simply recall that $\Omega_p^n X$ is the space of maps $(\overline{D}, \partial \overline{D}) \rightarrow (X, p)$. Finally, recall that we can take $\partial \overline{D}$ to be the 1-point compactification of D , which produces the n -sphere.

This prefactorization algebra is locally constant: given an inclusion of open disks $D \hookrightarrow D'$, the extension map is easily verified to be a weak homotopy equivalence. Additionally, we have isomorphisms

$$F(U_1 \amalg U_2) \cong F(U_1) \times F(U_2)$$

for disjoint opens. If we have a disk configuration $D_1, D_2 \subseteq D_3$, then we get a map $F(D_1) \times F(D_2) \rightarrow F(D_3)$. These maps correspond to the standard E_n -structure on the n -fold loop space. A nice example is that when $M = \mathbb{R}$, we recover the standard product on $\pi_0 \Omega_p X \cong \pi_1(X, p)$.

This prefactorization algebra does not always satisfy gluing. However, if X is sufficiently connected, F is indeed a factorization algebra.

6.4.2 Relationship with Hochschild homology

Evaluating a locally constant factorization algebra globally on a manifold M is known as *factorization homology* or *topological chiral homology*. It is sometimes denoted

$$\int_M \mathcal{F}.$$

The following result is interesting.

Theorem 6.4.3. *For A an E_1 -algebra (eg. an associative algebra, and A_∞ -algebra), there is a weak equivalence*

$$A^{fact}(S^1) \simeq HH_\bullet(A),$$

for A^{fact} the locally constant factorization algebra on $\mathbb{R}$ associated to A , and $HH_\bullet(A)$ the Hochschild homology of A , defined up to quasi-isomorphism by

$$HH_\bullet(A) \simeq A \otimes_{A \otimes A}^{\mathbb{L}} A.$$

Here, the extension of A^{fact} from $\mathbb{R}$ to S^1 is defined by covering S^1 with opens $U, V \cong \mathbb{R}$ whose intersection has $U \cap V \cong \mathbb{R} \amalg \mathbb{R}$, and then taking the homotopy colimit over these opens and their intersection.

There is a generalization of this result. There are a number of ways to extend A^{fact} from $\mathbb{R}$ to S^1 , by allowing “monodromy”. Pick an automorphism σ of A . Pick an orientation of S^1 , and a point $p \in S^1$. Let $(A^{fact})^\sigma$ denote the prefactorization algebra on S^1 such that $S^1 \setminus \{p\}$ agrees with A^{fact} , but where the structure maps use the automorphism σ . For instance, if L is a small interval to the left of p , and R is a small interval to the right of p , and M is an interval containing both L and R , then the structure map $A \otimes A \rightarrow A$ is

$$a \otimes b \mapsto a \cdot \sigma(b)$$

where the left copy of A corresponds to L , and so on. It’s natural to view the copy of A associated to an interval containing p as an (A, A) -bimodule, with a left A -action and a σ -twisted right A -action.

Theorem 6.4.4. *There is a weak equivalence $(A^{fact})^\sigma \simeq HH_\bullet(A, A^\sigma)$.*

6.4.3 Factorization envelopes

The connection between locally constant factorization algebras and E_n -algebras provides a beautiful universal property for factorization envelopes. Let $\mathfrak{g}$ be a differential graded Lie algebra, and define the (n) -factorization envelope

$$\mathbb{U}_n \mathfrak{g} : M \mapsto C_\bullet(\Omega_c^\bullet(M) \otimes \mathfrak{g}),$$

where M runs over the category $\mathcal{E}mb_n$ of smooth n -manifolds and smooth embeddings. This yields a functor

$$\mathbb{U}_n : \mathcal{DGLie}_k \rightarrow \mathcal{Alg}_{E_n}(\mathcal{Ch}_k),$$

where k is a field of characteristic 0. Here we can use the singular chains functor to turn E_n into a dg operad, which is a nice way of viewing E_n as an ∞ -operad. Here, we view $\mathbb{U}_n$ as an E_n -algebra, by Lurie's theorem (consider disks in coordinates; Lurie's theorem only appeals to disks; there's a bit more nuance here though).

There is another “forgetful functor” $U : \text{Alg}_{E_n}(\mathcal{C}_k) \rightarrow \mathcal{DGLie}_k$. For $n = 1$, we view A_∞ -algebras as dg associative algebras, and this functor is simply given by defining the Lie bracket as a commutator. For $n > 1$, such a functor exists since the cohomology of E_n -algebras are P_n -algebras (somehow), which clearly have a forgetful functor to $\mathcal{DGLie}_k$.

The fascinating part is that we have an ∞ -adjunction $\mathbb{U}_n \dashv U$!

Theorem 6.4.5. *Let k be a field of characteristic zero. Let $\mathcal{C}$ be a stable, k -linear, presentably symmetric monoidal ∞ -category. Then there is an ∞ -adjunction*

$$\mathbb{U}_n \dashv U : \mathcal{DGLie}_k \rightarrow \text{Alg}_{E_n}(\mathcal{C})$$

such that for any $x \in \mathcal{C}$, we have $\text{Free}_{E_n}(X) \simeq \mathbb{U}_n \text{Free}_{\text{Lie}}(\Sigma^{n-1}X)$.

Here, Σ is the suspension functor, which we recall for chain complexes is simply a shift.

This concludes the book's introduction to factorization homology, which was very hand-wavy. I have a paper which goes more into it, which I may read sometime.

6.5 Factorization algebras from cosheaves

This section will describe a natural class of factorization algebras. The main theorem is that given a nice cosheaf F of cochains on a manifold F , the functor $\text{Sym } F(-)$ is a factorization algebra. This is trivially a prefactorization algebra, so we just need to consider the descent axiom. The examples we are ultimately interested in are cosheaves of compactly supported sections of a graded vector bundle, so we focus on cosheaves of this form when working on this theorem.

6.5.1 Preliminary definitions

Definition 6.5.1. A *local cochain complex* on M is a graded vector bundle E on M with finite rank, whose smooth sections shall be denoted $\mathcal{E}$, equipped with a differential operator $d : \mathcal{E} \rightarrow \mathcal{E}$ satisfying $d^2 = 0$.

We use similar notation to previously. We denote by $E^\vee, \mathcal{E}^\vee$ the graded dual $(E^\vee)^n = (E^{-n})^*$; we denote by $\mathcal{E}_c$ the cosheaf of compactly supported sections; we denote by $\bar{\mathcal{E}} := \mathcal{E} \otimes_{C_M^\infty} \mathcal{D}$ the distributional sections (here $\mathcal{D}$ is distributions), and we denote by $E^!, \mathcal{E}^!$ the object

$$E^! = E^\vee \otimes \text{Dens}_M, \quad \mathcal{E}^! = \mathcal{E}^\vee \otimes_{C_M^\infty} \text{Dens}_M.$$

Note that $\bar{\mathcal{E}}_c$ is the continuous dual to $\mathcal{E}^!$, and $\mathcal{E}_c$ the continuous dual to $\bar{\mathcal{E}}^!$.

Lemma 6.5.2. *A local cochain complex $(\mathcal{E}, d)$ is a sheaf of differentiable cochain complexes. It is also a homotopy sheaf. Similarly, compactly supported sections $(\mathcal{E}_c, d)$ is a homotopy cosheaf of differentiable cochain complexes.*

Proof. We work with the cosheaf case; the sheaf case follows similarly. First, we recall how to define a differentiable vector space structure on $\mathcal{E}_c$; this follows from section 3.5.5. Fix an open subset

$U \subseteq M$, and let X be any manifold. Let $\pi_X : M \times X \rightarrow X$ be the natural projection, and recall that the pullback bundle $\pi_X^* E$ is defined to have fibers

$$(\pi_X^* E)_{(x,m)} = E_m.$$

Now define $C^\infty(M, \mathcal{E}(U))$ to consist of all smooth sections of $\pi_X^* E$ over $M \times U$ whose support maps properly to X . More precisely, the projection to X , restricted along the support of a section $\pi_X : \text{Supp } s \rightarrow X$ is a proper map, i.e. preimages of compact sets are compact.

The flat connection is defined in coordinates as follows. Given local coordinates $\{x^\mu, m^\mu\}$ defined on a neighborhood U of a point (x, m) , we can choose a local frame $\{e^i\}$ for E , and thus $\pi_X^* E$, on this open U . Then write a section s as

$$s_{(x,m)} = \sum_i s_i(m, x) e_x^i$$

for some component functions s_i . Define the flat connection

$$\nabla s = \sum_j \left(\sum_i \frac{\partial s_i}{\partial m^j} \otimes e_x^i \right) \otimes dm^j.$$

This yields a convenient vector space, and thus a differentiable vector space.

Let $\{U_i\}_{i \in I}$ be an open cover of some open $U \subseteq M$. Let $ext_i : \mathcal{E}_c(U_i) \rightarrow \mathcal{E}_c(U)$ denote the extension-by-zero maps defining the precosheaf structure of $\mathcal{E}_c$. Consider the map

$$\sum_i ext_i : \bigoplus_i \mathcal{E}_c(U_i) \rightarrow \mathcal{E}_c(U).$$

We can also consider the map

$$\sum_{i,j} ext_{i,j} : \bigoplus_{i,j} \mathcal{E}_c(U_i \cap U_j) \rightarrow \mathcal{E}_c(U).$$

Here, $ext_{i,j}$ is the extension-by-zero map $\mathcal{E}_c(U_i \cap U_j) \rightarrow \mathcal{E}_c(U)$. Now write $ext_{i,j}^i$ for the extension-by-zero map $\mathcal{E}_c(U_i \cap U_j) \rightarrow \mathcal{E}_c(U_i)$. We can immediately see that the diagrams

$$\begin{array}{ccc} \bigoplus_i \mathcal{E}_c(U_i) & \xrightarrow{\sum_i ext_i} & \mathcal{E}_c(U) \\ \uparrow \oplus_{i,j} ext_{i,j}^i & \nearrow \sum_{i,j} ext_{i,j} & \\ \bigoplus_{i,j} \mathcal{E}_c(U_i \cap U_j) & & \end{array} \quad \begin{array}{ccc} \bigoplus_j \mathcal{E}_c(U_j) & \xrightarrow{\sum_j ext_j} & \mathcal{E}_c(U) \\ \uparrow \oplus_{i,j} ext_{i,j}^j & \nearrow \sum_{i,j} ext_{i,j} & \\ \bigoplus_{i,j} \mathcal{E}_c(U_i \cap U_j) & & \end{array}$$

commute due to the nature of the extension-by-zero map. Additionally, the arrows $\sum_j ext_j$ and $\sum_i ext_i$ are the same morphism, just with a different letter used as the index for notational clarity. We claim that this arrow equalizes the relevant parallel arrows. By linearity of the extension-by-zero maps, and by explicit construction of the coproduct, it suffices to show this for a single section $s_{ij} \in \mathcal{E}_c(U_i \cap U_j)$. Let $p \in U$. We have

$$\left(\sum_i ext_i \right) \circ \left(\bigoplus_{i,j} ext_{i,j}^i \right) (s_{ij})_p = (ext_i \circ ext_{i,j}^i)(s_{ij})_p = \begin{cases} (s_{ij})_p, & p \in \text{Supp}(s) \\ 0, & \text{otherwise} \end{cases};$$

$$\left(\sum_i ext_i\right) \circ \left(\bigoplus_{i,j} ext_{i,j}^i\right) (s_{ij})_p = (ext_j \circ ext_{i,j}^j)(s_{ij})_p = \begin{cases} (s_{ij})_p, & p \in \text{Supp}(s) \\ 0, & \text{otherwise} \end{cases}.$$

Thus, the maps $\sum_i ext_i$ and $\sum_{i,j} ext_{i,j}$ form a cocone for the equalizer diagram, yielding a morphism

$$ext_I : \text{colim} \left(\bigoplus_{i,j \in I} \mathcal{E}_c(U_i \cap U_j) \rightrightarrows \bigoplus_{i \in I} \mathcal{E}_c(U_i) \right) \longrightarrow \mathcal{E}_c(U).$$

We will show this map is an isomorphism. We will use a partition of unity.

Recall that a partition of unity subordinate to the cover $\{U_i\}_{i \in I}$ is a collection $\{\rho_i\}_{i \in I}$ of nonnegative smooth functions $M \rightarrow \mathbb{R}$ such that

- every $x \in M$ has a neighborhood in which all but finitely many ρ_i vanish,
- the sum $\sum_i \rho_i = 1$ at all points, and
- $\text{Supp } \rho_i \subseteq U_i$ for all $i \in I$.

Every open cover has a smooth partition of unity subordinate to it.

Choose a smooth partition of unity $\{\rho_i\}_{i \in I}$ subordinate to the cover $\{U_i\}_{i \in I}$. Recall that $\mathcal{E}_c$ is a C_M^∞ -module, and thus we can define a morphism $\tilde{\rho}_i : \mathcal{E}_c(U) \rightarrow \mathcal{E}_c(U_i)$ by mapping $v \mapsto \rho_i v$. Here we use the fact that $\text{Supp } \rho_i \subseteq U_i$, meaning $\text{Supp}(\rho_i v) \subseteq U_i$, meaning it is supported in U_i . To show that the support is compact, we will argue that the intersection of compact and closed sets in any Hausdorff space is compact. It then follows since $\text{Supp}(\rho_i v) = \text{Supp}(\rho_i) \cap \text{Supp}(v)$.

Let X be Hausdorff, let $K \subseteq X$ be compact, and let $C \subseteq X$ be closed. In Hausdorff spaces, compact sets are closed, meaning K is closed. Thus, the intersection $K \cap C$ is closed in X , and thus closed in K in the subspace topology. Recall that all closed subsets of a compact space are compact, and recall that a subspace is compact with respect to the subspace topology if and only if it is compact as a subset of the full space. Therefore, $K \cap C$ is compact in K . Continuous images of compact sets are compact, and thus $K \cap C$ is compact in X .

Now consider the map

$$\sum_i \tilde{\rho}_i : \mathcal{E}_c(U) \rightarrow \bigoplus_i \mathcal{E}_c(U_i).$$

Consider the image of a section $v \in \mathcal{E}_c(U)$ under the map

$$\left(\sum_i ext_i\right) \circ \left(\sum_i \tilde{\rho}_i\right) : v \mapsto \sum_i \rho_i v \mapsto \sum_i ext_i(\rho_i v).$$

Of course, $ext_i(\rho_i v)$ is just equal to $\rho_i v$ on the support of $\text{Supp}(\rho_i v)$, and vanishing elsewhere, so we drop the ext_i for simplicity and consider $\rho_i v$ as a section over $\mathcal{E}_c(U)$ via this extension. Evaluating this at a point p , we see that not only does this linear combination exist (since all but finitely many ρ_i vanish at p), but we have

$$\sum_i (\rho_i v)_p = \left(\sum_i \rho_i(p)\right) v_p = 1 v_p = v_p$$

by the properties of a partition of unity. Thus,

$$\left(\sum_i ext_i\right) \circ \left(\sum_i \tilde{\rho}_i\right) = 1_{\mathcal{E}_c(U)}.$$

Denote the relevant colimit by Q , and use the fact that $\mathcal{DVS}$ is abelian to rewrite the colimit as the exact sequence

$$\bigoplus_{i,j} \mathcal{E}_c(U_i \cap U_j) \xrightarrow{f} \bigoplus_i \mathcal{E}_c(U_i) \xrightarrow{g} Q \longrightarrow 0,$$

where f is the difference of the two maps that the colimit coequalizes.

Since this sequence is exact, g is surjective. Thus $\text{Im } g \cong Q$. Applying exactness and the first isomorphism theorem yields

$$Q \cong \text{Im } g \cong \frac{\bigoplus_i \mathcal{E}_c(U_i)}{\text{Ker } g} \cong \frac{\bigoplus_i \mathcal{E}_c(U_i)}{\text{Im } f}.$$

We will identify Q with the quotient by $\text{Im } f$, and will identify g with the quotient morphism. We will show that the composition

$$g \circ \left(\sum_i \tilde{\rho}_i \right)$$

is a left inverse to ext_I . We will verify this on a convenient collection of elements that span $\bigoplus_i \mathcal{E}_c(U_i)$. Fix $k \in I$, and let $v = \{v_i\}_{i \in I} \in \bigoplus_i \mathcal{E}_c(U_i)$ have the property that $v_i = 0$ if $i \neq k$. Write $\tilde{v}$ for the image of v under $\sum_i \tilde{\rho}_i \circ \sum_i \text{ext}_i$. Thus,

$$\tilde{v} = \left(\sum_i \tilde{\rho}_i \right) \circ \left(\sum_i \text{ext}_i \right) (v) = \left(\sum_i \tilde{\rho}_i \right) (v_k) = \{\rho_i v_k\}_{i \in I}.$$

Thus, $\tilde{v}_i = \rho_i v_k$.

We claim that it suffices to show that $v = \tilde{v}$ in Q . To see this, first note that the collection of all such v clearly span $\bigoplus_i \mathcal{E}_c(U_i)$, and thus span the quotient by surjectivity of the quotient morphism. Additionally, let us recall our setup. The map ext_I is a morphism of cocones

$$\begin{array}{ccc} \bigoplus_{i,j \in I} \mathcal{E}_c(U_i \cap U_j) & \begin{array}{c} \xrightarrow{\iota_i} \\ \xrightarrow{\iota_j} \end{array} & \bigoplus_{i \in I} \mathcal{E}_c(U_i) \\ & \begin{array}{c} \searrow g \circ \iota \\ \swarrow g \end{array} & \\ & Q & \\ & \downarrow \text{ext}_I & \\ \sum_{i,j} \text{ext}_{i,j} & \mathcal{E}_c(U) & \sum_i \text{ext}_i \end{array}$$

Here we write ι for either ι_i or ι_j when composing with g , and we recall the definitions

$$\iota_k = \bigoplus_{i,j} \text{ext}_{i,j}^k$$

for $k \in \{i, j\}$. By directly constructing Q as the quotient $\bigoplus_i \mathcal{E}_c(U_i) / \text{Im}(\iota_i - \iota_j)$, we can obtain a direct formula for ext_I as the map $[\{s_i\}_{i \in I}] \mapsto \sum_i \text{ext}_i(s_i)$. Thus, we have actually already shown that $g \circ \sum_i \tilde{\rho}_i$ is a right-inverse for ext_I ! We will explore this later, to avoid getting distracted. For now, we must explain why showing $v = \tilde{v}$ in Q is sufficient.

We are trying to show the composite

$$\frac{\bigoplus_i \mathcal{E}_c(U_i)}{\text{Im } f} \xrightarrow{\text{ext}_I} \mathcal{E}_c(U) \xrightarrow{\sum_i \tilde{\rho}_i} \bigoplus_i \mathcal{E}_c(U_i) \xrightarrow{g} \frac{\bigoplus_i \mathcal{E}_c(U_i)}{\text{Im } f}$$

is the identity morphism. Consider the following diagram of solid arrows

$$\begin{array}{ccccc} \bigoplus_i \mathcal{E}_c(U_i) & \xrightarrow{\sum_i \text{ext}_i} & \mathcal{E}_c(U) & \xrightarrow{\sum_i \tilde{\rho}_i} & \bigoplus_i \mathcal{E}_c(U_i) \\ \downarrow g & \nearrow \text{ext}_I & & & \downarrow g \\ \frac{\bigoplus_i \mathcal{E}_c(U_i)}{\text{Im } f} & \xrightarrow{\quad \quad \quad 1 \quad \quad \quad} & \frac{\bigoplus_i \mathcal{E}_c(U_i)}{\text{Im } f} \end{array}$$

The goal is to show that the diagram commutes with the inclusion of the dotted arrow. However, commutativity of the top left triangle means that we need only verify that the outer square (with the inclusion of the dotted arrow) commutes (here we use the fact that cokernels are epi in abelian categories). Indeed, this simply means that given v , we must show

$$g(v) = \left(g \circ \sum_i \tilde{\rho}_i \circ \sum_i \text{ext}_i \right) (v) = g(\tilde{v}).$$

Since $g(v) = [v]$, this is precisely the statement that v and $\tilde{v}$ descend to the same class in the quotient. We now show this.

It suffices to show that $v - \tilde{v} \in \text{Im } f$. We recall that $v_i = 0$ if $i \neq k$, and $\tilde{v}_i = \rho_i v_k$. Consider $w \in \bigoplus_{i,j} \mathcal{E}_c(U_i \cap U_j)$ given by

$$w_{ik} = \rho_i v_k$$

for all i , and $w_{ij} = 0$ for $j \neq k$. Then

$$f(w) = \{f(w)_i\}_{i \in I} = \left\{ \sum_j \text{ext}_j w_{ij} - \text{ext}_j w_{ji} \right\}_{i \in I}.$$

For $i \neq k$, we have

$$f(w)_i = \sum_j w_{ij} - w_{ji} = \sum_j w_{ij} = w_{ik} = \rho_i v_k.$$

For $i = k$, we have

$$f(w)_k = \sum_j w_{kj} - w_{jk} = w_{kk} - \sum_j w_{jk} = \rho_k v_k - \sum_j \rho_j v_k = \rho_k v_k - v_k.$$

Here we have used the fact that the partition of unity always sums to 1. Finally, note that if $i \neq k$, then

$$\tilde{v}_i - v_i = \rho_i v_k - 0 = \rho_i v_k = f(w)_i,$$

and if $i = k$, then

$$\tilde{v}_k - v_k = \rho_k v_k - v_k = f(w)_k.$$

Thus, $\tilde{v} - v = f(w)$, meaning $\tilde{v} - v \in \text{Im } f$.

Now, we return to our discussion of why we have already shown that $g \circ \sum_i \tilde{\rho}_i$ is a right inverse. Recall that we showed

$$\left(\sum_i ext_i \right) \circ \left(\sum_i \tilde{\rho}_i \right) = 1_{\mathcal{E}_c(U)}.$$

Diagrammatically, we can obtain the diagram of solid arrows

$$\begin{array}{ccc} & 1 & \\ \text{---} \nearrow & & \searrow \text{---} \\ \mathcal{E}(U) & \xrightarrow{\sum_i \tilde{\rho}_i} \bigoplus_i \mathcal{E}_c(U_i) & \xrightarrow{\sum_i ext_i} \mathcal{E}_c(U) \\ & \downarrow g & \nearrow ext_I \\ & \bigoplus_i \mathcal{E}_c(U_i) & \\ & \text{Im } f & \end{array}$$

Note that the dashed arrow exists and is unique by the formula we found previously, and thus by commutativity of the diagram, we obtain

$$ext_I \circ g \circ \sum_i \tilde{\rho}_i = 1.$$

Thus, $\mathcal{E}_c$ is a cosheaf. A similar argument shows that $\mathcal{E}_c$ is a homotopy cosheaf, but I don't feel like working through it. ■

The factorization algebras we discuss in field theory are usually constructed from the symmetric algebra on the vector spaces $\mathcal{E}_c(U)$ and $\bar{\mathcal{E}}_c^!(U)$. First, define the completed (with respect to symmetric degree) symmetric algebra

$$\widehat{\text{Sym}}(V) = \prod_n \text{Sym}^n(V)$$

of a graded topological vector space V . The grading is by symmetric degree: an element of homogeneous degree n is a tuple which is everywhere vanishing, except for the n th degree. Essentially, it is graded in the exact same way as the usual symmetric algebra, but when viewed as a vector space and not a graded space, we are allowing infinite linear combinations due to the infinite slots in a tuple.

Note that since $\bar{\mathcal{E}}_c^!(U)$ is dual to $\mathcal{E}(U)$, we can view $\widehat{\text{Sym}}(\bar{\mathcal{E}}_c^!(U))$ as the algebra of formal power series on $\mathcal{E}(U)$. Here we note that

$$\bar{\mathcal{E}}_c^!(U) \cong \mathcal{E}^\vee \otimes_{C_M^\infty} \text{Dens}_M \otimes_{C_M^\infty} \mathcal{D}_c,$$

where compactly supported distributions are precisely those distributions that extend to C_M^∞ , hence $\mathcal{D}_c(U)$ is all continuous linear functionals on $C^\infty(U)$. I spent a ton of time trying to figure out what this duality was, and have concluded that $\mathcal{D}_c$ must denote the continuous linear dual of densities.

Given an inclusion $U \subseteq V$ of open sets, there are natural maps of dg commutative algebras

$$\begin{aligned} \text{Sym } \mathcal{E}_c(U) &\rightarrow \text{Sym } \mathcal{E}_c(V), \\ \widehat{\text{Sym}} \mathcal{E}_c(U) &\rightarrow \widehat{\text{Sym}} \mathcal{E}_c(V), \\ \text{Sym } \bar{\mathcal{E}}_c(U) &\rightarrow \text{Sym } \bar{\mathcal{E}}_c(V), \\ \widehat{\text{Sym}} \bar{\mathcal{E}}_c(U) &\rightarrow \widehat{\text{Sym}} \bar{\mathcal{E}}_c(V). \end{aligned}$$

6.5.2 The main theorem

Theorem 6.5.3. *Let M be a manifold.*

1. *Let E be a vector bundle on M . Then*

- (a) *$\mathrm{Sym} \mathcal{E}_c$ and $\mathrm{Sym} \bar{\mathcal{E}}_c$ are strict (non-homotopical) factorization algebras valued in $\mathcal{DVS}$.*
- (b) *$\widehat{\mathrm{Sym}} \mathcal{E}_c$ and $\widehat{\mathrm{Sym}} \bar{\mathcal{E}}_c$ are strict (non-homotopical) factorization algebras valued in the category of differentiable pro-vector spaces.*

2. *Let E be a local cochain complex on M . Then*

- (a) *$\mathrm{Sym} \mathcal{E}_c$ and $\mathrm{Sym} \bar{\mathcal{E}}_c$ are homotopy factorization algebras valued in $\mathrm{Ch}_{\mathcal{DVS}}$.*
- (b) *$\widehat{\mathrm{Sym}} \mathcal{E}_c$ and $\widehat{\mathrm{Sym}} \bar{\mathcal{E}}_c$ are homotopy factorization algebras valued in differentiable pro-cochain complexes.*

I am tired of typing, so I will omit typing this up. I worked through this proof on a whiteboard.

6.6 Factorization algebras from local Lie algebras

We showed that $\widehat{\mathrm{Sym}} \bar{\mathcal{E}}_c$ is a factorization algebra. Intuitively, this means “functions on $\mathcal{E}$ ” satisfy the descent condition. We will now try and do this when $\mathcal{E}$ has more structure. In particular, when $\mathcal{E}$ is a local dg Lie algebra. In particular, we might expect that the Chevalley-Eilenberg chains $C_\bullet \mathcal{L}$ and cochains $C^\bullet \mathcal{L}$ of a local dg Lie algebra $\mathcal{L}$ is a factorization algebra.

This is compounded from the fact that in the abelian case, we have that $\mathbb{U}\mathcal{L} = C_\bullet \mathcal{L}_c = \mathrm{Sym} \mathcal{L}_c[1]$, which we know is a factorization algebra. Indeed, this holds in the more general case of a dg Lie algebra. The following statement holds for L_∞ -algebras, which we introduce later.

Theorem 6.6.1. *Let $\mathcal{L}$ be a local dg Lie algebra on a manifold M . Then the prefactorization algebras*

$$\mathbb{U}\mathcal{L} : U \mapsto C_\bullet(\mathcal{L}_c(U))$$

$$\mathbb{O}\mathcal{L} : U \mapsto C^\bullet(\mathcal{L}(U))$$

are factorization algebras.

Proof. We start with the factorization envelope. We make the following remark: let $\mathfrak{g}$ be a dg Lie algebra. Then the Chevalley-Eilenberg chains $C_\bullet \mathfrak{g}$ have a natural filtration $F_n = \mathrm{Sym}^{\leq n}(\mathfrak{g}[1])$ compatible with the differential. It’s finally time to start nailing down specifics of spectral sequences associated to filtrations, so I comment a bit on this.

Recall a (decreasing) filtration $F^\bullet C$ of a cochain complex C is a sequence of subcomplexes $\dots \subseteq F^n C \subseteq F^{n-1} C \subseteq \dots \subseteq C$. There is a spectral sequence associated to this filtration, denoted $E_\bullet^{p,q}$. For each p, q , the 0th page is given by

$$E_0^{p,q} = \frac{F^p C^{p+q}}{F^{p+1} C^{p+q}}$$

and has 1st page

$$E_1^{p,q} = H^{p,q}(E_0^{p,\bullet}) = H^{p,q} \left(\frac{F^p C^{p+q}}{F^{p+1} C^{p+q}} \right).$$

(The complete i th page is the direct sum over all (p, q) -components). The comparison theorem states that if a map $f : C \rightarrow D$ between filtered cochain complexes induces an isomorphism on the n th

page with $n \geq 1$, then f is a quasi-isomorphism. All of this can be modified for increasing filtrations by having $F_n C \subseteq F_{n+1} C$ and setting

$$E_{p,q}^0 = \frac{F_p C^{p+q}}{F_{p-1} C^{p+q}}, \quad E_{p,q}^1 = H^{p,q}(E_{p,\bullet}^0).$$

Note that on $C_\bullet \mathfrak{g}$, F_n is indeed a filtration, since $\mathrm{Sym}^{\leq n}(\mathfrak{g}[1]) \subseteq \mathrm{Sym}^{\leq n+1}(\mathfrak{g}[1])$ tautologically. We have

$$\begin{aligned} E_0^{p,q} &= \frac{\mathrm{Sym}^{\leq p}(\mathfrak{g}[1]^{p+q})}{\mathrm{Sym}^{\leq p-1}(\mathfrak{g}[1]^{p+q})} = \frac{\bigoplus_{i=0}^p \mathrm{Sym}^i(\mathfrak{g}[1]^{p+q})}{\bigoplus_{i=0}^{p-1} \mathrm{Sym}^i(\mathfrak{g}[1]^{p+q})} \cong \mathrm{Sym}^p(\mathfrak{g}[1]^{p+q}) \\ \implies E_0 &= \bigoplus_{p,q} E_0^{p,q} = \bigoplus_{p,q} \mathrm{Sym}^p(\mathfrak{g}[1]^{p+q}) = \bigoplus_q \mathrm{Sym}(\mathfrak{g}[1]^q) = \mathrm{Sym}(\mathfrak{g}[1]). \end{aligned}$$

Here we have used the fact that Sym and $\bigoplus$ are colimits, and thus commute. Thus $E_1 = H_\bullet(\mathrm{Sym}(\mathfrak{g}[1]))$. Recall that the differential on the Chevalley-Eilenberg chains is the differential $d_{\mathfrak{g}}$ extended as a derivation, together with the Chevalley-Eilenberg differential d_{CE} , which is the Lie bracket, extended as a coderivation.

Note that $d_{\mathrm{CE}}(F_n) \subseteq F_{n-1}$ (eg. $[x, y]$ is a single element, so $xy \mapsto [x, y]$ decreases symmetric degree), and thus d_{CE} vanishes on $E_0^{p,q}$. Thus, the homology E_1 is computed just using the extension of $d_{\mathfrak{g}}$.

Now, we return to the actual proof. Consider the Čech complex of $\mathbb{U}\mathcal{L}$ with respect to a Weiss cover $\mathfrak{U}$ for an open set $U \subseteq M$. We can apply the given filtration to $\mathbb{U}\mathcal{L}(U)$, but it is unclear how to apply it to $\check{C}(\mathfrak{U}, \mathbb{U}\mathcal{L})$.

To see how this is done, recall

$$\check{C}(\mathfrak{U}, \mathbb{U}\mathcal{L})^\bullet = \bigoplus_{k \geq 0} \bigoplus_{|J|=k+1} \mathbb{U}\mathcal{L}(U_J)[k]^\bullet,$$

where $J = \{j_0, \dots, j_k\}$, and $U_J = U_{j_1} \cap \dots \cap U_{j_k}$, and $U_{j_i} \in \mathfrak{U}$ for all $j_i \in J$. We note that since $\mathbb{U}\mathcal{L} = C_\bullet \mathcal{L}_c$, we can rewrite

$$\check{C}(\mathfrak{U}, \mathbb{U}\mathcal{L})^\bullet \cong \bigoplus_{k \geq 0} \bigoplus_{|J|=k+1} \mathrm{Sym}(\mathcal{L}_c(U_J)[1])[k]^\bullet.$$

We can then see that the Čech complex admits the filtration

$$F_n \check{C}^\bullet = \bigoplus_{k \geq 0} \bigoplus_{|J|=k+1} \mathrm{Sym}^{\leq n}(\mathcal{L}_c(U_J)[1])[k]^\bullet.$$

I got confused here because I forgot that $C_\bullet \mathfrak{g}$ is a cochain complex, even though it is called the Chevalley-Eilenberg *chains*. To see this, recall that $d = d_{\mathfrak{g}[1]} + d_{\mathrm{CE}}$, where

$$d_{\mathfrak{g}[1]}(x_1 \cdots x_n) = -d_{\mathfrak{g}}[1](x_1 \cdots x_n) = -\sum_i \pm x_1 \cdots d_{\mathfrak{g}} x_i \cdots x_n,$$

which has an increased degree of 1 since $\mathfrak{g}$ is a cochain complex; and

$$d_{\mathrm{CE}}(x_1 \cdots x_n) = \sum_{1 \leq i < j \leq n} \pm [x_i, x_j] x_1 \cdots \widehat{x}_i \cdots \widehat{x}_j \cdots x_n.$$

It's a bit trickier to see that this has a higher degree, but recall that if $x_i \in \mathfrak{g}^{k_i}$, then $|x_i| = k_i - 1$ in $\mathfrak{g}[1]$. Thus since $[x_i, x_j] \in \mathfrak{g}^{k_i+k_j}$,

$$|[x_i, x_j]| = k_i + k_j - 1, \quad |x_i| + |x_j| = k_i - 1 + k_j - 1.$$

Thus d_{CE} also increases degree by 1. I don't know why I forgot this, I've literally derived this like 5 times.

Ok back to our regularly scheduled programming. We want the canonical map $\check{C}(\mathfrak{U}, \mathbb{U}\mathcal{L}) \rightarrow \mathbb{U}\mathcal{L}$ to respect the filtration, so that we can lower it to the level of the spectral sequence. Luckily, we can use the fact that we have an explicit description of the map: writing $\varphi : \check{C}(\mathfrak{U}, \mathbb{U}\mathcal{L}) \rightarrow \mathbb{U}\mathcal{L}$ and writing i_J for the pushforward $\mathbb{U}\mathcal{L}(U_J) \rightarrow \mathbb{U}\mathcal{L}(U)$, and π_0 for the projection onto the $k = 0$ component of the outer direct sum, we have

$$\varphi^n : \bigoplus_{k \geq 0} \bigoplus_{|J|=k+1} \mathbb{U}\mathcal{L}(U_J)[k]^n \xrightarrow{\pi_0} \bigoplus_{|J|=n+1} \mathbb{U}\mathcal{L}(U_J)^n \xrightarrow{\oplus_J i_J} \mathbb{U}\mathcal{L}(U)^n,$$

where we note that writing $\mathbb{U}\mathcal{L}(U_J)[k]^n$ as living in degree n is a convenient abuse of notation; it actually lives in degree $k + n$, allowing the given map to be defined. I find it more intuitive to write that than to work out what the degree n is.

This map clearly respects the symmetric degree. To see this, let $x \in \check{C}(\mathfrak{U}, \mathbb{U}\mathcal{L})$. Then x can be written as a linear combination of components

$$x_k \in \bigoplus_{|J|=k+1} \mathbb{U}\mathcal{L}(U_J)[k]^{n-k}.$$

We write x as a tuple $x = \{x_k\}_{k \geq 0}$ for simplicity. Recall that the Čech differential is given by $D = \bigoplus d_{\mathbb{U}\mathcal{L}} + d_e$, where d_e acts via the inclusion $U_J \subseteq U_{J \setminus \{j\}}$, $j \in J$. Using the fact that we already know this is a cochain map, we have

$$\pi_0(Dx) = d_{\mathbb{U}\mathcal{L}}x_0.$$

All $x_{k \geq 1}$ components vanish under this map, meaning that although their symmetric degree is not preserved, they still land in $\text{Sym}^0 \subseteq \text{Sym}^{\leq n}$. Finally, on the x_0 component, we just note from our earlier formula for the action of $d_{\mathfrak{g}}$ and d_{CE} that the dg Lie algebra differential and Chevalley-Eilenberg differential, extended to the symmetric algebra, preserves symmetric degree. Therefore, φ is a filtered map of filtered chain complexes.

By the comparison theorem, it suffices to show that φ induces an isomorphism on the first page. We recall that the first page of $\mathbb{U}\mathcal{L}(U)$ was computed to be $H^\bullet(\text{Sym } \mathcal{L}_c(U)[1])$, equipped with only the differential on $\mathcal{L}$. We also can compute the first page of $\check{C}(\mathfrak{U}, \mathbb{U}\mathcal{L})$. We start with the zeroth page:

$$\begin{aligned} E_0^{p,q} &= \frac{F_p \check{C}^{p+q}}{F_{p-1} \check{C}^{p+q}} = \frac{\bigoplus_{k \geq 0} \bigoplus_{|J|=k+1} \text{Sym}^{\leq p}(\mathcal{L}_c(U_J)[1])[k]^{p+q-k}}{\bigoplus_{k \geq 0} \bigoplus_{|J|=k+1} \text{Sym}^{\leq p-1}(\mathcal{L}_c(U_J)[1])[k]^{p+q-k}} \\ &\cong \bigoplus_{k \geq 0} \bigoplus_{|J|=k+1} \frac{\text{Sym}^{\leq p}(\mathcal{L}_c(U_J)[1])[k]^{p+q-k}}{\text{Sym}^{\leq p-1}(\mathcal{L}_c(U_J)[1])[k]^{p+q-k}} \cong \bigoplus_{k \geq 0} \bigoplus_{|J|=k+1} \text{Sym}^p(\mathcal{L}_c(U_J)[1])[k]^{p+q-k}. \end{aligned}$$

We then have

$$E_0 = \bigoplus_{p,q} \bigoplus_{k \geq 0} \bigoplus_{|J|=k+1} \text{Sym}^p(\mathcal{L}_c(U_J)[1])[k]^{p+q-k} \cong \bigoplus_q \bigoplus_{k \geq 0} \bigoplus_{|J|=k+1} \text{Sym}(\mathcal{L}_c(U_J)[1])[k]^{\bullet+q-k}$$

$$\cong \bigoplus_{k \geq 0} \bigoplus_{|J|=k+1} \text{Sym}(\mathcal{L}_c(U_J)[1])[k]^{\bullet-k}.$$

The first page is just the cohomology of this complex. Note that cohomology commutes with $\oplus$ and Sym , so we can write

$$E_1 \cong \bigoplus_{k \geq 0} \bigoplus_{|J|=k+1} \text{Sym}(H^\bullet(\mathcal{L}_c(U_J)[1]))[k]^{\bullet-k}.$$

Note that although the shift $[k]$ induces a change in differential $d \rightarrow (-1)^k d$, this doesn't change the cohomology of the complex. Thus

$$E_1 \cong \bigoplus_{k \geq 0} \bigoplus_{|J|=k+1} \text{Sym}(H^\bullet(\mathcal{L}_c(U_J)[1]))^\bullet.$$

It will be useful to decompose this into symmetric degrees, so we rewrite this as

$$E_1 \cong \bigoplus_{k \geq 0} \bigoplus_{|J|=k+1} H^\bullet \left(\bigoplus_{p \geq 0} \text{Sym}^p(\mathcal{L}_c(U_J))[1] \right) \cong H^\bullet \left(\bigoplus_{p \geq 0} \bigoplus_{k \geq 0} \bigoplus_{|J|=k+1} \text{Sym}^p(\mathcal{L}_c(U_J))[1] \right).$$

Observe that

$$\bigoplus_{k \geq 0} \bigoplus_{|J|=k+1} \text{Sym}^p(\mathcal{L}_c(U_J))[1] \cong \check{C}(\mathfrak{U}, \text{Sym}^p \mathcal{L}_c[1]).$$

Thus we obtain

$$E_1 \cong \bigoplus_{p \geq 0} \check{C}(\mathfrak{U}, \text{Sym}^p \mathcal{L}_c[1]).$$

By the comparison theorem, we thus need to consider the induced map

$$\varphi : \bigoplus_{p \geq 0} H^\bullet(\check{C}(\mathfrak{U}, \text{Sym}^p \mathcal{L}_c[1])) \rightarrow H^\bullet(\text{Sym}(\mathcal{L}_c(U)[1])) \cong \bigoplus_{p \geq 0} H^\bullet(\text{Sym}^p(\mathcal{L}_c(U)[1])).$$

The action of φ is still by $\bigoplus_J i_J \circ \pi_0$, since each internal Čech complex has a π_0 , so we can consider a global π_0 via the coproduct. These maps manifestly preserve symmetric degree (not cohomological degree, *symmetric* degree; easier to show if you commute Sym and $H^\bullet$ first). Thus, since φ respects the p -grading, we can decompose it along the direct sum to a collection of maps

$$\varphi^p : H^\bullet(\check{C}(\mathfrak{U}, \text{Sym}^p(\mathcal{L}_c[1]))) \rightarrow H^\bullet(\text{Sym}(\mathcal{L}_c(U)[1])).$$

Functoriality of $\oplus$ means that it suffices to show each φ^p is an isomorphism.

To do this, we re-interpret $\text{Sym}^p(\mathcal{L}_c(U)[1])$ as compactly supported sections on U^p . Recall that a local dg Lie algebra $\mathcal{L}$ is defined as arising as the sheaf of sections of some graded vector bundle L on M , where the differential and dg Lie bracket are defined at the level of $\mathcal{L}$. This means that we can rewrite $\mathcal{L}_c(U) := \Gamma_c(U, L)$, allowing for the very simple identification

$$\text{Sym}^p(\mathcal{L}_c(U)) \cong \Gamma_c(U^p, L^{\boxtimes p})^{S_p}.$$

Here, we recall that the exterior tensor product of the bundle is defined by $(L^{\boxtimes p})_x = L_{x_1} \otimes \cdots \otimes L_{x_p}$, for $x = \{x_1, \dots, x_p\} \in U^p$. We can then easily identify $\mathcal{L}_c(U)^{\otimes p} \cong \Gamma(U^n, L^{\boxtimes p})$ by viewing a product $v_1 \otimes \cdots \otimes v_p$ as having v_i be a section of the i th copy of L in the exterior tensor product. It then follows immediately that $\text{Sym}^p(\mathcal{L}_c(U))$ is just given by symmetrizing these sections. We prefer to write $\Gamma_c(-, L^{\boxtimes p})^{S_p} =: \mathcal{L}_c^{\boxtimes p}(-)^{S_p}$.

I'm tired of working through this proof and will come back to it at a later date. There are two ideas. 1: use $\mathrm{Sym}^p(\mathcal{L}_c[1]) \cong \Lambda^p(\mathcal{L}_c)[p]$, and 2: work on the level of the tensor algebra, allowing us to drop the S_p -invariants, and then show that the map descends to the symmetric algebra to become a quasi-isomorphism. Either way, the argument proceeds by using the Weiss cover to define an open cover of U^p by taking powers. We then use a partition of unity subordinate to U^p in a similar way to the previous lemma, and use that to build a contracting homotopy which is S_p -invariant. The argument for $\mathbb{O}\mathcal{L}$ is the same, we just use the opposite filtration on the product version of the symmetric algebra. ■

Chapter 7

Formal aspects of factorization algebras

7.1 Pushing forward factorization algebras

Let M, N be spaces admitting Weiss covers, and $f : M \rightarrow N$ a continuous map. Given a Weiss cover $\mathfrak{U} = \{U_\alpha\}_{\alpha \in A}$ of an open $U \subseteq N$, let $f^{-1}\mathfrak{U} = \{f^{-1}(U_\alpha)\}_{\alpha \in A}$ denote the preimage cover of $f^{-1}(U) \subseteq M$. Observe that $f^{-1}\mathfrak{U}$ is Weiss: given $\{x_1, \dots, x_n\} \in f^{-1}(U)$, we have $\{f(x_1), \dots, f(x_n)\} \subseteq U$, and thus is contained in some U_α since $\mathfrak{U}$ is Weiss. Thus, we have $\{x_1, \dots, x_n\} \subseteq f^{-1}(U_\alpha)$ for the same α .

Definition 7.1.1. With notation as above, if $\mathcal{F}$ is a factorization algebra on M , the pushforward factorization algebra $f_*\mathcal{F}$ on N is defined as

$$(f_*\mathcal{F})(U) = \mathcal{F}(f^{-1}(U)).$$

Note that if $f : M \rightarrow *$ is the unique map to the singleton, then $f_*\mathcal{F}(*) \cong \mathcal{F}(M)$ is just the global sections of $\mathcal{F}$. We call this the *factorization homology of $\mathcal{F}$ on M* .

7.2 Extension from a basis

7.2.1 Preliminary definitions

Let X be a space, and let $\mathfrak{U}$ be a basis for X that is closed under finite intersections. Then it is a well-known fact that there is an equivalence of categories between Shf_X and $\mathrm{Shf}_{\mathfrak{U}}$, where the latter denotes sheaves only defined on the basis $\mathfrak{U}$. There is a similar statement for factorizing bases and factorization algebras, when X is Hausdorff. Recall definition 6.1.15, which defines factorizing bases as bases that are Weiss covers, closed under unions of finite pairwise disjoint opens, and closed under finite intersections.

An example of a factorizing basis for $\mathbb{R}^n$ is the collection of all opens that are a disjoint union of finitely many convex open sets (convex sets contain line segments drawn between any two of their elements).

As noted before, assuming X is Hausdorff, a factorizing basis produces Weiss covers for all opens. Given $U \subseteq X$ open, define the Weiss cover $\mathfrak{U}_U$ by taking all pairwise disjoint unions of opens $V_i \subseteq U$ with $V_i \in \mathfrak{U}$. We use the Hausdorff condition to guarantee that any finite number of points can have disjoint neighborhoods, and we intersect each of those with the open U to ensure that they are subsets of U . Note that since $\mathfrak{U}$ is closed under finite intersection and finite pairwise disjoint union, we have $\mathfrak{U}_U \subseteq \mathfrak{U}$. Note that as we continue, the notation $\mathfrak{U}_U$ will only denote the collection of subsets of U contained in $\mathfrak{U}$, and the Weiss cover is given by unions of pairwise disjoint unions of elements in $\mathfrak{U}_U$.

Definition 7.2.1. Let $\mathfrak{U}$ be a factorizing basis for a Hausdorff space X , and let $\mathcal{C}$ be a symmetric monoidal Grothendieck abelian category. A $\mathfrak{U}$ -prefactorization algebra is a functor $\mathcal{F} : \mathfrak{U} \rightarrow \mathcal{C}$ with the following properties:

1. For every finite tuple of pairwise disjoint opens $\{U_{i_0}, \dots, U_{i_n}\}$, all contained in U_j , and all of these opens in $\mathfrak{U}$, there is a factorization product

$$\mathcal{F}(U_{i_0}) \otimes \dots \otimes \mathcal{F}(U_{i_n}) \rightarrow \mathcal{F}(U_j).$$

2. The factorization product is associative and equivariant.

We say $\mathcal{F}$ is a (homotopy) $\mathfrak{U}$ -factorization algebra if for every $U \in \mathfrak{U}$, and every Weiss cover $\mathfrak{V}$ of U with $\mathfrak{V} \subseteq \mathfrak{U}$, the natural map

$$\check{C}(\mathfrak{V}, \mathcal{F}) \rightarrow \mathcal{F}(U)$$

is a weak equivalence.

Note that we have not required that the factorization product be an equivalence, meaning we are not specializing to multiplicative factorization algebras here. Also note that we have not required $\mathcal{C}$ be a dg category, or any type of homotopic category, model category, or ∞ -category, meaning it may not have a notion of weak equivalence beyond that of an isomorphism. In this case, note that taking weak equivalences to be isomorphisms recovers the usual cosheaf condition from the Čech condition (this follows since the Čech nerve is 1-coskeletal, and (co)limits over $\Delta_{\leq 1}$ -indexed diagrams yield (co)equalizers for parallel arrows that happen to share a section).

The extension will rely on the existence of a lax symmetric monoidal functor $\text{sCh}_{\mathcal{A}} \rightarrow \text{Ch}_{\mathcal{A}}$, for sCh simplicial cochain complexes.

7.2.2 The statement

We will show that any $\mathfrak{U}$ -factorization algebra extends uniquely to a factorization algebra on the full space. For an open $V \subseteq X$, we denote $\mathfrak{U}_V$ for the collection of all the opens in $\mathfrak{U}$ that are contained inside V .

Let $\mathcal{F}$ be a $\mathfrak{U}$ -factorization algebra. Define a prefactorization algebra $\text{ext}(\mathcal{F})$ on X by

$$\text{ext}(\mathcal{F})(V) := \check{C}(\mathfrak{U}_V, \mathcal{F}).$$

Conversely, if $\mathcal{G}$ is a factorization algebra on X , let $\text{res}(\mathcal{G})$ denote its restriction to a $\mathfrak{U}$ -factorization algebra.

Proposition 7.2.2. *Given a $\mathfrak{U}$ -factorization algebra $\mathcal{F}$, its extension $\text{ext}(\mathcal{F})$ is a factorization algebra on X such that $\text{res}(\text{ext}(\mathcal{F}))$ is naturally isomorphic to $\mathcal{F}$. Conversely, if $\mathcal{G}$ is a factorization algebra on X , then $\text{ext}(\text{res}(\mathcal{G}))$ is naturally weakly equivalent to $\mathcal{G}$.*

7.2.3 The proof

Let $\mathfrak{U}$ be a factorizing basis for a Hausdorff space X , and let $\mathcal{F}$ be a $\mathfrak{U}$ -factorization algebra. We will construct a factorization algebra $\text{ext}(\mathcal{F})$ on X .

We use the following notation. A simplicial cochain complex will be denoted $A_{\bullet}$, where each A_n is a cochain complex. We obtain a double complex by taking the unnormalized cochains, and

denote the totalization thereof by $C(A_\bullet)$. We denote the totalization of the double complex given by the *normalized* chains by $C_N(A_\bullet)$. Finally, we denote by

$$sh_{A,B} : C_N(A_\bullet) \otimes C_N(B_\bullet) \rightarrow C_N(A_\bullet \otimes B_\bullet)$$

the Eilenberg-Zilber shuffle map, which yields a lax symmetric monoidal functor $s\mathcal{Ch}_\mathcal{A} \rightarrow \mathcal{Ch}_\mathcal{A}$.

To be more specific about definitions, define the unnormalized chains as

$$C(A_\bullet) := \begin{cases} A_{|m|}, & m \leq 0 \\ 0, & m > 0 \end{cases}$$

with differential

$$d : C(A_\bullet)^m \rightarrow C(A_\bullet)^{m+1}, \quad da = \sum_{k=0}^{|m|} (-1)^k d_k(a)$$

for d_k the k th face of $A_\bullet$. Note that this is indeed a chain complex since we take $C(A_\bullet)^{-m} = A_m$, so mapping $A_m \rightarrow A_{m-1}$ induces a map $C(A_\bullet)^{-m} \rightarrow C(A_\bullet)^{-m+1}$. We will always use the upright d for the differential, and the italic d_k for the k th face. This will be how we distinguish between the two.

The normalized chains are defined by

$$C_N(A_\bullet)^m := \bigcap_{k=0}^{|m|-1} \text{Ker}(d_k : A_m \rightarrow A_{m-1})$$

where $m \leq 0$, and vanishing on higher degrees. The differential is simply the remaining face map $d_{|m|}$. For example, the singular chains $C_\bullet(X)$ of a space X are the composition $C_N \circ \mathbb{Z}[-] \circ \text{Sing}$ of functors. Indeed, the Dold-Kan correspondence tells us precisely that $C_N : s\mathcal{A}b \simeq \mathcal{Ch}_{\mathcal{A}b}^{\leq 0}$ with $\pi_n(A_\bullet) \cong H^{-n}(C_N(A_\bullet))$. Moreover, this functor preserves all homotopies, meaning this is moreover a Quillen equivalence.

Next, recall that since simplicial objects are functor categories, tensor products and universals are computed pointwise. The Eilenberg-Zilber shuffle map is a natural quasi-isomorphism

$$sh : C(A_\bullet) \otimes C(B_\bullet) \rightarrow C(A_\bullet \otimes B_\bullet)$$

natural in $A_\bullet, B_\bullet$. The exact definition is not useful here; we simply note that this natural quasi-isomorphism provides the information that C is a lax-monoidal functor $s\mathcal{Ch}_\mathcal{A} \rightarrow \mathcal{Ch}_\mathcal{A}$.

Finally, we will need to reinterpret the Čech complex as a particular simplicial set. We define a variant of the Čech nerve, which I refer to as the *factorizing nerve of $\mathcal{F}$ with values in $\mathfrak{U}$* :

$$\mathcal{N}(\mathfrak{U}, \mathcal{F})_n = \bigoplus_{j_0 \leq \dots \leq j_n} \mathcal{F}(U_J)^\bullet.$$

The faces and degeneracies are defined in the obvious way. We now show that $C_N(\mathcal{N}(\mathfrak{U}, \mathcal{F})_\bullet) \cong \check{C}(\mathfrak{U}, \mathcal{F})$. A fact that is shown in the Dold-Kan correspondence is that if $D(A_\bullet)$ is the subcomplex of $C(A_\bullet)$ spanned by degenerate simplicies, then

$$C_N(A_\bullet) \cong \frac{C(A_\bullet)}{D(A_\bullet)}.$$

Note that trivial simplicies of $\mathcal{N}(\mathfrak{U}, \mathcal{F})_\bullet$ are simply those simplicies belonging to $f\mathcal{A}(U_J)$ where J contains repeated indicies. Thus

$$C_N(\mathcal{N}(\mathfrak{U}, \mathcal{F})_\bullet)^{-n} \cong \bigoplus_{|J|=n+1} \mathcal{F}(U_J)^\bullet.$$

Writing this in totality rather than appealing to degrees,

$$C_N(\mathcal{N}(\mathfrak{U}, \mathcal{F})_\bullet)^\bullet \cong \bigoplus_{k \geq 0} \bigoplus_{|J|=k+1} \mathcal{F}(U_J)[k]^\bullet,$$

where the shift is to slot $\mathcal{F}(U_J)^\bullet$ into the $-k$ th degree. This is precisely the Čech complex.

Now, we begin the proof. First, recall that we define

$$\text{ext}(\mathcal{F})(V) := \check{C}(\mathfrak{U}_V, \mathcal{F}).$$

This construction immediately provides a precosheaf on X valued in cochain complexes. We define the structure maps on the level of simplicial cochain complexes, since many properties are more immediate at that level. Given disjoint open subsets $V, V' \subseteq X$, we define the factorization product

$$m_{VV'} : \text{ext}(\mathcal{F})(V) \otimes \text{ext}(\mathcal{F})(V') \rightarrow \text{ext}(\mathcal{F})(V \cup V')$$

which we recall means a map of cochains

$$m_{VV'} : \check{C}(\mathfrak{U}_V, \mathcal{F}) \otimes \check{C}(\mathfrak{U}_{V'}, \mathcal{F}) \rightarrow \check{C}(\mathfrak{U}_{V \cup V'}, \mathcal{F}).$$

This recovers all binary multiplication maps by postcomposing with a unary map

$$\begin{array}{ccc} \text{ext}(\mathcal{F})(V) \otimes \text{ext}(\mathcal{F})(V') & \xrightarrow{\quad} & \text{ext}(\mathcal{F})(W) \\ & \searrow \quad \swarrow & \\ & \text{ext}(\mathcal{F})(V \cup V'). & \end{array}$$

We define the map $m_{VV',n}$ at each level of the simplicial cochain complex, and pass it to the level of cochains using the functor C_N . Let $\{U_{i_k}\}_{0 \leq k \leq n} \subseteq \mathfrak{U}_V$ and $\{U_{j_k}\}_{0 \leq k \leq n} \subseteq \mathfrak{U}_{V'}$, and write U_I, U_J for their intersections, respectively. We define the map

$$m_{VV',n} : \mathcal{N}(\mathfrak{U}_V, \mathcal{F})_n \otimes \mathcal{N}(\mathfrak{U}_{V'}, \mathcal{F})_n \rightarrow \mathcal{N}(\mathfrak{U}_{V \cup V'}, \mathcal{F})_n$$

at the level of the direct summands $\mathcal{F}(U_I), \mathcal{F}(U_J)$. In particular, we define a map

$$\mathcal{F}(U_{i_0} \cap \cdots \cap U_{i_n}) \otimes \mathcal{F}(U_{j_0} \cap \cdots \cap U_{j_n}) \rightarrow \mathcal{F}((U_{i_0} \cap \cdots \cap U_{i_n}) \cup (U_{j_0} \cap \cdots \cap U_{j_n})).$$

Note that

$$(U_{i_0} \cap \cdots \cap U_{i_n}) \cup (U_{j_0} \cap \cdots \cap U_{j_n}) = (U_{i_0} \cup U_{j_0}) \cap \cdots \cap (U_{i_n} \cup U_{j_n})$$

by de Morgan's laws. Additionally, since $U_{i_k} \cap U_{j_k} = \emptyset$ since $V \cap V' = \emptyset$, since $\mathfrak{U}$ is a factorizing basis and $U_{i_k}, U_{j_k} \in \mathfrak{U}$, we have $U_{i_k} \cup U_{j_k} \in \mathfrak{U}$. Additionally, all finite intersections between U_i 's and U_j 's are in $\mathfrak{U}$, by the closure property. Thus,

$$(U_{i_0} \cap \cdots \cap U_{i_n}), (U_{j_0} \cap \cdots \cap U_{j_n}), (U_{i_0} \cup U_{j_0}) \cap \cdots \cap (U_{i_n} \cup U_{j_n}) \in \mathfrak{U}.$$

Since $\mathcal{F}$ is a $\mathfrak{U}$ -factorization algebra, there already exists a map

$$\mathcal{F}(U_{i_0} \cap \cdots \cap U_{i_n}) \otimes \mathcal{F}(U_{j_0} \cap \cdots \cap U_{j_n}) \rightarrow \mathcal{F}(U_{i_0} \cup U_{j_0}) \cap \cdots \cap (U_{i_n} \cup U_{j_n})$$

since all elements are in the factorizing basis. Thus, we define this map on all direct summands, and define $m_{VV',n}$ as the direct summand of these maps. Moreover, these levelwise maps manifestly

commute with faces and degeneracies (using functoriality and coherence data of the factorization product of $\mathcal{F}$ on $\mathfrak{U}$), so they assemble into a map of simplicial cochains. Finally, we recover the map $m_{VV'}$ on the level of cochains is recovered as the composition

$$\begin{array}{c} C_N(\mathcal{N}(\mathfrak{U}_V, \mathcal{F})_\bullet) \otimes C_N(\mathcal{N}(\mathfrak{U}_{V'}, \mathcal{F})_\bullet) \\ \downarrow sh \\ C_N(\mathcal{N}(\mathfrak{U}_V, \mathcal{F})_\bullet \otimes \mathcal{N}(\mathfrak{U}_{V'}, \mathcal{F})_\bullet) \\ \downarrow C_N(m_{VV'}, \bullet) \\ C_N(\mathcal{N}(\mathfrak{U}_{V \cup V'}, \mathcal{F})_\bullet). \end{array}$$

It is evident from this construction how to construct the factorization product on n disjoint opens. Associativity and equivariance of the factorization product are true on $\mathfrak{U}$ by definition, and thus are clearly true at the simplicial level for all opens. Thus, they are indeed true.

It remains only to be shown that $\mathcal{F}$ respects gluing/descent/whatever. The standard results for Čech cohomology apply, since our gluing axiom is the standard cosheaf axiom, just with respect to a particular class of covers. We will note these as we continue.

For $W \subseteq X$ open and $\mathfrak{W} = \{W_j\}_{j \in J}$ a Weiss cover of W , there are two important associated covers we will use.

- $\mathfrak{U}_W = \{U_i \subseteq W \mid U_i \in \mathfrak{U}\};$
- $\mathfrak{U}_{\mathfrak{W}} = \{U_i \mid \exists j \text{ s.t. } U_i \subseteq W_j\}.$

The first is just the Weiss cover of W induced by $\mathfrak{U}$. The latter consists of all opens in $\mathfrak{U}$ subordinate to $\mathfrak{W}$. Both are Weiss covers.

Lemma 7.2.3. *There is a natural quasi-isomorphism*

$$f : \check{C}(\mathfrak{W}, \text{ext}(\mathcal{F})) \simeq \check{C}(\mathfrak{U}_{\mathfrak{W}}, \mathcal{F})$$

for every Weiss cover $\mathfrak{W}$ of an open W .

Proof. This proof boils down to combinatorics of the covers. The book provides extra notation to clarify what is happenin. Given an n -fold intersection $W_{j_0} \cap \cdots \cap W_{j_n}$, we denote the intersection by $W_{\vec{j}}$. This departure from our previous notation is not only because the book uses this notation, but because we index $\mathfrak{W}$ by J , so using the notation W_J would be a bad conflict of notation. Denote by $\widehat{J^{n+1}}$ the subset of J^{n+1} whose tuples have no repeated entries. Then $\vec{j} \in \widehat{J^{n+1}}$.

First, we exhibit the map f of cochain complexes. The source complex $\check{C}(\mathfrak{W}, \text{ext}(\mathcal{F}))$ is defined by

$$\check{C}(\mathfrak{W}, \text{ext}(\mathcal{F})) = \bigoplus_{n \geq 0} \bigoplus_{\vec{j} \in \widehat{J^{n+1}}} \check{C}(\mathfrak{U}_{W_{\vec{j}}}, \mathcal{F})[n].$$

Each direct summand is another direct sum of terms of the form $\mathcal{F}(U_{\vec{i}})$ for $\vec{i} \in \widehat{I^{k+1}}$, $U_i \subseteq W$ and $U_i \in \mathfrak{U}$. The target complex is

$$\check{C}(\mathfrak{U}_{\mathfrak{W}}, \mathcal{F}) = \bigoplus_{n \geq 0} \bigoplus_{\vec{i} \in \widehat{I^{n+1}}} \mathcal{F}(U_{\vec{i}})[n],$$

where it is understood that each $U_i \in \mathfrak{U}_{\mathfrak{W}}$. We can show that each term $\mathcal{F}(U_{\vec{i}})$ from the source complex appears exactly once in the target complex. To show this, start by recalling $U_i \subseteq W$. ■

7.3 Pullback along an open immersion

Factorization algebras do not always pull back in a simple way. However, factorization algebras clearly restrict: given an open subset $U \subseteq X$ and a factorization algebra $\mathcal{F}$ on X , we can define a restriction factorization algebra $\mathcal{F}|_U$ on U . We can generalize this to open immersions.

For simplicity, let Y, X be smooth manifolds of the same dimension (since these are the only examples we will care about), and let $f : Y \rightarrow X$ be a local diffeomorphism. We will restrict to multiplicative factorization algebras. Equip X with a factorizing basis by fixing a Riemannian metric, and letting $\mathfrak{C} = \{C_i\}_{i \in I}$ be all strongly convex open sets in X (the unique minimizing geodesic between any two points is wholly contained in the open, and no other geodesic is contained in the open). This is a basis since it contains all balls, and is closed under intersection. Thus the Weiss cover $\overline{\mathfrak{C}}$ generated by $\mathfrak{C}$ is a factorizing basis.

Now define $\mathfrak{C}'$ to be all opens $U \subseteq Y$ such that $f(U) \in \mathfrak{C}$ (here we recall $f : Y \rightarrow X$). Then this also has a factorizing basis $\overline{\mathfrak{C}'}$ generated by $\mathfrak{C}'$.

Given a multiplicative factorization algebra $\mathcal{F}$ on X , let $\mathcal{F}_{\mathfrak{C}}$ denote the associated multiplicative factorization algebra on the basis $\overline{\mathfrak{C}}$. Define a multiplicative factorization algebra $f^*\mathcal{F}_{\mathfrak{C}'}$ on the basis $\overline{\mathfrak{C}'}$ as follows. For all $U \in \mathfrak{C}'$, let

$$(f^*\mathcal{F}_{\mathfrak{C}'})(U) = \mathcal{F}(f(U)).$$

Any $U \in \overline{\mathfrak{C}'}$ can be written as a finite disjoint union of opens $V_i \in \mathfrak{C}'$. By multiplicativeness, we have

$$(f^*\mathcal{F}_{\mathfrak{C}'})(V_1 \cup \dots \cup V_n) \simeq (f^*\mathcal{F}_{\mathfrak{C}'})(V_1) \otimes \dots \otimes (f^*\mathcal{F}_{\mathfrak{C}'})(V_n).$$

Thus, we define these two to be strictly equal to each other. It now suffices only to define the precosheaf structure, which is given by that of $\mathcal{F}$: for $U \subseteq V$ in $\mathfrak{C}'$, define

$$(f^*\mathcal{F}_{\mathfrak{C}'})(V) \rightarrow (f^*\mathcal{F}_{\mathfrak{C}'})(U)$$

as the map $\mathcal{F}(f(V)) \rightarrow \mathcal{F}(f(U))$ (here we recall the definition of $f^*\mathcal{F}_{\mathfrak{C}'}$). By multiplicativity, this determines all factorization products on $\overline{\mathfrak{C}'}$. By the previous section, $f^*\mathcal{F}_{\mathfrak{C}'}$ extends to a factorization algebra $f^*\mathcal{F}$ which is unique up to isomorphism.

7.4 Descent along a torsor

Let G be a discrete group acting properly discontinuously on a space X , so that $X \rightarrow X/G$ is a principal G -bundle. Then there is an equivalence of categories between G -equivariant multiplicative factorization algebras on X , and multiplicative factorization algebras on X/G , given in one direction by the pullback of the quotient.

Chapter 8

Factorization algebras: examples

8.1 Some examples of computations

The examples we now provide are very appealing in the sense that computing the global sections (i.e. the factorization homology) is straightforward, and does not require invoking the gluing axiom. Instead, we can use the theorems in the preceding sections to compute global sections. By analogy, consider how de Rham cohomology, which exploits partitions of unity and analysis, compares to the derived functor of global sections for the constant sheaf.

We focus in this section on examples we have already seen.

8.1.1 Enveloping algebras

Let $\mathfrak{g}$ be a Lie algebra and $\mathfrak{g}^{\mathbb{R}}$ the local Lie algebra $\Omega_{\mathbb{R}}^{\bullet} \otimes \mathfrak{g}$ on the real line $\mathbb{R}$. Recall that the factorization envelope $\mathbb{U}\mathfrak{g}^{\mathbb{R}}$ is simply the universal enveloping algebra $U\mathfrak{g}$. This factorization algebra can also be defined on the circle

Proposition 8.1.1. *There is a weak equivalence*

$$\mathbb{U}\mathfrak{g}^{\mathbb{R}}(S^1) \simeq C_{\bullet}(\mathfrak{g}, U\mathfrak{g}^{ad}) \simeq HH_{\bullet}(U\mathfrak{g})$$

where in the middle we mean the Lie algebra homology complex for $U\mathfrak{g}$ as a $\mathfrak{g}$ -module via the adjoint action

$$v \cdot a = va - av,$$

where $v \in \mathfrak{g}$ and $a \in U\mathfrak{g}$.

Proof. The second map is a standard result in Hochschild homology, so we prove the first. The wonderful fact here is that we need not use the Čech complex. De Rham's theorem provides a quasi-isomorphism between singular cochains and the de Rham complex. However, it is possible to show that S^1 is *formal*, i.e. there is a quasi-isomorphism $C^{\bullet}(S^1) \simeq H^{\bullet}(S^1)$ which preserves the dg algebra structure. Thus, there is a quasi-isomorphism $\Omega^{\bullet}(S^1) \simeq H^{\bullet}(S^1)$. Thus, there is a quasi-isomorphism of dg Lie algebras

$$\Omega^{\bullet}(S^1) \otimes \mathfrak{g} \simeq H^{\bullet}(S^1) \otimes \mathfrak{g} \cong (\mathbb{R} \oplus \mathbb{R}[-1]) \otimes \mathfrak{g} \cong \mathfrak{g} \oplus \mathfrak{g}[-1],$$

where, on the right, $\mathfrak{g}$ acts by the shifted adjoint action of $\mathfrak{g}[-1]$. Then of course

$$C_{\bullet}(\mathfrak{g} \oplus \mathfrak{g}[-1]) \cong C_{\bullet}(\mathfrak{g}, \text{Sym } \mathfrak{g}).$$

As vector spaces, $U\mathfrak{g} \cong \text{Sym } \mathfrak{g}$, so direct computation shows that the $\mathfrak{g}$ -module structure on $\text{Sym } \mathfrak{g}$ obtained from the Chevalley-Eilenberg homology complex is precisely the adjoint action of $\mathfrak{g}$ on $U\mathfrak{g}$. ■

8.1.2 The free scalar field in dimension 1

Recall that the factorization algebra of quantum observables Obs^q on the free scalar field theory on $\mathbb{R}$ was the Weyl algebra. Thus we expect that the factorization homology of Obs^q on a circle should have some relationship with the Hochschild homology of the Weyl algebra, but we will see that this relation depends on the mass of the theory, and the radius of the circle.

Let $S_L^1 = \mathbb{R}/\mathbb{Z}L$ be a circle with a rotation-invariant metric, radius r , and a total length $L = 2\pi r$. We work with $\mathbb{C}$ -linear observables, rather than $\mathbb{R}$ -linear.

Proposition 8.1.2. *If the mass $m = 0$, then $H^k(\text{Obs}^q(S_L^1))$ is $\mathbb{C}[\hbar]$ for $k = 0, -1$, and vanishes for all other k .*

If the mass satisfies $mL = in$ for some $n \in \mathbb{Z}$, then $H^k(\text{Obs}^q(S_L^1))$ is $\mathbb{C}[\hbar]$ for $k = 0, -2$, and vanishes for all other k .

For other values of m , the cohomology is $\mathbb{C}[\hbar]$ concentrated in degree 0.

The proof follows much of what I've seen in my quantum physics experience. For example, we use the topological basis of exponentials $e^{ikx/L}$, $k \in \mathbb{Z}$, and use

$$(\Delta + m^2)e^{ikx/L} = \left(\frac{k^2}{L^2} + m^2\right)e^{ikx/L}$$

to note that the complex $\mathcal{E}$ of fields has trivial cohomology when m^2L^2 is not a square integer. Then

$$H^\bullet(\text{Obs}^q) = H^\bullet(\text{Sym}(\mathcal{E}_c[1] \oplus \mathbb{C}\hbar)) \cong \text{Sym}(H^\bullet(\mathcal{E}_c[1]) \oplus H^\bullet(\mathbb{C}\hbar)) \cong \text{Sym}(\mathbb{C}\hbar) \cong \mathbb{C}[\hbar].$$

In the $m = 0$ case, we fairly easily use the constant functions in each degree of the cohomology of $\mathcal{L} = \mathcal{E} \oplus \mathbb{C}\hbar[-1]$ to compute $H^\bullet(C_\bullet \mathcal{L})$. A similar argument works for $m = in/L$, $n \in \mathbb{Z}$, by using the 2-dimensional kernel and cokernel of $\Delta + m^2$, and writing them as being spanned by $\pm e^{inx/L}$.

Part IV

Volume 2

Chapter 9

Introduction to classical field theory

I wanted to begin some light reading of concepts covered in volume 2, such as L_∞ -algebras, elliptic moduli problems, etc. Since I'm only reading for a few days, I am not making a new project for this, and am keeping it in this file. We will mostly be following part 1 of [CG20], as well as their appendices. We will be using significant supplemental material since they provide only a brief overview of many important topics.

We start by following chapter 2 of [CG20] for motivation. However, since the current focus is not on understanding the mathematical physics, but rather the mathematical tools, we will keep this chapter very brief.

We will be working perturbatively throughout this book. Our goal is to describe the derived space of solutions to the Euler-Lagrange equations. Since these are usually a system of PDEs on a manifold M , and those PDEs are usually elliptic in a particular sense, the space we want to describe will be called a *formal elliptic moduli problem*. Since one can solve the Euler-Lagrange equations on any open subset $U \subseteq M$, the correct object to use is a sheaf $\mathcal{EL}$ of formal elliptic moduli problems on M .

We then think of observables as some kind of function on $\mathcal{EL}$, so that

$$\mathrm{Obs}^d(U) = \mathcal{O}(\mathcal{EL}(U)).$$

We will be precise about what this means later. The factorization algebra structure arises from this idea. In finite dimensions, a formal moduli problem that arises as the derived critical locus of a function is equipped with a symplectic form of cohomological degree -1 . This symplectic form is quite useful, and in fact we later *define* a classical field theory to be a formal elliptic moduli problem equipped with a symplectic form of cohomological degree -1 . We can then show how to construct such a theory from a local action functional satisfying certain non-degeneracy properties. Finally, we look at a particular construction on an elliptic moduli problem $\mathcal{F}$ similar to that of the cotangent bundle. This is motivated by symplectic geometry, and turns out to be physically interesting.

In finite dimensions, the symplectic form for a formal moduli problem $\mathcal{F}$ yields a Poisson bracket of degree 1 for $\mathcal{O}(\mathcal{F})$, hence $\mathcal{O}(\mathcal{F})$ is a P_0 -algebra. We show that something similar happens in infinite dimensions: any classical field theory has a P_0 -structure on its factorization algebra of observables. This is the main payoff of ellipticity of the Euler-Lagrange equations.

Chapter 10

L_∞ algebras and moduli problems

10.1 Lie algebras and L_∞ algebras

Given a dg Lie algebra $\mathfrak{g}$, we have extensively used the Chevalley-Eilenberg chains and cochains to construct many important factorization algebras. It is necessary to make a homotopic generalization of the notion of a Lie algebra for interacting theories, and the category $\mathbf{Ch}_{\mathbb{K}}$ (here $\mathbb{K}$ is a field of characteristic 0) is a great place to work, since it is an ∞ -category via the dg nerve.

Definition 10.1.1. An L_∞ algebra over a ring R is a graded projective R -module $\mathfrak{g}$ equipped with multilinear maps

$$\ell_n : \mathfrak{g} \otimes_R \cdots \otimes_R \mathfrak{g} \rightarrow \mathfrak{g}[n-2]$$

for all $n \geq 1$, referred to as *brackets*, satisfying the following properties.

1. Each bracket ℓ_n is graded-alternating, meaning

$$\ell_n(x_1, \dots, x_i, x_{i+1}, \dots, x_n) = -(-1)^{|x_i||x_{i+1}|} \ell_n(x_1, \dots, x_{i+1}, x_i, \dots, x_n).$$

2. Each bracket ℓ_n satisfies the n -Jacobi rule: for all tuples $(x_1, \dots, x_n) \in \mathfrak{g}$,

$$0 = \sum_{k=1}^n (-1)^k \sum_{\substack{i_1 < \dots < i_k \\ j_{k+1} < \dots < j_n \\ \{i_1, \dots, j_n\} = \{1, \dots, n\}}} \varepsilon_{i_1, \dots, j_n} \ell_{n-k+1}(\ell_k(x_{i_1}, \dots, x_{i_k}), x_{j_{k+1}}, \dots, x_{j_n}).$$

Here, ε yields the proper sign for the alternating Koszul sign rule $ab = -(-1)^{|a||b|}ba$.

Part (2) is a tad confusing so I shall explain it. To be more explicit about the actual equation itself, I will write down some terms from the sum. We will let $x_1, x_2, x_3 \in \mathfrak{g}$, and work with the 3-Jacobi rule. First, we set $k = 1$. Then we have $\{i_1, j_2, j_3\} = \{1, 2, 3\}$. If we set $i_1 = 1$, $j_2 = 2$, and $j_3 = 3$, then we simply have

$$(-1)^k \varepsilon \ell_3(\ell_1(x_1), x_2, x_3).$$

Since there are no crossings, we have $\varepsilon = -(-1)^1 = 1$. Thus this term evaluates to simply

$$\ell_3(\ell_1(x_1), x_2, x_3).$$

If we were to switch x_1 and x_2 by setting $i_1 = 2$ and $j_2 = 1$, then we would have $\varepsilon = (-1)^{|x_1||x_2|}$, yielding the term

$$(-1)^k \varepsilon \ell_3(\ell_1(x_2), x_1, x_3) = -(-1)^{|x_1||x_2|} \ell_3(\ell_1(x_2), x_1, x_3).$$

From here, it should be clear how the rest of the terms are calculated.

We can use this to compute low dimensional examples of the brackets. For example, the 1-Jacobi rule simply says

$$0 = \sum_{k=1}^1 (-1)^k \sum_{i_1=1} \varepsilon_{i_1} \ell_{1-k+1}(\ell_k(x_{i_1})) = -(-1)^1 \ell_1(\ell_1(x_1)).$$

Since there are no crossings, $\varepsilon = 1$, so this evaluates to

$$0 = \ell_1(\ell_1(x_1)),$$

or more simply, $\ell_1 \circ \ell_1 = 0$. Thus, ℓ_1 is a differential. Let's temporarily write $d = \ell_1$ and $[-, -] = \ell_2$. We now consider the 2-Jacobi rule:

$$0 = \sum_{k=1}^2 (-1)^k \sum_{\substack{i_1 < \dots < i_k \\ j_{k+1} < \dots < j_n \\ \{i_1, \dots, j_n\} = \{1, \dots, n\}}} \varepsilon_{i_1, \dots, j_n} \ell_{n-k+1}(\ell_k(x_{i_1}, \dots, x_{i_k}), x_{j_{k+1}}, \dots, x_{j_n}).$$

There are two possible values for k , so we work with each one individually. First, consider the terms where $k = 1$. Let $x_1, x_2 \in \mathfrak{g}$. The list of all possible permutations, and the corresponding sign value, is

$$\begin{array}{ll} (x_1, x_2) & \varepsilon = -(-1)^1 = 1 \\ (x_2, x_1) & \varepsilon = -(-1)^{|x_1||x_2|}. \end{array}$$

The $k = 1$ component of the summand ranges over $i_1 = 1$ and $i_1 = 2$, hence it ranges over both permutations. Since $(-1)^k = -1$, the $k = 1$ component is simply

$$-[dx_1, x_2] + (-1)^{|x_1||x_2|}[dx_2, x_1].$$

The $k = 2$ component of the summand forces $i_1 = 1$ and $i_2 = 2$ due to the requirement that $i_1 < i_2$. Thus, it is simply

$$(-1)^2(-(-1)^1)\ell_{2-2+1}(\ell_2(x_1, x_2)) = d[x_1, x_2].$$

Putting it together, we have

$$0 = d[x_1, x_2] - [dx_1, x_2] + (-1)^{|x_1||x_2|}[dx_2, x_1].$$

We can write this in a more conventional form. First, we move $d[x_1, x_2]$ over to the other side to get

$$-d[x_1, x_2] = -[dx_1, x_2] + (-1)^{|x_1||x_2|}[dx_2, x_1].$$

We then multiply through by -1 to obtain

$$d[x_1, x_2] = [dx_1, x_2] - (-1)^{|x_1||x_2|}[dx_2, x_1].$$

Recall that ℓ_2 is graded alternating, meaning $[x, y] = -(-1)^{|x||y|}[y, x]$. Since $|dx_2| = |x_2| + 1$, we have that

$$-(-1)^{|x_1||x_2|} = -(-1)^{|x_1||x_2|}(-(-1)^{|x_1||x_2|+|x_1|})[x_1, dx_2] = (-1)^{2|x_1||x_2|+|x_1|}[x_1, dx_2] = (-1)^{|x_1|}[x_1, dx_2].$$

Thus the 2-Jacobi rule reads

$$d[x_1, x_2] = [dx_1, x_2] + (-1)^{|x_1|}[x_1, dx_2].$$

This is precisely the graded Leibniz rule. Additionally, note that due to the definition of the differential on the graded tensor product $\mathfrak{g} \otimes_R \mathfrak{g}$, this is also the requirement that ℓ_2 be a cochain map $\mathfrak{g} \otimes_R \mathfrak{g} \rightarrow \mathfrak{g}$.

Now we consider the 3-Jacobi rule. Let $x_1, x_2, x_3 \in \mathfrak{g}$. Then the possible permutations, and their corresponding ε value are

$$\begin{array}{ll} (x_1, x_2, x_3) & \varepsilon = -(-1)^1 = 1 \\ (x_1, x_3, x_2) & \varepsilon = -(-1)^{|x_2||x_3|} \\ (x_2, x_1, x_3) & \varepsilon = -(-1)^{|x_1||x_2|} \\ (x_2, x_3, x_1) & \varepsilon = (-1)^{|x_1||x_2|}(-1)^{|x_1||x_3|} \\ (x_3, x_2, x_1) & \varepsilon = -(-1)^{|x_1||x_2|}(-1)^{|x_1||x_3|}(-1)^{|x_2||x_3|} \\ (x_3, x_1, x_2) & \varepsilon = (-1)^{|x_1||x_3|}(-1)^{|x_2||x_3|}. \end{array}$$

Note that not all of these are possible, since we must have $i_1 < i_2 < i_3$ and likewise for the j 's. In fact, the only possible permutations are

$$\begin{aligned} (i_1, i_2, i_3) &= (i_1, i_2, j_3) = (i_1, j_2, j_3) = (x_1, x_2, x_3), & (i_1, i_2, j_3) &= (x_1, x_3, x_2), \\ (i_1, j_2, j_3) &= (x_2, x_1, x_3), & (i_1, i_2, j_3) &= (x_2, x_3, x_1), & (i_1, j_2, j_3) &= (x_3, x_1, x_2). \end{aligned}$$

Essentially, all permutations except (x_3, x_2, x_1) are possible. We once again work termwise under k . Start with $k = 3$. Then we must have 3 i 's, meaning there is only one possible permutation, $(i_1, i_2, i_3) = (x_1, x_2, x_3)$. Thus the $k = 3$ term is

$$(-1)^3(-(-1)^1)\ell_{3-3+1}(\ell_3(x_1, x_2, x_3)) = -d\ell_3(x_1, x_2, x_3).$$

The bracket for the $k = 2$ terms is $\ell_{3-2+1} \circ \ell_2 = \ell_2(\ell_2, -)$. Thus plugging in the terms with two i 's yields the $k = 2$ terms:

$$[[x_1, x_2], x_3] - (-1)^{|x_2||x_3|}[[x_1, x_3], x_2] + (-1)^{|x_1|(|x_2|+|x_3|)}[[x_2, x_3], x_1].$$

We will later be able to rearrange the brackets to get rid of these signs. For now, we compute the $k = 1$ terms. These all pick up a factor of -1 from $(-1)^k = -1$, which cancels the leading minus sign in all ε 's except for the (x_1, x_2, x_3) permutation, where $\varepsilon = 1$. Moreover, $\ell_{3-k+1} = \ell_3$, so the operator is simply $\ell_3(d, -, -)$. Taking all permutations with two j 's yields

$$-\ell_3(dx_1, x_2, x_3) + (-1)^{|x_1||x_2|}\ell_3(dx_2, x_1, x_3) - (-1)^{|x_3|(|x_1|+|x_2|)}\ell_3(dx_3, x_1, x_2).$$

We can now write the full 3-Jacobi rule:

$$\begin{aligned} 0 &= -d\ell_3(x_1, x_2, x_3) + [[x_1, x_2], x_3] - (-1)^{|x_2||x_3|}[[x_1, x_3], x_2] + (-1)^{|x_1|(|x_2|+|x_3|)}[[x_2, x_3], x_1] \\ &\quad - \ell_3(dx_1, x_2, x_3) + (-1)^{|x_1||x_2|}\ell_3(dx_2, x_1, x_3) - (-1)^{|x_3|(|x_1|+|x_2|)}\ell_3(dx_3, x_1, x_2). \end{aligned}$$

We will isolate the terms with ℓ_2 brackets from the terms with ℓ_3 brackets to begin putting this in a more conventional form. We also use $[x_1, x_3] = -(-1)^{|x_1||x_3|}[x_3, x_1]$. We rewrite

$$\begin{aligned} &[[x_1, x_2], x_3] + (-1)^{|x_3|(|x_1|+|x_2|)}[[x_3, x_1], x_2] + (-1)^{|x_1|(|x_2|+|x_3|)}[[x_2, x_3], x_1] \\ &= d\ell_3(x_1, x_2, x_3) + \ell_3(dx_1, x_2, x_3) - (-1)^{|x_1||x_2|}\ell_3(dx_2, x_1, x_3) + (-1)^{|x_3|(|x_1|+|x_2|)}\ell_3(dx_3, x_1, x_2). \end{aligned}$$

We will hereafter refer to this expression as the 3-Jacobi rule, and will make specific reference to its left hand side and right hand side.

We claim that this equation encodes the usual graded Jacobi identity up to homotopy, and this homotopy is encoded in the bracket ℓ_3 . Thus, we should first check that when $\ell_3 = 0$, we recover the usual graded Jacobi identity

$$(-1)^{|x||z|}[x, [y, z]] + (-1)^{|y||x|}[y, [z, x]] + (-1)^{|z||y|}[z, [x, y]] = 0.$$

Setting $\ell_3 = 0$ yields

$$[[x_1, x_2], x_3] + (-1)^{|x_3|(|x_1|+|x_2|)}[[x_3, x_1], x_2] + (-1)^{|x_1|(|x_2|+|x_3|)}[[x_2, x_3], x_1] = 0.$$

Note that $[[x, y]] = |x| + |y|$ since ℓ_2 has cohomological degree 0. Thus,

$$[x, [y, z]] = -(-1)^{|x|(|y|+|z|)}[[y, z], x].$$

We can apply this to the first and third term of our equation, allowing us to write

$$-(-1)^{|x_3|(|x_1|+|x_2|)}[x_3, [x_1, x_2]] + (-1)^{|x_3|(|x_1|+|x_2|)}[[x_3, x_1], x_2] - [x_1, [x_2, x_3]] = 0.$$

We did not apply this to the second term, because we would like to treat that term separately. We have

$$(-1)^{|x_3|(|x_1|+|x_2|)}[[x_3, x_1], x_2] = -(-1)^{|x_3|(|x_1|+|x_2|)+|x_2|(|x_3|+|x_1|)}[x_2, [x_3, x_1]] = -(-1)^{|x_1|(|x_3|+|x_2|)}[x_2, [x_3, x_1]].$$

The last equality is given by expanding out the exponent and removing the factor of $2|x_2||x_3|$. Substituting this in and distributing in all the exponents, we have

$$-(-1)^{|x_3||x_1|+|x_3||x_2|}[x_3, [x_1, x_2]] - (-1)^{|x_1|(|x_3|+|x_2|)}[x_2, [x_3, x_1]] - [x_1, [x_2, x_3]] = 0.$$

Note that both terms with an exponent contain an $|x_1||x_3|$ term. We can now simply multiply through by $-(-1)^{|x_3||x_1|}$ to obtain

$$(-1)^{|x_2||x_3|}[x_3, [x_1, x_2]] + (-1)^{|x_1||x_2|}[x_2, [x_3, x_1]] + (-1)^{|x_1||x_3|}[x_2, [x_3, x_1]] = 0.$$

This is precisely the graded Jacobi identity. Thus, if $\mathfrak{g}$ is an L_∞ algebra with $\ell_n = 0$ for all $n \geq 3$, then $\mathfrak{g}$ is precisely a differential graded Lie R -algebra.

Now, we will consider the case where $\ell_3 \neq 0$. We claim that the datum of ℓ_3 allows us to pass to cohomology, and to have the graded Jacobi identity hold in cohomology. First, we note that if $x = d\eta$ is exact and y is closed, then by the graded Leibniz rule,

$$[x, y] = [d\eta, y] = d[\eta, y] - (-1)^{|\eta|}[\eta, dy] = d[\eta, y] - 0 = d[\eta, y].$$

Thus, $[x, y]$ is exact. This is the standard argument that a dgla's cohomology inherits a well-defined bracket: given any exact form, the bracket with that exact form is necessarily killed.

We can apply similar logic to the 3-Jacobi rule. Let x_1, x_2, x_3 now be closed elements, such that they define a cohomology class. The right hand side of the 3-Jacobi rule reads

$$d\ell_3(x_1, x_2, x_3) + \ell_3(dx_1, x_2, x_3) - (-1)^{|x_1||x_2|}\ell_3(dx_2, x_1, x_3) + (-1)^{|x_3|(|x_1|+|x_2|)}\ell_3(dx_3, x_1, x_2).$$

Since each x_i is closed, $dx_i = 0$ for all i , so the latter three ℓ_3 -brackets necessarily vanish, leaving just

$$d\ell_3(x_1, x_2, x_3).$$

Note that this is an exact element. Thus, when evaluating on closed elements, we have

$$[[x_1, x_2], x_3] + (-1)^{|x_3|(|x_1|+|x_2|)}[[x_3, x_1], x_2] + (-1)^{|x_1|(|x_2|+|x_3|)}[[x_2, x_3], x_1] = d\ell_3(x_1, x_2, x_3).$$

Alternatively, using the more conventional graded Jacobi identity (and recalling that to derive it, we had to multiply through by $(-1)^{|x_1||x_3|}$), we can write

$$(-1)^{|x_1||x_3|}[x_1, [x_2, x_3]] + (-1)^{|x_1||x_2|}[x_2, [x_3, x_1]] + (-1)^{|x_2||x_3|}[x_3, [x_1, x_2]] = d\ell_3(x_1, x_2, x_3).$$

Regardless of how we write it, the point is that the left hand side is an exact element, and thus vanishes in cohomology. Therefore, the Jacobi identity is satisfied in cohomology.

Finally, we can prove that ℓ_3 is a homotopy which guarentees this relation. This is a trivial application of basic results, which we now recall. First, a cochain map $f \simeq 0$ if and only if there is a homotopy h with $D(h) = f$, for D the differential on the internal hom $\underline{\mathcal{C}h}_R(\text{dom } f, \text{cod } f)$. In this case, we use $\underline{\mathcal{C}h}_R(\mathfrak{g}^{\otimes 3}, \mathfrak{g})$. Next, recall that since a homotopy h has cohomological degree -1 , we have $D(h) = d_{\mathfrak{g}} \circ h + h \circ d_{\mathfrak{g}^{\otimes 3}}$. Finally, recall that given differential graded R -modules M, N , the differential is given by

$$d_{M \otimes_R N}(m \otimes n) = d_M(m) \otimes n + (-1)^{|m|} m \otimes d_N(n).$$

Applying this here, we have

$$D(\ell_3)(x_1, x_2, x_3) = d(\ell_3(x_1, x_2, x_3)) + \ell_3(dx_1 \otimes x_2 \otimes x_3 + (-1)^{|x_1|} x_1 \otimes dx_2 \otimes x_3 + (-1)^{|x_1|+|x_2|} x_1 \otimes x_2 \otimes dx_3).$$

Since ℓ_3 is 3-multilinear, and thus linear in triple tensors, this can be rewritten as

$$D(\ell_3)(x_1, x_2, x_3) = d\ell_3(x_1, x_2, x_3) + \ell_3(dx_1, x_2, x_3) + (-1)^{|x_1|} \ell_3(x_1, dx_2, x_3) + (-1)^{|x_1|+|x_2|} \ell_3(x_1, x_2, dx_3).$$

We claim that we can exactly recover the right hand side of the 3-Jacobi rule from this expression. We have already recovered the first two terms, so we simply need to match the signs on the remaining two. For the first one, we have

$$(-1)^{|x_1|} \ell_3(x_1, dx_2, x_3) = -(-1)^{|x_1|+|x_1|+|x_1||x_2|} \ell_3(dx_2, x_1, x_3) = -(-1)^{|x_1||x_2|} \ell_3(dx_2, x_1, x_3),$$

which is exactly the term that shows up in the 3-Jacobi rule. Finally, the last remaining term

$$(-1)^{|x_1|+|x_2|} \ell_3(x_1, x_2, dx_3) = (-1)^{|x_1|+|x_2|+|x_2||x_3|+|x_2|+|x_1||x_3|+|x_1|} \ell_3(dx_3, x_1, x_2) = (-1)^{|x_3|(|x_1|+|x_2|)} \ell_3(dx_3, x_1, x_2),$$

which is the exact final term in the 3-Jacobi rule. Therefore, ℓ_3 is a *cochain homotopy* expressing the fact that the usual graded Jacobi identity is null-homotopic. We have now finally done good on our motivation of L_∞ algebras as homotopy Lie algebras.

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